

Vel Tech Group of Institutions

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Department: CSE-AIML

Batch: 2028

Degree: B.Tech - Computer Science and Engineering with AIML

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2028_NeoColab_Competitive Coding II_L9

VTU_Week 1_COD_SB

Attempt : 1

Total Mark : 50

Marks Obtained : 50

Section 1 : Coding

1. Problem Statement

Arjun is working on a project where he needs to analyze a series of integers within a specified range $[A, B]$. The project involves identifying certain numbers that are not divisible by any smaller positive integers other than 1 and themselves. These numbers are crucial for the next step of his analysis, as they have unique mathematical properties.

Arjun has been asked to implement an efficient way to find all such numbers in a given range. Given that the range can be large, he needs to ensure the solution is optimized.

Can you help him implement this task using the Simple Sieve of Eratosthenes?

Input Format

The first line of input consists of an integer A, representing the starting range.

The second line of input consists of an integer B, representing the ending range.

Output Format

The output prints a space-separated list of numbers that are not divisible by any smaller positive integers other than 1 and themselves. within the range [A, B].

Refer to the sample output for formatting specifications

Sample Test Case

Input: 2

10

Output: 2 3 5 7

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
#include <stdbool.h>
```

```
int main() {
```

```
    int A, B;
```

```
    scanf("%d", &A);
```

```
    scanf("%d", &B);
```

```
    bool prime[B + 1];
```

```
    for(int i = 0; i <= B; i++) {
```

```
        prime[i] = true;
```

```
    }
```

```
    prime[0] = false;
```

```
    prime[1] = false;
```

```
    for(int i = 2; i * i <= B; i++) {
```

```
        if(prime[i]) {
```

```
            for(int j = i * i; j <= B; j += i) {
```

```
                prime[j] = false;
```

```
            }
```

```
}  
}  
  
for(int i = A; i <= B; i++) {  
    if(prime[i]) {  
        printf("%d ", i);  
    }  
}  
  
return 0;  
}
```

Status : Correct

Marks : 10/10

2. Problem Statement

Joanna is developing a utility tool for a math club's quiz event. One of the rounds involves finding the greatest common divisor (GCD) of a list of numbers. Instead of computing it manually for each pair, she wants to automate the task using code.

She decides to use Euclid's Algorithm, a well-known and efficient method to compute the GCD of two numbers. For a list of numbers, she will apply the algorithm iteratively across all elements in the array.

Help Joanna implement a program that takes an array of positive integers and computes the GCD of the entire array using Euclid's Algorithm.

Make sure the program also handles special cases:

If the array is empty, return -1. If the array has only one number, return that number as the GCD.

Example

Input:

3

12 18 24

Output:

6

Explanation:

$\text{GCD}(12, 18) = 6$
 $\text{GCD}(6, 24) = 6$
Final result = 6

Input Format

The first line of input contains an integer n , representing the number of elements in the array.

The second line contains n space-separated positive integers, each representing an element of the array.

Output Format

The output consists of a single line:

- If the array contains two or more elements, print a single integer representing the GCD of all array elements.
- If the array contains only one element, print that element as the GCD.
- If the array is empty, print -1 as a convention to indicate undefined GCD.

Refer to the sample outputs for the formatting specifications.

Sample Test Case

Input: 2

8 12

Output: 4

Answer

```
// You are using GCC
#include <stdio.h>
```

```
int gcd(int a, int b) {
    while (b != 0) {
```

```
int temp = b;
b = a % b;
a = temp;
}
return a;
}

int main() {
    int n;
    scanf("%d", &n);

    if (n == 0) {
        printf("-1");
        return 0;
    }

    int arr[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    if (n == 1) {
        printf("%d", arr[0]);
        return 0;
    }

    int result = arr[0];
    for (int i = 1; i < n; i++) {
        result = gcd(result, arr[i]);
    }

    printf("%d", result);

    return 0;
}
```

Status : Correct

Marks : 10/10

3. Problem Statement

Raychel is working on a calendar synchronization tool where she needs to determine how often two repeating tasks will coincide. Each task occurs after a fixed number of days, and she must find the least number of days after which both tasks align again — in other words, the least common multiple (LCM) of the two intervals.

Raychel knows that the most efficient way to compute the LCM is by first calculating the greatest common divisor (GCD) using Euclid's Algorithm, and then using that result to derive the LCM. She also wants to make sure the program works for large inputs without running into overflow errors.

Help Raychel implement a program that calculates the LCM of two non-negative integers by first using Euclid's Algorithm to compute their GCD, and then deriving the LCM from it.

Example

Input:

15 20

Explanation:

$\text{GCD}(15, 20) = 5$
 $\text{LCM} = (15 / 5) \times 20 = 3 \times 20 = 60$

Output:

60

Input Format

The input consists of a single line containing two space-separated integers 'a' and 'b', representing the two input numbers.

Output Format

The output consists of a single line:

- If both 'a' and 'b' are non-zero, print a single integer – the least number that is divisible by both.
- If either 'a' or b is '0', print 0 as the result since no common multiple exists in such cases.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 15 20

Output: 60

Answer

```
// You are using GCC
#include <stdio.h>
```

```
long long gcd(long long a, long long b) {
    while (b != 0) {
        long long temp = b;
        b = a % b;
        a = temp;
    }
    return a;
}
```

```
int main() {
    long long a, b;
    scanf("%lld %lld", &a, &b);
```

```
    if (a == 0 || b == 0) {
        printf("0");
        return 0;
    }
```

```
    long long g = gcd(a, b);
    long long lcm = (a / g) * b;
```

```
    printf("%lld", lcm);
```

```
    return 0;
```

}

Status : Correct

Marks : 10/10

4. Problem Statement:

A software company is developing a mathematical tool for students to help them understand factorization and modular arithmetic. As part of the tool, they need a function that calculates the product of all positive divisors (factors) of a given integer, with the result taken modulo 1000000007. The tool is intended to assist in solving mathematical problems and providing insights into number theory.

Your task is to implement a function that computes the product of all factors of a given integer n , and then display the result modulo 1000000007.

Examples :

Input: 12

Output: 1728

$$1 * 2 * 3 * 4 * 6 * 12 = 1728$$

Input Format

The input consists of a single integer n .

Output Format

The output displays a single integer which is the product of all factors of n modulo 1000000007.

Refer to the sample output for better understanding.

Sample Test Case

Input: 12

Output: 1728

Answer

```
// You are using GCC
#include <stdio.h>

int main() {
    int n;
    long long product = 1;
    const long long MOD = 1000000007;

    scanf("%d", &n);

    for (int i = 1; i <= n; i++) {
        if (n % i == 0) {
            product = (product * i) % MOD;
        }
    }

    printf("%lld", product);

    return 0;
}
```

Status : Correct

Marks : 10/10

5. Problem Statement

Pradeep Narwal is analyzing multiple numerical ranges to identify prime numbers within each range. He is given several ranges defined by a starting number A and an ending number B. Using the Sieve of Eratosthenes, Pradeep wants to efficiently determine all prime numbers that lie within each range.

Write a program to ensure that all prime numbers between A and B (inclusive) are printed for every given range.

Input Format

The first line of input consists of an integer Q, representing the number of ranges.

The next Q lines each contain two integers A and B , representing the start and end of each range.

Output Format

For each range, print all prime numbers between A and B (inclusive) in a single line.

Prime numbers must be printed in ascending order, separated by a single space.

If a range contains no prime numbers, print an empty line.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 2

1 5

10 15

Output: 2 3 5

11 13

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
#include <stdbool.h>
```

```
#define MAX 1000000
```

```
int main() {
```

```
    int Q;
```

```
    scanf("%d", &Q);
```

```
    bool prime[MAX + 1];
```

```
    for (int i = 0; i <= MAX; i++)
```

```
        prime[i] = true;
```

```
    prime[0] = prime[1] = false;
```

```
for (int i = 2; i * i <= MAX; i++) {  
    if (prime[i]) {  
        for (int j = i * i; j <= MAX; j += i) {  
            prime[j] = false;  
        }  
    }  
}  
  
while (Q--) {  
    int A, B;  
    scanf("%d %d", &A, &B);  
  
    for (int i = A; i <= B; i++) {  
        if (prime[i]) {  
            printf("%d ", i);  
        }  
    }  
    printf("\n");  
}  
  
return 0;  
}
```

Status : Correct

Marks : 10/10

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2028_NeoColab_Competitive Coding II_L9

VTU_Week 1_COD_CY

Attempt : 1

Total Mark : 40

Marks Obtained : 40

Section 1 : Coding

1. Problem Statement

In the field of cryptography, especially in public-key algorithms like RSA, it is crucial to identify pairs of numbers that are coprime — i.e., they do not share any common divisors other than 1.

Before generating secure keys, cryptographic systems often verify whether two large numbers are coprime. This process uses a classical method known as Euclid's Algorithm, which efficiently determines the greatest common divisor (GCD).

You are given a list of number pairs. Your task is to determine whether each pair of integers is coprime using Euclid's Algorithm. If the GCD of the two numbers is 1, output "Coprime". Otherwise, output "Not Coprime".

Example

Input:

3

15 28

12 35

8 8

Explanation:

$\text{GCD}(15, 28) = 1$ Coprime

$\text{GCD}(12, 35) = 1$ Coprime

$\text{GCD}(8, 8) = 8$ Not Coprime

Output:

Coprime

Coprime

Not Coprime

Input Format

- The first line contains an integer 't' — the number of pairs to check.
- Each of the next 't' lines contains two space-separated integers 'a' and 'b'.

Output Format

- For each pair, output a single line:
- "Coprime" if the two numbers are coprime
- "Not Coprime" otherwise

- Each output must appear on a new line, in the same order as the input pairs.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 1

17 29

Output: Coprime

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
long long gcd(long long a, long long b) {  
    while (b != 0) {  
        long long temp = b;  
        b = a % b;  
        a = temp;  
    }  
    return a;  
}
```

```
int main() {  
    int t;  
    long long a, b;  
  
    scanf("%d", &t);  
  
    for (int i = 0; i < t; i++) {  
        scanf("%lld %lld", &a, &b);  
  
        if (gcd(a, b) == 1)  
            printf("Coprime\n");  
        else  
            printf("Not Coprime\n");  
    }  
  
    return 0;  
}
```

Status : Correct

Marks : 10/10

2. Problem Statement

Sai Kishore is working on a tool that identifies special numbers in a range. These numbers must have only two divisors: 1 and themselves. His task is to count how many of these special numbers exist below a given threshold N . Since N can be as large as one million, Sai Kishore needs an efficient way to find these numbers in a given range.

Sai Kishore decides to use a technique called Sieve of Eratosthenes, which is well-suited for efficiently identifying these special numbers.

Can you help him implement a program that counts how many such numbers exist below N ?

Example:

Input:

10

Output:

4

Explanation:

The numbers less than 10 that have only two divisors (1 and themselves) are: 2, 3, 5, and 7. Thus, the output is 4.

Input Format

The input consists of a single integer N , representing the upper bound for the search.

Output Format

The output should be a single integer, representing the count of special numbers less than N .

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 10

Output: 4

Answer

```
// You are using GCC
#include <stdio.h>
#include <stdbool.h>

int main() {
    int N;
    scanf("%d", &N);

    bool prime[N];

    for(int i = 0; i < N; i++)
        prime[i] = true;

    prime[0] = false;
    prime[1] = false;

    for(int i = 2; i * i < N; i++) {
        if(prime[i]) {
            for(int j = i * i; j < N; j += i) {
                prime[j] = false;
            }
        }
    }

    int count = 0;
    for(int i = 2; i < N; i++) {
        if(prime[i])
            count++;
    }

    printf("%d", count);

    return 0;
}
```

Status : Correct

Marks : 10/10

3. Problem Statement

Arjun is analyzing a numerical range to find special pairs of prime numbers. He is given a range defined by two integers L and R, and a target value T. Within the range [L,R], Arjun wants to identify all pairs of prime numbers whose sum equals T.

Write a program to ensure that all such valid prime pairs are printed in ascending order.

Input Format

The first line contains three space-separated integers:

- L starting value of the range
- R ending value of the range
- T target sum

Output Format

The output prints each valid pair of prime numbers whose sum equals T, one pair per line.

Each line should contain two space-separated integers representing a valid prime pair.

If no such pairs exist, no output should be printed.

Refer to the sample output for the formatting specifications

Sample Test Case

Input: 1 40 10

Output: 3 7

5 5

Answer

```
// You are using GCC
#include <stdio.h>
#include <stdbool.h>

#define MAX 1000000
```

```

int main() {
    int L, R, T;
    scanf("%d %d %d", &L, &R, &T);

    bool prime[MAX + 1];

    for(int i = 0; i <= MAX; i++)
        prime[i] = true;

    prime[0] = prime[1] = false;

    for(int i = 2; i * i <= MAX; i++) {
        if(prime[i]) {
            for(int j = i * i; j <= MAX; j += i) {
                prime[j] = false;
            }
        }
    }

    for(int p = L; p <= R; p++) {
        int q = T - p;

        if(q < p) continue; // avoid duplicate pairs
        if(q >= L && q <= R && prime[p] && prime[q]) {
            printf("%d %d\n", p, q);
        }
    }

    return 0;
}

```

Status : Correct

Marks : 10/10

4. Problem Statement

Bella is developing a number puzzle game where each level involves simplifying equations using number theory. In one level, she comes across two large positive integers and is asked to reduce them using their greatest common divisor (GCD).

However, the puzzle has an extra twist:

She must express the GCD as a linear combination of the two numbers – that is, she needs to find two integers x and y such that: $\text{GCD}(a, b) = a * x + b * y$

Help Bella write a program that, given two integers a and b , finds:

The GCD of a and b The coefficients ' x ' and ' y ' that satisfy the linear combination condition

Example

Input

48 18

Output:

6 -1 3

Explanation:

The GCD of 48 and 18 is 6.

Using the Extended Euclidean Algorithm, we can find integers $x = -1$ and $y = 3$ such that:

$$48 \times (-1) + 18 \times 3 = -48 + 54 = 6$$

This satisfies Bézout's Identity: $\text{GCD}(48, 18) = 48x + 18y = 6$.

Input Format

- The input consists of a single line with two space-separated integers ' a ' and ' b '

Output Format

The output prints three space-separated integers in a single line:

- The first integer is the GCD of a and b .
- The second integer is the coefficient x such that $a * x + b * y = \text{GCD}$.
- The third integer is the coefficient y such that $a * x + b * y = \text{GCD}$.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 45 21

Output: 3 1 -2

Answer

// You are using GCC

#include <stdio.h>

```
long long gcdExtended(long long a, long long b, long long *x, long long *y) {
```

```
    if (b == 0) {
```

```
        *x = 1;
```

```
        *y = 0;
```

```
        return a;
```

```
    }
```

```
    long long x1, y1;
```

```
    long long gcd = gcdExtended(b, a % b, &x1, &y1);
```

```
    *x = y1;
```

```
    *y = x1 - (a / b) * y1;
```

```
    return gcd;
```

```
}
```

```
int main() {
```

```
    long long a, b, x, y;
```

```
    scanf("%lld %lld", &a, &b);
```

```
    long long g = gcdExtended(a, b, &x, &y);
```

```
    printf("%lld %lld %lld", g, x, y);
```

```
    return 0;
```

```
}
```

Status : Correct

Marks : 10/10

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2028_NeoColab_Competitive Coding II_L9

VTU_Week 1_COD_PAH

Attempt : 1

Total Mark : 40

Marks Obtained : 39

Section 1 : Coding

1. Problem Statement

Haritha is working on a project to identify pairs of numbers in a given range $[L, R]$ where each number in the pair is divisible only by 1 and itself.

She needs to find pairs of such numbers that differ by exactly 2. These pairs of numbers should be displayed efficiently within the range.

Help her to implement the task.

Input Format

The input consists of two integers L and R separated by a new line.

Output Format

The output prints each pair of numbers within the range $[L, R]$ on a new line, in the format: a b where a and b are numbers that differ by exactly 2 and are

divisible only by 1 and themselves.

Refer to the sample output for formatting specifications

Sample Test Case

Input: 30

60

Output: 41 43

Answer

// You are using GCC

```
#include <stdio.h>
```

```
#include <stdbool.h>
```

```
#define MAX 1000000
```

```
int main() {
```

```
    int L, R;
```

```
    scanf("%d", &L);
```

```
    scanf("%d", &R);
```

```
    bool prime[MAX + 1];
```

```
    for(int i = 0; i <= MAX; i++)
```

```
        prime[i] = true;
```

```
    prime[0] = false;
```

```
    prime[1] = false;
```

```
    for(int i = 2; i * i <= MAX; i++) {
```

```
        if(prime[i]) {
```

```
            for(int j = i * i; j <= MAX; j += i)
```

```
                prime[j] = false;
```

```
        }
```

```
    }
```

```
    for(int i = L; i + 2 <= R; i++) {
```

```
        if(prime[i] && prime[i + 2]) {
```

```
            printf("%d %d\n", i, i + 2);
```

```
}  
}  
return 0;  
}
```

Status : Correct

Marks : 10/10

2. Problem Statement

Neha is designing a secret code system for her cybersecurity class. She discovers that certain numbers with exactly two distinct divisors (1 and themselves) create the most secure keys when their reversed digit sum is divisible by a given validation number K.

Neha needs to find all such numbers up to a specified limit N that meet both conditions. These numbers are crucial for the security of her code system. Given the large range of numbers, she needs an efficient solution to identify them.

Help her write a program to find all such numbers $\leq N$ that meet this criteria.

Input Format

The input consists of two space separated integers N and K

Output Format

The output prints all numbers that meet both conditions. Each number should be printed space-separated.

Refer to the sample output for formatting specifications

Sample Test Case

Input: 10 1

Output: 2 3 5 7

Answer

```

#include <stdio.h>
#include <stdbool.h>

// Function to check if a number is prime
bool isPrime(int num) {
    if (num <= 1) return false;
    for (int i = 2; i * i <= num; i++) {
        if (num % i == 0) return false;
    }
    return true;
}

// Function to calculate the reversed digit sum
int reversedDigitSum(int num) {
    int sum = 0;
    while (num > 0) {
        sum += num % 10;
        num /= 10;
    }
    return sum;
}

int main() {
    int N, K;
    scanf("%d %d", &N, &K);

    for (int i = 2; i <= N; i++) {
        if (isPrime(i) && reversedDigitSum(i) % K == 0) {
            printf("%d ", i);
        }
    }

    return 0;
}

```

Status : Correct

Marks : 10/10

3. Problem Statement

Ashwin is working on a system to identify numbers within a given range [L, R] that have a special property: they are divisible only by 1 and themselves.

Ashwin needs to find pairs of such numbers where both numbers are rearrangements of each other's digits. Ashwin must efficiently identify all such pairs in the given range.

Help him to implement the task.

Input Format

The input consists of two space separated integers L and R

Output Format

The output prints each pair of numbers within the range [L, R] that satisfy both conditions: they are divisible only by 1 and themselves, and they are rearrangements of each other's digits. Each pair is printed in the format: p1 p2, where p1 and p2 are numbers that are rearrangements of each other's digits.

Refer to the sample output for formatting specifications

Sample Test Case

Input: 10 100

Output: 13 31

17 71

37 73

79 97

Answer

```
#include <stdio.h>
```

```
#include <stdbool.h>
```

```
#include <string.h>
```

```
// Function to check if a number is prime
```

```
bool isPrime(int num) {
```

```
    if (num <= 1) return false;
```

```
    for (int i = 2; i * i <= num; i++) {
```

```
        if (num % i == 0) return false;
```

```
    }
```

```
    return true;
```

```
}
```

```
// Function to sort digits of a number
```

```
int sortDigits(int num) {  
    char str[10];  
    sprintf(str, "%d", num);  
    int len = strlen(str);  
    for (int i = 0; i < len; i++) {  
        for (int j = i + 1; j < len; j++) {  
            if (str[i] > str[j]) {  
                char temp = str[i];  
                str[i] = str[j];  
                str[j] = temp;  
            }  
        }  
    }  
    return atoi(str);  
}
```

```
int main() {  
    int L, R;  
    scanf("%d %d", &L, &R);  
  
    for (int i = L; i <= R; i++) {  
        if (isPrime(i)) {  
            int sortedI = sortDigits(i);  
            for (int j = i + 1; j <= R; j++) {  
                if (i < j && isPrime(j) && sortDigits(j) == sortedI) {  
                    printf("%d %d ", i, j);  
                }  
            }  
        }  
    }  
    printf("\n");  
    return 0;  
}
```

Status : Partially correct

Marks : 9/10

4. Problem Statement

Suresh is studying prime numbers and is interested in identifying a special

category of primes. Given a number N, he wants to find all prime numbers less than or equal to N whose sum of digits is also a prime number. To solve this efficiently, Suresh decides to use the Sieve of Eratosthenes for prime number generation.

Write a program to ensure that all such prime numbers are printed in increasing order.

Input Format

The input consists of single integer N, representing the upper limit of the range.

Output Format

The output prints all prime numbers less than or equal to N whose digit sum is also a prime number.

The numbers must be printed in ascending order, separated by a single space.

If no such numbers exist, print nothing.

Refer to the sample output for the formatting specifications

Sample Test Case

Input: 50

Output: 2 3 5 7 11 23 29 41 43 47

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
#include <stdbool.h>
```

```
#define MAX 1000000
```

```
int digitSum(int n) {  
    int sum = 0;  
    while(n > 0) {  
        sum += n % 10;  
        n /= 10;  
    }  
}
```

```

    return sum;
}

int main() {
    int N;
    scanf("%d", &N);

    bool prime[MAX + 1];

    for(int i = 0; i <= N; i++)
        prime[i] = true;

    prime[0] = prime[1] = false;

    for(int i = 2; i * i <= N; i++) {
        if(prime[i]) {
            for(int j = i * i; j <= N; j += i)
                prime[j] = false;
        }
    }

    for(int i = 2; i <= N; i++) {
        if(prime[i]) {
            int s = digitSum(i);

            if(s >= 2 && prime[s]) {
                printf("%d ", i);
            }
        }
    }

    return 0;
}

```

Status : Correct

Marks : 10/10

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2028_NeoColab_Competitive Coding II_L9

VTU_Week 2_COD

Attempt : 1

Total Mark : 50

Marks Obtained : 50

Section 1 : Coding

1. Problem Statement

Neha is organizing a sequence of integers where some elements are zeroes. She wants to rearrange the sequence so that all zeroes are moved to the end while keeping the relative order of the non-zero elements intact. Help Neha perform this task efficiently using the two-pointer technique to modify the sequence in place.

Input Format

The first line of input consists of an integer n , representing the size of the array `nums[]`.

The second line contains n integers, representing the elements of the array `nums[]`.

Output Format

The output prints the modified array `nums[]`, with all zeroes moved to the end and the relative order of the non-zero elements maintained.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 5

0 1 0 3 12

Output: 1 3 12 0 0

Answer

```
// You are using GCC
#include <stdio.h>
```

```
int main() {
    int n;
    scanf("%d", &n);

    int arr[100];

    for(int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    int j = 0;

    for(int i = 0; i < n; i++) {
        if(arr[i] != 0) {
            int temp = arr[i];
            arr[i] = arr[j];
            arr[j] = temp;
            j++;
        }
    }

    for(int i = 0; i < n; i++) {
        printf("%d ", arr[i]);
    }
}
```

```
    return 0;  
}
```

Status : Correct

Marks : 10/10

2. Problem Statement

Arjun has a list of integers representing a specific sequence, and he wants to find its next permutation in lexicographical order. If the sequence is the largest permutation possible, he should rearrange it into the smallest permutation. Help Arjun solve this problem efficiently using the two-pointer technique to rearrange the sequence in place with constant extra memory.

Input Format

The first line of input consists of an integer n , representing the size of the array `nums[]`.

The second line contains n integers, representing the elements of the array `nums[]`.

Output Format

The output prints "Next permutation:" followed by n space-separated integers on the next line, representing the next lexicographically greater permutation of the given sequence.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 3

1 2 3

Output: Next permutation:

1 3 2

Answer

```
// You are using GCC  
#include <stdio.h>
```

```
void reverse(int arr[], int start, int end) {  
    while(start < end) {  
        int temp = arr[start];  
        arr[start] = arr[end];  
        arr[end] = temp;  
        start++;  
        end--;  
    }  
}
```

```
int main() {  
    int n;  
    scanf("%d", &n);  
  
    int nums[10];  
    for(int i = 0; i < n; i++) {  
        scanf("%d", &nums[i]);  
    }  
  
    int i = n - 2;  
  
    while(i >= 0 && nums[i] >= nums[i + 1]) {  
        i--;  
    }  
  
    if(i >= 0) {  
        int j = n - 1;  
        while(nums[j] <= nums[i]) {  
            j--;  
        }  
  
        int temp = nums[i];  
        nums[i] = nums[j];  
        nums[j] = temp;  
    }  
  
    reverse(nums, i + 1, n - 1);  
  
    printf("Next permutation:\n");  
    for(int k = 0; k < n; k++) {  
        printf("%d ", nums[k]);  
    }  
}
```



```
} return 0;
```

Status : Correct

Marks : 10/10

3. Problem Statement

Priya is searching for a specific word (needle) in a large text (haystack). She needs to find the index of the first occurrence of the word in the text. If the word is not present, she should return -1. Help Priya solve this efficiently using the two-pointer technique.

Input Format

The first line of input consists of a string haystack, representing the text in which the word is to be searched.

The second line of input consists of a string needle, representing the word to search for in the text.

Output Format

The output prints "The index of the first occurrence of needle in haystack is: X", where X represents the index value.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: sadbutsad
sad

Output: The index of the first occurrence of needle in haystack is: 0

Answer

```
// You are using GCC
#include <stdio.h>
#include <string.h>

int main() {
```

```

char haystack[31], needle[31];

scanf("%s", haystack);
scanf("%s", needle);

int n = strlen(haystack);
int m = strlen(needle);
int index = -1;

for(int i = 0; i <= n - m; i++) {
    int j = 0;

    while(j < m && haystack[i + j] == needle[j]) {
        j++;
    }

    if(j == m) {
        index = i;
        break;
    }
}

printf("The index of the first occurrence of needle in haystack is: %d", index);

return 0;
}

```

Status : Correct

Marks : 10/10

4. Problem Statement

Ramesh is working with an array of integers and needs to remove all occurrences of a specific value from it.

To accomplish this efficiently, he uses the Two-Pointer technique, where one pointer scans the array and another pointer tracks the position to place valid elements. Given an integer array `nums` and an integer `val`, Ramesh must modify the array in-place so that all elements equal to `val` are removed while preserving the order of the remaining elements.

Write a program to ensure that the number of remaining elements and the

modified array are printed correctly.

Input Format

The first line contains an integer n, representing the number of elements in the array.

The second line contains n space-separated integers, representing the elements of the array.

The third line contains an integer val, representing the value to be removed from the array.

Output Format

The first line prints "Number of elements after removal:" followed by count of elements not equal to val.

The second line prints "Modified array:"

Then print the modified array elements (up to index k) separated by a single space.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 4

3 2 2 3

3

Output: Number of elements after removal: 2

Modified array:

2 2

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
int main() {
```

```
    int n;
```

```
    scanf("%d", &n);
```

```

int nums[20];
for(int i = 0; i < n; i++) {
    scanf("%d", &nums[i]);
}

int val;
scanf("%d", &val);

int k = 0;

for(int i = 0; i < n; i++) {
    if(nums[i] != val) {
        nums[k] = nums[i];
        k++;
    }
}

printf("Number of elements after removal: %d\n", k);
printf("Modified array:\n");

for(int i = 0; i < k; i++) {
    printf("%d ", nums[i]);
}

return 0;
}

```

Status : Correct

Marks : 10/10

5. Problem Statement

Anil is working with a sorted array of integers and wants to remove duplicate elements efficiently. To solve this problem, he uses the Two-Pointer technique, where one pointer scans the array and another pointer keeps track of the position to store unique elements. Given a sorted integer array `nums`, Anil must modify the array in-place so that each unique element appears only once.

Write a program to ensure that the number of unique elements and the

modified array are printed correctly.

Input Format

The first line contains an integer n, representing the number of elements in the array.

The second line contains n space-separated integers, representing the elements of the sorted array.

Output Format

The first line prints "Number of unique elements: k"

On the next line, print "Modified array:"

Then, print the first k elements of the modified array, separated by a single space.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 8

-1 -1 0 0 1 1 2 2

Output: Number of unique elements: 4

Modified array:

-1 0 1 2

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
int main() {
```

```
    int n;
```

```
    scanf("%d", &n);
```

```
    int nums[20];
```

```
    for(int i = 0; i < n; i++) {
```

```
        scanf("%d", &nums[i]);
```

```
}  
int k = 1;  
  
for(int i = 1; i < n; i++) {  
    if(nums[i] != nums[k - 1]) {  
        nums[k] = nums[i];  
        k++;  
    }  
}  
  
printf("Number of unique elements: %d\n", k);  
printf("Modified array:\n");  
  
for(int i = 0; i < k; i++) {  
    printf("%d ", nums[i]);  
}  
  
return 0;  
}
```

Status : Correct

Marks : 10/10

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2028_NeoColab_Competitive Coding II_L9

VTU_Week 2_CY

Attempt : 1

Total Mark : 40

Marks Obtained : 40

Section 1 : Coding

1. Problem Statement

Karthik is organizing marbles of three different colors: red, white, and blue. Each color is represented by an integer—0 for red, 1 for white, and 2 for blue. Karthik wants to sort the marbles so that all marbles of the same color are adjacent, and they are arranged in the order red, white, and blue. Help Karthik solve this problem efficiently using the two-pointer technique.

Input Format

The first line of input consists of an integer n , representing the number of marbles in the array `nums[]`.

The second line contains n integers, representing the marbles' colors, where each integer is either 0, 1, or 2.

Output Format

The output prints the sorted array, where all marbles are grouped by color in the order red (0), white (1), and blue (2).

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 6

2 0 2 1 1 0

Output: 0 0 1 1 2 2

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
int main() {
```

```
    int n;
```

```
    scanf("%d", &n);
```

```
    int nums[100];
```

```
    for(int i = 0; i < n; i++) {
```

```
        scanf("%d", &nums[i]);
```

```
    }
```

```
    int low = 0, mid = 0, high = n - 1;
```

```
    while(mid <= high) {
```

```
        if(nums[mid] == 0) {
```

```
            int temp = nums[low];
```

```
            nums[low] = nums[mid];
```

```
            nums[mid] = temp;
```

```
            low++;
```

```
            mid++;
```

```
        }
```

```
        else if(nums[mid] == 1) {
```

```
            mid++;
```

```
        }
```

```
        else {
```

```
            int temp = nums[mid];
```

```
            nums[mid] = nums[high];
```



```

        nums[high] = temp;
        high--;
    }
}

for(int i = 0; i < n; i++) {
    printf("%d ", nums[i]);
}

return 0;
}

```

Status : Correct

Marks : 10/10

2. Problem Statement

Kavya is an avid reader who loves finding hidden patterns in texts. One day, she stumbled upon a string of characters and wondered about the longest palindromic substring it contains. Help Kavya identify this substring using an efficient algorithm based on the two-pointer technique.

Input Format

The input consists of a single string *s*.

Output Format

The output displays "Longest Palindromic Substring: " followed by the longest palindrome found.

Refer the sample output for format specifications.

Sample Test Case

Input: babad

Output: Longest Palindromic Substring: bab

Answer

```

// You are using GCC
#include <stdio.h>

```

```
#include <string.h>
```

```
int main() {  
    char s[25];  
    scanf("%s", s);
```

```
  
    int n = strlen(s);  
    int start = 0, maxLen = 1;
```

```
    for(int i = 0; i < n; i++) {
```

```
        // Odd length palindrome
```

```
        int l = i, r = i;
```

```
        while(l >= 0 && r < n && s[l] == s[r]) {
```

```
            if(r - l + 1 > maxLen) {
```

```
                start = l;
```

```
                maxLen = r - l + 1;
```

```
            }
```

```
            l--;
```

```
            r++;
```

```
        }
```

```
        // Even length palindrome
```

```
        l = i;
```

```
        r = i + 1;
```

```
        while(l >= 0 && r < n && s[l] == s[r]) {
```

```
            if(r - l + 1 > maxLen) {
```

```
                start = l;
```

```
                maxLen = r - l + 1;
```

```
            }
```

```
            l--;
```

```
            r++;
```

```
        }
```

```
    }
```

```
    printf("Longest Palindromic Substring: ");
```

```
    for(int i = start; i < start + maxLen; i++) {
```

```
        printf("%c", s[i]);
```

```
    }
```

```
    return 0;
```

```
}
```

Status : Correct

Marks : 10/10

3. Problem Statement

Rohit is analyzing an array of integers to find special combinations of numbers. Using the Two-Pointer technique, he wants to identify all unique triplets in the array whose sum is equal to 0. Given an integer array, Rohit must find and print all such triplets, ensuring that no duplicate triplets are included in the result.

Write a program to ensure that all valid triplets with sum zero are printed correctly.

Example 1

Input:

6

-1 0 1 2 -1 -4

Output:

-1 -1 2

-1 0 1

Explanation:

$\text{nums}[0] + \text{nums}[1] + \text{nums}[2] = (-1) + 0 + 1 = 0.$

$\text{nums}[1] + \text{nums}[2] + \text{nums}[4] = 0 + 1 + (-1) = 0.$

$\text{nums}[0] + \text{nums}[3] + \text{nums}[4] = (-1) + 2 + (-1) = 0.$

The distinct triplets are $[-1, 0, 1]$ and $[-1, 1, 2]$.

Notice that the order of the output and the order of the triplets does not matter.

Example 2

Input:

3

0 1 1

Output:

No triplets found with sum 0.

Explanation:

The only possible triplet does not sum up to 0.

Example 3

Input:

3

0 0 0

Output:

0 0 0

Explanation:

The only possible triplet sums up to 0.

Input Format

The first line contains an integer n , representing the number of elements in the array.

The second line contains n space-separated integers, representing the elements of the array.

Output Format

The output prints each unique triplet whose sum is 0 on a separate line.

Each triplet must be printed in ascending order (as they appear after sorting).

If no such triplets exist, print "No triplets found with sum 0."

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 6

-1 0 1 2 -1 -4

Output: -1 -1 2

-1 0 1

Answer

```
// You are using GCC
#include <iostream>
#include <vector>
#include <algorithm>
using namespace std;
```

```
int main() {
    int n;
    cin >> n;

    vector<int> nums(n);
    for(int i = 0; i < n; i++) {
        cin >> nums[i];
    }

    sort(nums.begin(), nums.end());

    bool found = false;

    for(int i = 0; i < n - 2; i++) {

        if(i > 0 && nums[i] == nums[i - 1])
            continue;

        int left = i + 1;
        int right = n - 1;

        while(left < right) {
            int sum = nums[i] + nums[left] + nums[right];

            if(sum == 0) {
                cout << nums[i] << " " << nums[left] << " " << nums[right] << " " << endl;
```

```

        found = true;

        while(left < right && nums[left] == nums[left + 1])
            left++;
        while(left < right && nums[right] == nums[right - 1])
            right--;

        left++;
        right--;
    }
    else if(sum < 0)
        left++;
    else
        right--;
}
}

if(!found) {
    cout << "No triplets found with sum 0.";
}

return 0;
}

```

Status : Correct

Marks : 10/10

4. Problem Statement

Rahul is given an array of integers and a target value T. Using the Two-Pointer technique, he wants to find two distinct elements in the array such that the sum of the pair is closest to the target value. To apply this technique efficiently, the array must first be sorted.

Write a program to ensure that the pair of numbers whose sum is closest to T is identified and printed correctly.

Input Format

The first line contains two space-separated integers n and T, where:

- n is the number of elements in the array

- T is the target sum

The second line contains n space-separated integers, representing the elements of the array.

Output Format

The output prints two space separated integers representing the pair whose sum is closest to the target.

The output pair must be printed in ascending order.

If multiple pairs have the same closest sum, print the first pair found by the two-pointer traversal.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 6 50
10 22 28 29 30 40

Output: 10 40

Answer

```
// You are using GCC
#include <stdio.h>
#include <stdlib.h>
```

```
int compare(const void *a, const void *b) {
    return (*(int*)a - *(int*)b);
}
```

```
int main() {
    int n, T;
    scanf("%d %d", &n, &T);
```

```
    int arr[20];
    for(int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }
```

```
qsort(arr, n, sizeof(int), compare);

int left = 0, right = n - 1;
int best1 = arr[left], best2 = arr[right];
int minDiff = abs((arr[left] + arr[right]) - T);

while(left < right) {
    int sum = arr[left] + arr[right];
    int diff = abs(sum - T);

    if(diff < minDiff) {
        minDiff = diff;
        best1 = arr[left];
        best2 = arr[right];
    }

    if(sum < T)
        left++;
    else
        right--;
}

printf("%d %d", best1, best2);

return 0;
}
```

Status : Correct

Marks : 10/10

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2028_NeoColab_Competitive Coding II_L9

VTU_Week 2_PAH

Attempt : 1

Total Mark : 40

Marks Obtained : 38.5

Section 1 : Coding

1. Problem Statement

At AquaPark, the maintenance team has a row of vertical water tanks of varying heights arranged side by side along a riverbank. The team wants to identify two tanks that, together with the riverbank, can hold the maximum volume of water between them when the space between them is filled. Each tank's height represents its capacity, and the distance between tanks determines the width of the container formed.

Your task is to read the heights of the tanks and determine the maximum amount of water (volume) that can be contained between any two tanks.

Input Format

The first line of input contains an integer n representing the number of water tanks.

The second line of input contains n space-separated integers representing the heights of the tanks.

Output Format

The output prints a single integer representing the maximum volume of water that can be contained between two tanks.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 5

1 2 3 4 5

Output: 6

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
int min(int a, int b) {  
    return (a < b) ? a : b;  
}
```

```
int main() {  
    int n;  
    scanf("%d", &n);
```

```
    int height[n];  
    for(int i = 0; i < n; i++) {  
        scanf("%d", &height[i]);  
    }
```

```
    int left = 0, right = n - 1;  
    int maxVolume = 0;
```

```
    while(left < right) {  
        int width = right - left;  
        int h = min(height[left], height[right]);  
        int volume = width * h;
```

```
    if(volume > maxVolume)
        maxVolume = volume;

    if(height[left] < height[right])
        left++;
    else
        right--;
}

printf("%d", maxVolume);

return 0;
}
```

Status : Correct

Marks : 10/10

2. Problem Statement

Dhanya is tasked with writing a program to merge data from two sensors into a single unified dataset. Each sensor produces an array of data readings over time. The goal is to create a program that takes the data arrays from two sensors, merges them, and produces a single sorted array of merged data readings.

Can you assist Dhanya in this using two-pointer technique?

Input Format

The first line of input consists of an integer n , representing the number of readings produced by the first sensor.

The second line consists of an integer m , representing the number of readings produced by the second sensor.

The third line consists of n space-separated integers, representing the readings from the first sensor.

The fourth line consists of m space-separated integers, representing the readings from the second sensor.

Output Format

The output prints the merged and sorted sensor readings in ascending order, separated by space.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 3

4

1 2 8

3 5 6 7

Output: 1 2 3 5 6 7 8

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
int main() {
```

```
    int n, m;
```

```
    scanf("%d", &n);
```

```
    scanf("%d", &m);
```

```
    int a[20], b[20], merged[40];
```

```
    for(int i = 0; i < n; i++)
```

```
        scanf("%d", &a[i]);
```

```
    for(int i = 0; i < m; i++)
```

```
        scanf("%d", &b[i]);
```

```
    int i = 0, j = 0, k = 0;
```

```
    while(i < n && j < m) {
```

```
        if(a[i] <= b[j]) {
```

```
            merged[k++] = a[i++];
```

```
        } else {
```

```
            merged[k++] = b[j++];
```

```
        }
```

```

    }
    while(i < n) {
        merged[k++] = a[i++];
    }

    while(j < m) {
        merged[k++] = b[j++];
    }

    for(int x = 0; x < k; x++) {
        printf("%d ", merged[x]);
    }

    return 0;
}

```

Status : Correct

Marks : 10/10

3. Problem Statement

At ShopEase, two branches maintain sorted lists of product IDs available in their inventories. The regional manager wants to find the list of product IDs that are common to both branches to plan joint promotions.

Your task is to read the sorted lists of product IDs from both branches, find their intersection and print the common IDs in sorted order. If there are no common products, print "No Common Intersection".

Input Format

The first line of input contains an integer m representing the number of products in the first branch.

The second line of input contains m space-separated integers representing the sorted product IDs in the first branch.

The third line of input contains an integer n representing the number of products in the second branch.

The fourth line of input contains n space-separated integers representing the

sorted product IDs in the second branch.

Output Format

The output prints the common product IDs in sorted order, space-separated.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 5

101 102 103 104 105

6

102 104 106 108 110 115

Output: 102 104

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
int main() {
```

```
    int m, n;
```

```
    scanf("%d", &m);
```

```
    int a[1000], b[1000];
```

```
    for(int i = 0; i < m; i++)
```

```
        scanf("%d", &a[i]);
```

```
    scanf("%d", &n);
```

```
    for(int i = 0; i < n; i++)
```

```
        scanf("%d", &b[i]);
```

```
    int i = 0, j = 0;
```

```
    int found = 0;
```

```
    while(i < m && j < n) {
```

```
        if(a[i] == b[j]) {
```

```
            printf("%d ", a[i]);
```

```
            found = 1;
```

```

        i++;
        j++;
    }
    else if(a[i] < b[j]) {
        i++;
    }
    else {
        j++;
    }
}

if(!found) {
    printf("No Common Intersection");
}

return 0;
}

```

Status : Correct

Marks : 10/10

4. Problem Statement

At FreshMart Grocery, the cashier needs to quickly find pairs of products whose prices add up to a specific gift card value, so they can suggest bundles to customers using their gift cards efficiently.

You are given a sorted list of product prices. Your task is to find all unique pairs of products whose prices sum exactly to the given gift card value. If no such pairs exist, print "No Pairs Found".

Input Format

The first line of input contains an integer n representing the number of products.

The second line of input contains n space-separated integers representing the sorted prices of the products.

The third line of input contains an integer k representing the gift card value.

Output Format

The output prints each pair of product prices that sum to k on a separate line,

with the two prices separated by a space.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 6
2 3 4 5 6 8
10

Output: 2 8
4 6

Answer

```
// You are using Java
import java.util.*;
```

```
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        long[] arr = new long[n];

        for(int i = 0; i < n; i++) {
            arr[i] = sc.nextLong();
        }

        long k = sc.nextLong();

        int left = 0, right = n - 1;
        boolean found = false;

        while(left < right) {
            long sum = arr[left] + arr[right];

            if(sum == k) {
                System.out.println(arr[left] + " " + arr[right]);
                found = true;
                left++;
                right--;
            }
        }
    }
}
```



```
    }  
    else if(sum < k) {  
        left++;  
    }  
    else {  
        right--;  
    }  
}  
  
if(!found) {  
    System.out.print("No Pairs Found");  
}  
}  
}
```

Status : Partially correct

Marks : 8.5/10

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2028_NeoColab_Competitive Coding II_L9

VTU_Week 3_MCQ

Attempt : 1

Total Mark : 15

Marks Obtained : 15

Section 1 : MCQ

1. The main idea behind the Knuth-Morris-Pratt algorithm is to:

Answer

Avoid redundant checking of characters that have already been matched

Status : Correct

Marks : 1/1

2. The Boyer-Moore algorithm uses which two rules to determine the shift amount?

Answer

Good suffix rule and bad character rule

Status : Correct

Marks : 1/1

3. The Boyer-Moore algorithm is particularly efficient when

Answer

The pattern contains repetitive elements

Status : Correct

Marks : 1/1

4. In the KMP algorithm, if the prefix table at position i has a value of k , what does it imply?

Answer

The length of the longest proper prefix which is also a suffix up to position i is k .

Status : Correct

Marks : 1/1

5. The concept of prefix and suffix is used in which of the following algorithms?

Answer

Knuth-Morris-Pratt (KMP) algorithm

Status : Correct

Marks : 1/1

6. What is the main purpose of Manacher's Algorithm?

Answer

To find the longest palindromic substring in linear time

Status : Correct

Marks : 1/1

7. In the Z-algorithm, the Z-array is used to store:

Answer

The length of the longest substring starting at each position that matches the prefix

Status : Correct

Marks : 1/1

8. Which application is most efficiently solved using Z-algorithm?

Answer

Finding all occurrences of a pattern in a text

Status : Correct

Marks : 1/1

9. In Z-algorithm, when $Z[k]$ is greater than 0, what does it imply?

Answer

Substring starting at k matches the prefix for length $Z[k]$

Status : Correct

Marks : 1/1

10. Which of the following best explains why Manacher's algorithm is preferable for palindrome problems over expand-around-center approaches?

Answer

Avoids recomputation of overlapping palindromes by mirroring results around the center

Status : Correct

Marks : 1/1

11. In bioinformatics applications, KMP is often used instead of Boyer-Moore. Why?

Answer

DNA has only 4 characters, reducing Boyer-Moore's skipping advantage

Status : Correct

Marks : 1/1

12. In KMP, the LPS array for pattern "ABABAC" is:

Answer

[0,0,1,2,3,0]

Status : Correct

Marks : 1/1

13. Which is true about Manacher's algorithm with respect to handling even-length palindromes?

Answer

Uses a modified string with separators between characters

Status : Correct

Marks : 1/1

14. How is Z-algorithm used for pattern search in text efficiently?

Answer

Concatenate P + "\$" + T and compute Z-array

Status : Correct

Marks : 1/1

15. Which of the following strings has the largest number of palindromic substrings using Manacher's algorithm?

Answer

"aaaaa"

Status : Correct

Marks : 1/1

Vel Tech Group of Institutions

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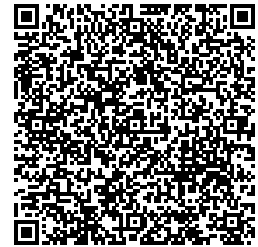
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2028_NeoColab_Competitive Coding II_L9

VTU_Week 3_COD

Attempt : 1

Total Mark : 50

Marks Obtained : 50

Section 1 : Coding

1. Problem Statement

Anil is working on a project that involves counting occurrences of a specific pattern within a given text. He needs a program that efficiently finds and counts all occurrences of a pattern within a given text using the Knuth-Morris-Pratt (KMP) algorithm.

Your task is to help Anil by developing a program that takes a text and a pattern as input and counts the number of times the pattern appears in the text using the KMP algorithm.

Input Format

The first line contains a string txt, representing the text in which Anil wants to search for the pattern.

The second line contains a string `pat`, representing the pattern Anil wants to search for in the text.

Output Format

The output displays a single integer, representing the count of occurrences of the pattern `pat` in the text `txt`.

If the pattern is not found, nothing will be printed.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: AABAACAADAABAABA
AABA

Output: 3

Answer

// You are using GCC

#include <stdio.h>

#include <string.h>

```
void computeLPS(char *pat, int M, int *lps) {
```

```
    int len = 0;
```

```
    lps[0] = 0;
```

```
    int i = 1;
```

```
    while (i < M) {
```

```
        if (pat[i] == pat[len]) {
```

```
            len++;
```

```
            lps[i] = len;
```

```
            i++;
```

```
        } else {
```

```
            if (len != 0) {
```

```
                len = lps[len - 1];
```

```
            } else {
```

```
                lps[i] = 0;
```

```
                i++;
```

```
            }
```

```
        }
```

```

    }
}

int KMPSearch(char *txt, char *pat) {
    int N = strlen(txt);
    int M = strlen(pat);

    int lps[M];
    computeLPS(pat, M, lps);

    int i = 0, j = 0;
    int count = 0;

    while (i < N) {
        if (pat[j] == txt[i]) {
            i++;
            j++;
        }

        if (j == M) {
            count++;
            j = lps[j - 1];
        } else if (i < N && pat[j] != txt[i]) {
            if (j != 0)
                j = lps[j - 1];
            else
                i++;
        }
    }

    return count;
}

```

```

int main() {
    char txt[51], pat[26];

    scanf("%s", txt);
    scanf("%s", pat);

    int result = KMPSearch(txt, pat);

    if (result > 0)

```



```
printf("%d", result);  
return 0;  
}
```

Status : Correct

Marks : 10/10

2. Problem Statement

Rohan is developing a text analysis tool to efficiently find repeated patterns inside large documents. To speed up the searching process, he uses the Boyer–Moore string matching algorithm, specifically relying on the Bad Character Heuristic as the main data structure/technique.

Given a text string and a pattern string, write a program to ensure that the task is done by counting how many times the pattern occurs in the text.

Input Format

The first line contains a string representing the text (may contain spaces and special characters).

The second line contains a string representing the pattern to be searched.

Output Format

The output prints a single integer representing the number of occurrences of the pattern in the text.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: ababababa
ab

Output: 4

Answer

// You are using GCC

```

#include <stdio.h>
#include <string.h>

#define MAX_CHAR 256

// Function to build bad character table
void badCharHeuristic(char *str, int size, int badchar[MAX_CHAR]) {
    int i;

    // Initialize all occurrences as -1
    for (i = 0; i < MAX_CHAR; i++)
        badchar[i] = -1;

    // Fill the actual value of last occurrence
    for (i = 0; i < size; i++)
        badchar[(int)str[i]] = i;
}

// Function to count pattern occurrences
int countPattern(char *txt, char *pat) {
    int m = strlen(pat);
    int n = strlen(txt);

    int badchar[MAX_CHAR];
    badCharHeuristic(pat, m, badchar);

    int count = 0;
    int s = 0; // shift of the pattern

    while (s <= (n - m)) {
        int j = m - 1;

        // Keep reducing index j while characters match
        while (j >= 0 && pat[j] == txt[s + j])
            j--;

        // Pattern found
        if (j < 0) {
            count++;

            // Shift pattern so next character aligns
            s += (s + m < n) ? m - badchar[(int)txt[s + m]] : 1;
        }
    }
}

```

```

    } else {
        // Shift using bad character rule
        int shift = j - badchar[(int)txt[s + j]];
        s += (shift > 1) ? shift : 1;
    }
}

return count;
}

int main() {
    char text[101];
    char pattern[101];

    // Read full line (including spaces)
    fgets(text, sizeof(text), stdin);
    fgets(pattern, sizeof(pattern), stdin);

    // Remove newline characters if present
    text[strcspn(text, "\n")] = '\0';
    pattern[strcspn(pattern, "\n")] = '\0';

    int result = countPattern(text, pattern);

    printf("%d", result);

    return 0;
}

```

Status : Correct

Marks : 10/10

3. Problem Statement

Kiran is working on a string analysis module where he needs to identify repeating patterns efficiently. To achieve this, he uses the LPS (Longest Prefix Suffix) array from the KMP string matching algorithm as the main data structure.

Given a string, write a program to ensure that the task is done by finding the length of the longest proper prefix that is also a suffix, ensuring the

prefix and suffix do not overlap.

Example

Input:

aabcdaabc

Output:

4

Explanation:

The string "aabc" is the longest prefix which is also a suffix. So the length of the string "aabc" is printed as 4.

Input Format

The input consists of a string S, representing the input string.

Output Format

The output prints an integer representing the length of the longest proper prefix which is also a suffix.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: aabcdaabc

Output: 4

Answer

```
#include <stdio.h>
#include <string.h>
```

```
int main() {
    char s[101];
    scanf("%s", s);
```

```

int n = strlen(s);
int lps[101];

lps[0] = 0;
int len = 0;
int i = 1;

// Build LPS array
while (i < n) {
    if (s[i] == s[len]) {
        len++;
        lps[i] = len;
        i++;
    } else {
        if (len != 0) {
            len = lps[len - 1];
        } else {
            lps[i] = 0;
            i++;
        }
    }
}

int res = lps[n - 1];

// Ensure no overlap
while (res > n / 2) {
    res = lps[res - 1];
}

printf("%d", res);

return 0;
}

```

Status : Correct

Marks : 10/10

4. Problem Statement

Liam is a software engineer who loves analyzing strings for patterns. One day, while reviewing usernames in a system, he noticed that some

usernames read the same backward and forward (palindromes). He wants to count all palindromic sequences of characters in a username efficiently.

To accomplish this, Liam decides to use Manacher's Algorithm, which finds all palindromic substrings in linear time.

Given a string representing the username, help Liam count all the possible palindromic substrings using Manacher's Algorithm.

Input Format

The input consists of a single line containing the string S.

The string consists of lowercase English letters only.

Output Format

The output prints a single integer, representing the total number of palindromic substrings present in the username.

Refer to the sample input and output for format specifications.

Sample Test Case

Input: abaa

Output: 6

Answer

```
// You are using GCC
#include <stdio.h>
#include <string.h>
```

```
int main() {
    char s[101];
    scanf("%s", s);
```

```
    int n = strlen(s);
```

```
    // Transform string: ^#a#b#a#a#$
    char t[205];
```

```

int j = 0;
t[j++] = '^';
for (int i = 0; i < n; i++) {
    t[j++] = '#';
    t[j++] = s[i];
}
t[j++] = '#';
t[j++] = '$';
t[j] = '\0';

int len = j;
int p[205] = {0};

int center = 0, right = 0;

// Manacher's Algorithm
for (int i = 1; i < len - 1; i++) {
    int mirror = 2 * center - i;

    if (i < right)
        p[i] = (right - i < p[mirror]) ? right - i : p[mirror];

    // Expand palindrome
    while (t[i + (1 + p[i])] == t[i - (1 + p[i])]) {
        p[i]++;
    }

    // Update center and right
    if (i + p[i] > right) {
        center = i;
        right = i + p[i];
    }
}

// Count palindromes
int count = 0;
for (int i = 1; i < len - 1; i++) {
    count += (p[i] + 1) / 2;
}

printf("%d", count);

```

```
    return 0;  
}
```

Status : Correct

Marks : 10/10

5. Problem Statement

Aryan is working on a text processing project where he needs to analyze strings to find patterns efficiently. He decides to use the Z-algorithm, which for each position in a string, computes the length of the longest substring starting from that position which is also a prefix of the string.

Your task is to implement a program that computes the Z-array for a given string.

Input Format

The input consists of a single line containing the string S.

S may contain only uppercase and lowercase English letters.

Output Format

The output prints the Z-array of S as a sequence of integers separated by spaces.

The i-th element of the Z-array represents the length of the longest substring starting at position i which matches the prefix of S.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: aabxaab

Output: 0 1 0 0 3 1 0

Answer

```
// You are using GCC  
#include <stdio.h>
```



```
#include <string.h>
```

```
int main() {  
    char s[101];  
    scanf("%s", s);
```

```
  
    int n = strlen(s);  
    int Z[101];
```

```
  
    Z[0] = 0;
```

```
  
    int L = 0, R = 0;
```

```
    for (int i = 1; i < n; i++) {  
        if (i > R) {  
            L = R = i;  
            while (R < n && s[R] == s[R - L]) {  
                R++;  
            }  
            Z[i] = R - L;  
            R--;  
        } else {  
            int k = i - L;  
  
            if (Z[k] < R - i + 1) {  
                Z[i] = Z[k];  
            } else {  
                L = i;  
                while (R < n && s[R] == s[R - L]) {  
                    R++;  
                }  
                Z[i] = R - L;  
                R--;  
            }  
        }  
    }  
}
```

```
// Print result
```

```
for (int i = 0; i < n; i++) {  
    printf("%d ", Z[i]);  
}
```

```
} return 0;
```

Status : Correct

Marks : 10/10

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2028_NeoColab_Competitive Coding II_L9

VTU_Week 3_CY

Attempt : 1

Total Mark : 40

Marks Obtained : 40

Section 1 : Coding

1. Problem Statement

Dhiya is a data analyst working for a data compression company. One of her tasks is to analyze logs that are compressed using a simple format where each character is followed by the number of times it appears. For example, "A3B2" represents the string "AAABB".

Her goal is to identify whether a specific pattern exists in the original, uncompressed message and, if so, determine all starting and ending indices of the pattern's occurrence in the expanded string. To perform this efficiently, Dhiya decides to first expand the compressed string and then apply pattern matching.

Your task is to help Dhiya by building a program using Boyer-Moore pattern matching algorithm that performs this operation.

Input Format

The first line of input contains a string *s* consisting of characters followed by digits representing the compressed text.

The second line of input contains a string *p* representing the pattern string.

Output Format

The first line of output prints "Expanded Text: *X*" where *X* represents the expanded text.

The second line of output prints "Matches found (start_index end_index):" if match found.

For each match of the pattern in the expanded text, print the starting and ending indices (0-based) separated by a space (Print one pair per line).

If no match is found, print "No match found."

Refer to the sample output for formatting specifications.

Sample Test Case

Input: A3B2C4D1

CCC

Output: Expanded Text: AAABBCCCCD

Matches found (start_index end_index):

5 7

6 8

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
#include <string.h>
```

```
#include <ctype.h>
```

```
#define MAX 2000
```

```
// Function to expand compressed string
```

```
void expand(char *s, char *res) {
```

```

int i = 0, k = 0;

while (s[i]) {
    char ch = s[i++];
    int num = 0;

    while (isdigit(s[i])) {
        num = num * 10 + (s[i] - '0');
        i++;
    }

    for (int j = 0; j < num; j++) {
        res[k++] = ch;
    }
    res[k] = '\0';
}

```

```

// Build bad character table
void buildBadChar(char *pat, int badChar[]) {
    int m = strlen(pat);

    for (int i = 0; i < 256; i++)
        badChar[i] = -1;

    for (int i = 0; i < m; i++)
        badChar[(int)pat[i]] = i;
}

```

```

int main() {
    char s[1001], p[1001];
    char text[MAX];

    scanf("%s", s);
    scanf("%s", p);

    // Expand string
    expand(s, text);

    printf("Expanded Text: %s\n", text);

    int n = strlen(text);
}

```

```

int m = strlen(p);
int badChar[256];
buildBadChar(p, badChar);

int found = 0;

// Boyer-Moore search
int shift = 0;
while (shift <= (n - m)) {
    int j = m - 1;

    while (j >= 0 && p[j] == text[shift + j])
        j--;

    if (j < 0) {
        if (!found) {
            printf("Matches found (start_index end_index):\n");
            found = 1;
        }

        printf("%d %d\n", shift, shift + m - 1);

        shift += (shift + m < n) ? m - badChar[text[shift + m]] : 1;
    } else {
        int bc = badChar[text[shift + j]];
        shift += (j - bc > 1) ? j - bc : 1;
    }
}

if (!found) {
    printf("No match found.");
}

return 0;
}

```

Status : Correct

Marks : 10/10

2. Problem Statement

Sanya is developing a text analysis tool and wants to identify the longest palindromic sequence in a given string. A palindrome is a string that reads the same forwards and backwards.

Your task is to write a program that takes a string as input and returns the longest palindromic substring it contains. If there are multiple substrings of the same maximum length, returning any one of them is acceptable.

Input Format

The input consists of a single line containing the string S.

S may consist of uppercase and lowercase English letters.

Output Format

The output prints a single string representing the longest palindromic substring found in S.

Refer to the sample input and output for format specifications.

Sample Test Case

Input: babad

Output: bab

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
#include <string.h>
```

```
// Function to expand around center
```

```
void expand(char *s, int left, int right, int *start, int *maxLen) {  
    int n = strlen(s);
```

```
    while (left >= 0 && right < n && s[left] == s[right]) {
```

```
        if (right - left + 1 > *maxLen) {
```

```
            *maxLen = right - left + 1;
```

```
            *start = left;
```

```
        }
```

```

        left--;
        right++;
    }
}

int main() {
    char s[101];
    scanf("%s", s);

    int n = strlen(s);
    int start = 0, maxLen = 1;

    for (int i = 0; i < n; i++) {
        // Odd length
        expand(s, i, i, &start, &maxLen);

        // Even length
        expand(s, i, i + 1, &start, &maxLen);
    }

    // Print result
    for (int i = start; i < start + maxLen; i++) {
        printf("%c", s[i]);
    }

    return 0;
}

```

Status : Correct

Marks : 10/10

3. Problem Statement

Rita is working on string pattern matching using the Z-Algorithm. She wants to find all starting indices where a given pattern string occurs in a text string efficiently.

Write a program that takes a text string and a pattern string as input and outputs a list of indices (0-based) where the pattern occurs. If the pattern does not occur in the text, output an empty list.

Input Format

The first line contains the text string txt.

The second line contains the pattern string pat.

Output Format

The output prints a list of integers representing the starting indices of all occurrences of the pattern in the text.

If no match is found, print [].

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: abesdu
edu

Output: []

Answer

```
// You are using GCC
#include <stdio.h>
#include <string.h>
```

```
int main() {
    char text[1001], pat[101];
    scanf("%s", text);
    scanf("%s", pat);

    char concat[1205];
    sprintf(concat, "%s%s", pat, text);

    int n = strlen(concat);
    int Z[1205];

    Z[0] = 0;
    int L = 0, R = 0;

    // Build Z array
```

```

for (int i = 1; i < n; i++) {
    if (i > R) {
        L = R = i;
        while (R < n && concat[R] == concat[R - L]) {
            R++;
        }
        Z[i] = R - L;
        R--;
    } else {
        int k = i - L;
        if (Z[k] < R - i + 1) {
            Z[i] = Z[k];
        } else {
            L = i;
            while (R < n && concat[R] == concat[R - L]) {
                R++;
            }
            Z[i] = R - L;
            R--;
        }
    }
}
}

```

```

int patLen = strlen(pat);
int found = 0;

```

```

printf("[");
for (int i = patLen + 1; i < n; i++) {
    if (Z[i] == patLen) {
        if (found) printf(", ");
        printf("%d", i - patLen - 1);
        found = 1;
    }
}
printf("]");

return 0;
}

```

Status : Correct

Marks : 10/10

4. Problem Statement

Neha is building a plagiarism-detection feature where she must compare two text strings to find similarities. To efficiently detect overlaps between the strings, she uses the KMP (Knuth–Morris–Pratt) string matching algorithm, relying on the LPS (Longest Prefix Suffix) array as the main data structure.

Given two strings, write a program to ensure that the task is done by finding the longest common substring present in both strings. If no common substring exists, the program should clearly indicate that result.

Input Format

The first line contains a string representing the first text.

The second line contains a string representing the second text.

Output Format

The output prints the longest common substring if it exists.

If there is no common substring, print "There is no common substring".

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: abcdef
abc

Output: abc

Answer

```
// You are using GCC
#include <stdio.h>
#include <string.h>
```

```
int main() {
    char s1[101], s2[101];
    scanf("%s", s1);
```

```

scanf("%s", s2);

int n = strlen(s1);
int m = strlen(s2);

int dp[101][101] = {0};

int maxLen = 0;
int endIndex = -1;

for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= m; j++) {
        if (s1[i - 1] == s2[j - 1]) {
            dp[i][j] = dp[i - 1][j - 1] + 1;

            if (dp[i][j] > maxLen) {
                maxLen = dp[i][j];
                endIndex = i - 1;
            }
        } else {
            dp[i][j] = 0;
        }
    }
}

if (maxLen == 0) {
    printf("There is no common substring");
} else {
    for (int i = endIndex - maxLen + 1; i <= endIndex; i++) {
        printf("%c", s1[i]);
    }
}

return 0;
}

```

Status : Correct

Marks : 10/10

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2028_NeoColab_Competitive Coding II_L9

VTU_Week 3_PAH

Attempt : 1

Total Mark : 40

Marks Obtained : 40

Section 1 : Coding

1. Problem Statement

Suresh is working on a text-processing system where he needs to efficiently count how many times a smaller pattern appears inside a larger string. To achieve fast pattern matching, he uses the Z-Algorithm, with the Z-array as the main data structure.

Given multiple test cases, each containing a text string and a pattern string, write a program to ensure that the task is done by computing the number of occurrences of the pattern in the text using the Z-algorithm approach.

For example:

If S = "ababa", and P = "ab", then "ab" occurs twice in "ababa".

Input Format

The first line of input contains T, representing the number of test cases.

For each test case:

Line 1 Contains two space separated integers N and M representing the length of the text and pattern respectively.

Line 2: Contains string S of length N representing the text.

Line 3: A string P of length M representing the pattern.

Output Format

For each test case, print a single integer on a new line indicating the number of times the pattern occurs in the text.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 1
10 2
ababababa
ab

Output: 4

Answer

```
// You are using GCC
#include <stdio.h>
#include <string.h>
```

```
int main() {
    int T;
    scanf("%d", &T);

    while (T--) {
        int N, M;
        scanf("%d %d", &N, &M);
```

```
char S[105], P[105];
scanf("%s", S);
scanf("%s", P);
```

```
char concat[210];
sprintf(concat, "%s%s", P, S);
```

```
int len = strlen(concat);
int Z[210];
```

```
Z[0] = 0;
int L = 0, R = 0;
```

```
// Build Z-array
for (int i = 1; i < len; i++) {
    if (i > R) {
        L = R = i;
        while (R < len && concat[R] == concat[R - L]) {
            R++;
        }
        Z[i] = R - L;
        R--;
    } else {
        int k = i - L;
        if (Z[k] < R - i + 1) {
            Z[i] = Z[k];
        } else {
            L = i;
            while (R < len && concat[R] == concat[R - L]) {
                R++;
            }
            Z[i] = R - L;
            R--;
        }
    }
}
```

```
int count = 0;
```

```
// Count matches
for (int i = M + 1; i < len; i++) {
    if (Z[i] == M) {
```

```
        count++;
    }
}

printf("%d\n", count);
}

return 0;
}
```

Status : Correct

Marks : 10/10

2. Problem Statement

Arjun is working as an NLP engineer to improve the smart assistant chatbot in his company. His team wants the chatbot to understand user emotions better, especially when users repeat or emphasize words.

They noticed that palindromic patterns (like "wow", "pop", "madam") in messages can help detect such emotions or expressions.

To help with this, Arjun is asked to write a program that finds out how many palindromic substrings are centered at each character in the user's message.

Since the message could be long, Arjun decides to use Manacher's Algorithm for fast and efficient processing.

Example:

Input

bananas

Output

1 1 3 5 3 1 1

Explanation

At the first character 'b', the only palindrome is "b" itself length 1
At 'a', only "a" is a palindrome length 1
At 'n', we get "ana" which is a palindrome length 3
At 'a', we get "anana" which is a palindrome length 5
At 's', we get "sanas" which is a palindrome length 3
At 'a', we get "asana" which is a palindrome length 1
At 'n', we get "anana" which is a palindrome length 1

centered at 'n' length 3At the middle 'a', we get "anana" which is a bigger palindrome length 5At the next 'n', again "ana" is a palindrome length 3At the second 'a', only "a" length 1Finally at 's', only "s" length 1

Input Format

The first line of input contains a string s, representing a message typed by a user in the chatbot conversation.

Output Format

The output prints the length of the longest palindromic substring centered at each character in the input string, separated by spaces.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: ab

Output: 1 1

Answer

```
// You are using GCC
#include <stdio.h>
#include <string.h>
```

```
int main() {
    char s[1001];
    scanf("%s", s);
```

```
    int n = strlen(s);
    int d1[1001]; // odd-length palindrome radius
```

```
    int l = 0, r = -1;
```

```
    // Manacher for odd-length palindromes
    for (int i = 0; i < n; i++) {
        int k;
```

```
        if (i > r)
            k = 1;
```

```

else
    k = (d1[l + r - i] < (r - i + 1)) ? d1[l + r - i] : (r - i + 1);

    while (i - k >= 0 && i + k < n && s[i - k] == s[i + k]) {
        k++;
    }

    d1[i] = k;

    if (i + k - 1 > r) {
        l = i - k + 1;
        r = i + k - 1;
    }
}

// Print result
for (int i = 0; i < n; i++) {
    int length = 2 * d1[i] - 1;
    printf("%d ", length);
}

return 0;
}

```

Status : Correct

Marks : 10/10

3. Problem Statement

Given a text txt [0 . . . N-1] and a pattern pat [0 . . . M-1], write a function search (char pat [], char txt []) that prints all occurrences of pat [] in txt []. You may assume that N > M. Use zero-based indexing while returning the indices.

Example:

Input: txt [] = "THIS IS A TEST TEXT", pat [] = "TEST"

Output: Pattern found at index 10

Input: txt [] = "AABAACAADAABAABA", pat [] = "AABA"

Output: Pattern found at index 0, Pattern found at index 9, Pattern found at index 12

Input Format

The first line of input consists of a string txt, in which the pattern is to be searched.

The second line of input consists of a string pat, the pattern that needs to be found in the text.

Output Format

For each occurrence of the pattern in the text, print "Pattern found at index X" in a new line, where X the indices where the pattern is found within the text string.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: THIS IS A TEST TEXT
TEST

Output: Pattern found at index 10

Answer

```
// You are using GCC
#include <stdio.h>
#include <string.h>

// Build LPS array
void computeLPS(char *pat, int M, int lps[]) {
    int len = 0;
    lps[0] = 0;

    int i = 1;
    while (i < M) {
        if (pat[i] == pat[len]) {
            len++;
            lps[i] = len;
        }
    }
}
```

```

        i++;
    } else {
        if (len != 0) {
            len = lps[len - 1];
        } else {
            lps[i] = 0;
            i++;
        }
    }
}
}
}

```

// KMP Search

```

void search(char *pat, char *txt) {
    int M = strlen(pat);
    int N = strlen(txt);

```

```

    int lps[100];
    computeLPS(pat, M, lps);

```

```

    int i = 0, j = 0;

```

```

    while (i < N) {
        if (pat[j] == txt[i]) {
            i++;
            j++;
        }
    }

```

```

    if (j == M) {
        printf("Pattern found at index %d\n", i - j);
        j = lps[j - 1];
    } else if (i < N && pat[j] != txt[i]) {
        if (j != 0)
            j = lps[j - 1];
        else
            i++;
    }
}

```

```

int main() {
    char txt[100], pat[20];

```

```

// Read full line (including spaces)
fgets(txt, sizeof(txt), stdin);
fgets(pat, sizeof(pat), stdin);

// Remove newline characters
txt[strcspn(txt, "\n")] = '\0';
pat[strcspn(pat, "\n")] = '\0';

search(pat, txt);

return 0;
}

```

Status : Correct

Marks : 10/10

4. Problem Statement

Ritika is a software engineer working on an AI-based spam detection system for a popular email client. To efficiently scan email content, she uses the Boyer–Moore string matching algorithm, specifically the Bad Character Heuristic as the main data structure/technique.

Given an email message and a list of spam keywords, write a program to ensure that the task is done by detecting all keywords present in the email (case-insensitive), reporting their first occurrence positions, and computing a spam score based on the number of detected keywords.

Input Format

The first line of input contains a string representing the email content (may contain spaces and special characters).

The second line contains an integer n , representing the number of spam keywords.

The next n lines each contain a spam keyword.

Output Format

The first line prints a string "Detected Spam Keywords:"

For each detected keyword, print "<index>: "<keyword>" at position <starting_index>" where indexing starts from 0.

The final line prints "Spam Score: x/y" where x is the number of detected keywords and y is the total number of keywords.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: Buy this book now and get another free!

3

buy

now

free

Output: Detected Spam Keywords:

1: "buy" at position 0

2: "now" at position 14

3: "free" at position 34

Spam Score: 3/3

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
#include <string.h>
```

```
#include <ctype.h>
```

```
#define MAX 1005
```

```
// Convert string to lowercase
```

```
void toLowerCase(char *s) {
```

```
    for (int i = 0; s[i]; i++) {
```

```
        s[i] = tolower(s[i]);
```

```
    }
```

```
}
```

```
// Build bad character table
```

```
void buildBadChar(char *pat, int badChar[]) {
```

```
    for (int i = 0; i < 256; i++)
```

```
        badChar[i] = -1;
```

```
for (int i = 0; pat[i]; i++)  
    badChar[(int)pat[i]] = i;  
}
```

// Boyer-Moore search (returns first occurrence)

```
int boyerMoore(char *txt, char *pat) {  
    int n = strlen(txt);  
    int m = strlen(pat);
```

```
    int badChar[256];  
    buildBadChar(pat, badChar);
```

```
    int shift = 0;
```

```
    while (shift <= n - m) {  
        int j = m - 1;
```

```
        while (j >= 0 && pat[j] == txt[shift + j])  
            j--;
```

```
        if (j < 0) {  
            return shift; // first occurrence  
        } else {  
            int bc = badChar[(int)txt[shift + j]];  
            shift += (j - bc > 1) ? j - bc : 1;
```

```
        }  
    }  
    return -1;  
}
```

```
int main() {  
    char text[MAX];  
    fgets(text, sizeof(text), stdin);  
    text[strlen(text)] = '\0';
```

```
    int n;  
    scanf("%d", &n);  
    getchar(); // consume newline
```

```
    char keywords[105][105];  
    for (int i = 0; i < n; i++) {
```

```

    fgets(keywords[i], sizeof(keywords[i]), stdin);
    keywords[i][strcspn(keywords[i], "\n")] = '\0';
}

// Create lowercase copy of text
char textLower[MAX];
strcpy(textLower, text);
toLowerCase(textLower);

int foundCount = 0;
int positions[105];

// Store results
for (int i = 0; i < n; i++) {
    char patLower[105];
    strcpy(patLower, keywords[i]);
    toLowerCase(patLower);

    positions[i] = boyerMoore(textLower, patLower);

    if (positions[i] != -1)
        foundCount++;
}

if (foundCount == 0) {
    printf("No spam keywords detected.\n");
    printf("Spam Score: 0/%d", n);
    return 0;
}

printf("Detected Spam Keywords:\n");

int index = 1;
for (int i = 0; i < n; i++) {
    if (positions[i] != -1) {
        printf("%d: \"%s\" at position %d\n", index++, keywords[i], positions[i]);
    }
}

printf("Spam Score: %d/%d", foundCount, n);
return 0;

```


}

Status : Correct

Marks : 10/10

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2028_NeoColab_Competitive Coding II_L9

VTU_Week 4_MCQ

Attempt : 1

Total Mark : 15

Marks Obtained : 10

Section 1 : MCQ

1. The user performs the following operations on a stack of size 5 then, which of the following is the correct statement for the 1stack?

```
push(1);  
pop();  
push(2);  
push(3);  
pop();  
push(2);  
pop();  
pop();  
push(4);  
pop();  
pop();  
push(5);
```

Answer

Overflow occurs

Status : Wrong

Marks : 0/1

2. If the sequence of operations - push (1), push (2), pop, push (1), push (2), pop, pop, pop, push (2), pop are performed on a stack, the sequence of popped out values

Answer

2,2,1,1,2

Status : Correct

Marks : 1/1

3. What is the result of the following operation?

Top (Push (S, X))

Answer

X

Status : Correct

Marks : 1/1

4. Stack A has the entries a, b, c (with a on top). Stack B is empty. An entry popped out of stack A can be printed immediately or pushed to stack B. An entry popped out of the stack B can only be printed. In this arrangement, which of the following permutations of a, b, c are not possible?

Answer

b a c

Status : Wrong

Marks : 0/1

5. Pushing an element into the stack already has five elements. The stack size is 5, then the stack becomes

Answer

Overflow

Status : Correct

Marks : 1/1

6. In a Stack, if a user tries to remove an Element from Empty stack it is called

Answer

Underflow

Status : Correct

Marks : 1/1

7. The user performs the following operations on the stack of size 5 then at the end of the last operation, the total number of elements present in the stack is

```
push(1);  
pop();  
push(2);  
push(3);  
pop();  
push(4);  
pop();  
pop();  
push(5);
```

Answer

2

Status : Wrong

Marks : 0/1

8. What will be the output after performing these sequence of operations

```
push(20);  
push(4);  
top();  
pop();
```

```
pop();  
pop();  
push(5);  
top();
```

Answer

Stack Underflow

Status : Correct

Marks : 1/1

9. A normal queue, if implemented using an array of size MAX_SIZE, gets full when _____.

Answer

rear = MAX_SIZE – 1

Status : Correct

Marks : 1/1

10. If the elements 'A', 'B', 'C', and 'D' are placed in a queue and are deleted one at a time, in what order will they be removed?

Answer

ABCD

Status : Correct

Marks : 1/1

11. After performing this set of operations, what does the final queue contain?

```
InsertFront(10);  
InsertFront(20);  
InsertRear(30);  
DeleteFront();  
InsertRear(40);  
InsertRear(10);  
DeleteRear();  
InsertRear(15);  
display();
```

Answer

20 30 40 15

Status : Wrong

Marks : 0/1

12. What is the purpose of defining the constant SIZE in a queue implementation using an array?

Answer

To limit the maximum number of elements in the queue

Status : Correct

Marks : 1/1

13. What does the condition front = NULL indicate in inserting an element into a queue?

Answer

The queue is empty

Status : Correct

Marks : 1/1

14. Which of the following is a valid condition for a successful dequeue operation in the array implementation of a queue?

Answer

None of the mentioned options

Status : Correct

Marks : 1/1

15. The essential condition that is checked before insertion in a queue is?

Answer

Underflow

Status : Wrong

Marks : 0/1

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2028_NeoColab_Competitive Coding II_L9

VTU_Week 4_COD

Attempt : 1

Total Mark : 50

Marks Obtained : 50

Section 1 : Coding

1. Problem Statement

Neo is experimenting with text transformations and decides to use a stack to reverse the sequence of words in his sentence. He provides a line of text and expects the program to utilize a stack to reorder the words so that the last word appears first and the first word appears last.

Write a program to ensure this is done correctly using a stack-based approach.

Example

Input: I am Neo

Output: Neo am I

Input Format

The input consists of a single line of input containing a sentence with words separated by spaces.

Output Format

The output prints a single line of output containing the words of the sentence reversed in order, separated by a single space.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: Hello World

Output: World Hello

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
#include <string.h>
```

```
int main() {  
    char str[105];  
    fgets(str, sizeof(str), stdin);
```

```
    // Remove newline  
    str[strcspn(str, "\n")] = '\0';
```

```
    char stack[100][100]; // stack to store words  
    int top = -1;
```

```
    char word[100];  
    int j = 0;
```

```
    // Extract words and push to stack  
    for (int i = 0; ; i++) {  
        if (str[i] != ' ' && str[i] != '\0') {  
            word[j++] = str[i];  
        } else {  
            word[j] = '\0';
```



```

        if (j > 0) {
            strcpy(stack[++top], word);
            j = 0;
        }
        if (str[i] == '\0')
            break;
    }
}

// Pop and print
for (int i = top; i >= 0; i--) {
    printf("%s", stack[i]);
    if (i != 0)
        printf(" ");
}

return 0;
}

```

Status : Correct

Marks : 10/10

2. Problem Statement

Give a string, remove duplicate letters from the string *s* such that each letter appears just once using Stack. The lexicographical order of your result must be the least of all feasible outcomes.

Input Format

The input consists of a string.

Output Format

The output displays the string in the lexicographical order with unique characters.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: bcabc

Output: abc

Answer

// You are using GCC

#include <stdio.h>

#include <string.h>

#include <stdbool.h>

int main() {

char s[30];

scanf("%s", s);

int freq[26] = {0};

bool visited[26] = {false};

int n = strlen(s);

// Count frequency

for (int i = 0; i < n; i++) {

freq[s[i] - 'a']++;

}

char stack[30];

int top = -1;

for (int i = 0; i < n; i++) {

char c = s[i];

int idx = c - 'a';

freq[idx]--;

// If already in stack skip

if (visited[idx])

continue;

// Maintain lexicographical order

while (top >= 0 &&

stack[top] > c &&

freq[stack[top] - 'a'] > 0) {

visited[stack[top] - 'a'] = false;

top--;

```

    }

    stack[++top] = c;
    visited[idx] = true;
}

// Print result
for (int i = 0; i <= top; i++) {
    printf("%c", stack[i]);
}

return 0;
}

```

Status : Correct

Marks : 10/10

3. Problem Statement

Mira is learning about palindromes and wants to build a program that can check whether a given string is a palindrome using a stack-based approach. A palindrome reads the same backward as forward.

Write a program to ensure that the task is done by pushing characters to a stack and comparing them in reverse order to determine if the string is a palindrome.

Input Format

The input contains a single line containing a string with space-separated characters.

Output Format

The output consists of ,

If the string is a palindrome, print "It's a palindrome!"

Otherwise, print "Not a palindrome"

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: madam

Output: It's a palindrome!

Answer

```
// You are using GCC
#include <stdio.h>
#include <string.h>
#include <ctype.h>

int main() {
    char input[105];
    fgets(input, sizeof(input), stdin);

    // Remove newline
    input[strcspn(input, "\n")] = '\0';

    char processed[105];
    int k = 0;

    // Remove spaces and convert to lowercase
    for (int i = 0; input[i]; i++) {
        if (isalpha(input[i])) {
            processed[k++] = tolower(input[i]);
        }
    }
    processed[k] = '\0';

    // Push to stack
    char stack[105];
    int top = -1;

    for (int i = 0; i < k; i++) {
        stack[++top] = processed[i];
    }

    // Compare with original processed string
    int isPalindrome = 1;
    for (int i = 0; i < k; i++) {
```

```

    if (processed[i] != stack[top--]) {
        isPalindrome = 0;
        break;
    }
}

if (isPalindrome)
    printf("It's a palindrome!");
else
    printf("Not a palindrome");

return 0;
}

```

Status : Correct

Marks : 10/10

4. Problem Statement

Gokul is working on a program to rearrange elements in a queue using an array. Given a queue of integers, he needs to modify it by interleaving its first half with the second half. Write a program to help Gokul achieve this task.

Note: Interleaving involves merging elements from two separate halves in an alternating fashion to create a new sequence.

Example

Input

6

1 2 3 7 9 5

Output

1 7 2 9 3 5

Explanation

The interleave rearranges elements in a queue by alternating between two halves. For instance, given input [1, 2, 3, 7, 9, 5], it splits into [1, 2, 3] and [7, 9, 5], and then interleaves them to form [1, 7, 2, 9, 3, 5].

9, 5], then interleaves them as [1, 7, 2, 9, 3, 5]. It achieves this by splitting the queue into two halves and then merging them interleaved.

Input Format

The first line of the input consists of an integer n , representing the number of elements in the queue.

The second line consists of n space-separated integers, representing the queue elements.

Output Format

The output prints n space-separated integers, representing the rearranged queue after interleaving its first half with the second half.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 6

1 2 3 7 9 5

Output: 1 7 2 9 3 5

Answer

```
// You are using GCC
#include <stdio.h>
```

```
int main() {
    int n;
    scanf("%d", &n);

    int arr[20];
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }
```

```
    int half = n / 2;
```

```
    // Interleave and print
    for (int i = 0; i < half; i++) {
```

```
        printf("%d %d ", arr[i], arr[i + half]);  
    }  
  
    return 0;  
}
```

Status : Correct

Marks : 10/10

5. Problem Statement

Sanjay is learning about queues in his data structures class and he wants to write a program that counts the number of distinct elements in the queue. Your task is to guide him in this.

Example

Input:

7

45 33 29 45 45 46 39

Output:

5

Explanation:

The distinct elements in the queue are 45, 33, 29, 46, 39. So, the count is 5.

Input Format

The first line of input consists of an integer M, representing the number of elements in the queue.

The second line consists of M space-separated integers, representing the queue elements.

Output Format

The output prints an integer, representing the number of distinct elements in the queue.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 5
30 24 43 31 30

Output: 4

Answer

```
// You are using GCC
#include <stdio.h>
```

```
int main() {
    int n;
    scanf("%d", &n);

    int arr[100];
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    int count = 0;

    for (int i = 0; i < n; i++) {
        int isDuplicate = 0;

        // Check if element appeared before
        for (int j = 0; j < i; j++) {
            if (arr[i] == arr[j]) {
                isDuplicate = 1;
                break;
            }
        }

        if (!isDuplicate) {
            count++;
        }
    }

    printf("%d", count);
}
```



```
} return 0;
```

Status : Correct

Marks : 10/10

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2028_NeoColab_Competitive Coding II_L9

VTU_Week 4_CY

Attempt : 1

Total Mark : 40

Marks Obtained : 40

Section 1 : Coding

1. Problem Statement

A financial analytics company is developing a tool to analyze stock price trends. The tool needs to identify the Next Greater Price for each stock price in a given list of daily prices. For a given stock price, the Next Greater Price is defined as the first price appearing after it in the list that is greater. If no such price exists, the result should be -1.

Note: The Next greater Element for an element x is the first greater element on the right side of x in the array. Elements for which no greater element exist, consider the next greater element as -1.

Examples:

Input Format

The first line of input consists of an integer n, representing the size of the array.

The second line consists of n space-separated stack of integers.

Output Format

The output prints "Next Greater Elements: " followed by an array of integers, representing the next greater elements in the stack.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 4

4 5 2 25

Output: Next Greater Elements: 5 25 25 -1

Answer

// You are using GCC

#include <stdio.h>

```
int main() {
```

```
    int n;
```

```
    scanf("%d", &n);
```

```
    int arr[20], result[20];
```

```
    for (int i = 0; i < n; i++) {
```

```
        scanf("%d", &arr[i]);
```

```
    }
```

```
    int stack[20];
```

```
    int top = -1;
```

```
    // Traverse from right to left
```

```
    for (int i = n - 1; i >= 0; i--) {
```

```
        // Remove smaller elements
```

```
        while (top >= 0 && stack[top] <= arr[i]) {
```

```
            top--;
```

```
        }
```

```

// If stack empty no greater element
if (top == -1)
    result[i] = -1;
else
    result[i] = stack[top];

// Push current element
stack[++top] = arr[i];
}

// Print result
printf("Next Greater Elements: ");
for (int i = 0; i < n; i++) {
    printf("%d ", result[i]);
}

return 0;
}

```

Status : Correct

Marks : 10/10

2. Problem Statement:

Riya is working on a text-cleaning utility where certain unwanted patterns must be removed from user input. She uses a stack data structure to process the string character by character.

Given a string containing alphabets, digits, and special characters, write a program to ensure that the task is done by removing every digit along with the closest non-digit character immediately to its left. The remaining characters should be preserved in their original order and printed as the final cleaned string.

Example:

Input:

s = "cb34"

Output:

'''

Explanation:

Start with an empty stack.

Push c stack: ['c']

Push b stack: ['c','b']

Encounter 3 pop 'b' stack: ['c']

Encounter 4 pop 'c' stack: []

Result: empty string

Input Format

The input consists of a string *s* without spaces, consisting of letters, digits, and special characters.

Output Format

The output prints the resulting string after performing the required removals.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: abc

Output: abc

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
#include <string.h>
```

```
#include <ctype.h>
```

```
int main() {
```

```
    char s[105];
```

```
    scanf("%s", s);
```

```
    char stack[105];
```

```
    int top = -1;
```

```

for (int i = 0; s[i]; i++) {
    if (isdigit(s[i])) {
        // Remove closest non-digit (if exists)
        if (top >= 0) {
            top--;
        }
    } else {
        // Push non-digit character
        stack[++top] = s[i];
    }
}

// Print result
for (int i = 0; i <= top; i++) {
    printf("%c", stack[i]);
}

return 0;
}

```

Status : Correct

Marks : 10/10

3. Problem Statement

Kavya is developing a real-time analytics system that processes a continuous stream of numerical data. To efficiently track trends, she uses a Queue data structure to maintain incoming values and compute statistics incrementally.

Given a sequence of integers arriving one by one, write a program to ensure that the task is done by calculating and printing the moving average of all elements received so far after each new input. The average should be displayed up to two decimal places.

Example

Input

3

1 2 3

Output

1.00 1.50 2.00

Explanation

The moving average of the first element is 1.00.

The next elements are 1 and 2, the moving average is $(1 + 2) / 2 = 1.50$.

The next elements are 1, 2, and 3, the moving average is $(1 + 2 + 3) / 3 = 2.00$.

Input Format

The first line contains an integer n, representing the number of elements in the data stream.

The second line contains n space-separated integers representing the incoming values.

Output Format

The output prints n floating-point numbers, each representing the moving average after processing the corresponding element.

Each value should be printed with two decimal places, separated by spaces.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 3

1 2 3

Output: 1.00 1.50 2.00

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
int main() {
```

```
int n;
scanf("%d", &n);

int x;
double sum = 0;

for (int i = 1; i <= n; i++) {
    scanf("%d", &x);
    sum += x;

    double avg = sum / i;
    printf("%.2f ", avg);
}

return 0;
}
```

Status : Correct

Marks : 10/10

4. Problem Statement

In a small business, Jane is tasked with organizing customer feedback scores for a new product. Each customer has rated the product, and Jane needs to arrange these ratings in a specific order to analyze the feedback more effectively.

Specifically, she wants to rearrange the ratings such that the most frequent ratings appear first. If two ratings have the same frequency, the smaller rating should come first. To accomplish this, assist Jane with a program to process a queue of customer ratings and rearrange them based on their frequencies.

Input Format

The first line contains an integer n , which represents the number of customer ratings.

The second line contains n integers, each representing a customer rating.

Output Format

The output displays the rearranged ratings in the order of their frequency (most frequent first).

If frequencies are the same, the smaller rating should come first.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 7

2 3 2 3 2 3 2

Output: 2 2 2 2 3 3 3

Answer

```
// You are using GCC
#include <stdio.h>
```

```
int main() {
    int n;
    scanf("%d", &n);

    int arr[25];
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    int freq[25];

    // Calculate frequency for each element
    for (int i = 0; i < n; i++) {
        freq[i] = 0;
        for (int j = 0; j < n; j++) {
            if (arr[i] == arr[j]) {
                freq[i]++;
            }
        }
    }

    // Sort based on frequency and value
    for (int i = 0; i < n - 1; i++) {
```

```

for (int j = i + 1; j < n; j++) {
    if (freq[i] < freq[j] ||
        (freq[i] == freq[j] && arr[i] > arr[j])) {

        // swap arr
        int temp = arr[i];
        arr[i] = arr[j];
        arr[j] = temp;

        // swap freq
        temp = freq[i];
        freq[i] = freq[j];
        freq[j] = temp;
    }
}

// Print result
for (int i = 0; i < n; i++) {
    printf("%d ", arr[i]);
}

return 0;
}

```

Status : Correct

Marks : 10/10

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2028_NeoColab_Competitive Coding II_L9

VTU_Week 4_PAH

Attempt : 1

Total Mark : 40

Marks Obtained : 40

Section 1 : Coding

1. Problem Statement

Rohit is learning combinatorics and wants to generate all possible selections from a given set of numbers. To systematically explore every combination, he uses a Stack data structure along with a bitwise technique.

Given a set of integers, write a program to ensure that the task is done by generating and printing all possible subsets (the power set) of the given set, including the empty set. Each subset must be printed in set notation.

Example

Input:

3

1 2 3

Output:

{}

{1}

{2}

{1, 2}

{3}

{1, 3}

{2, 3}

{1, 2, 3}

Input Format

The first line of input consists of an integer n , representing the number of elements in the set.

The second line consists of n space-separated integers representing the elements of the set.

Output Format

The output prints all possible subsets of the given set.

Each subset should be printed on a new line in the format {element1, element2, ...}.

The empty subset should be printed as {}.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 3

1 2 3

Output: {}

{1}
{2}
{1, 2}
{3}
{1, 3}
{2, 3}
{1, 2, 3}

Answer

// You are using GCC
#include <stdio.h>

```
int main() {
    int n;
    scanf("%d", &n);

    int arr[15];
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    int total = 1 << n; // 2^n subsets

    for (int i = 0; i < total; i++) {
        printf("{");

        int first = 1;

        for (int j = 0; j < n; j++) {
            if (i & (1 << j)) {
                if (!first)
                    printf(", ");
                printf("%d", arr[j]);
                first = 0;
            }
        }

        printf("}\n");
    }

    return 0;
}
```

Status : Correct

Marks : 10/10

2. Problem Statement

Given an array of integers, the task is to find the maximum absolute difference between the nearest left and the right smaller element of every element in the array.

Note: If there is no smaller element on right side or left side of any element then we take zero as the smaller element. For example for the leftmost element, the nearest smaller element on the left side is considered as 0. Similarly, for rightmost elements, the smaller element on the right side is considered as 0.

Examples:

Input: arr[] = {2, 1, 8}

Output: 1

Left smaller LS[] {0, 0, 1}

Right smaller RS[] {1, 0, 0}

Maximum Diff of $\text{abs}(\text{LS}[i] - \text{RS}[i]) = 1$

Input: arr[] = {2, 4, 8, 7, 7, 9, 3}

Output: 4

Left smaller LS[] = {0, 2, 4, 4, 4, 7, 2}

Right smaller RS[] = {0, 3, 7, 3, 3, 3, 0}

Maximum Diff of $\text{abs}(\text{LS}[i] - \text{RS}[i]) = 7 - 3 = 4$

Input: arr[] = {5, 1, 9, 2, 5, 1, 7}

Output: 1

Input Format

The first line contains an integer n, representing the size of the array.

The second line contains n space-separated integers, representing the array elements.

Output Format

The output prints a single integer, representing the maximum absolute difference between the left and right smaller elements for any element in the array.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 3

2 1 8

Output: 1

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
int main() {
```

```
    int n;
```

```
    scanf("%d", &n);
```

```
    int arr[20];
```

```
    for (int i = 0; i < n; i++) {
```

```
        scanf("%d", &arr[i]);
```

```
    }
```

```
    int maxDiff = 0;
```

```
    for (int i = 0; i < n; i++) {
```

```
        int left = 0, right = 0;
```

```
        // Find nearest smaller on LEFT
```

```
        for (int j = i - 1; j >= 0; j--) {
```

```
            if (arr[j] < arr[i]) {
```

```
                left = arr[j];
```

```
                break;
```

```

    }
}

// Find nearest smaller on RIGHT
for (int j = i + 1; j < n; j++) {
    if (arr[j] < arr[i]) {
        right = arr[j];
        break;
    }
}

int diff = abs(left - right);
if (diff > maxDiff) {
    maxDiff = diff;
}
}

printf("%d", maxDiff);

return 0;
}

```

Status : Correct

Marks : 10/10

3. Problem Statement

A logistics company uses a queue data structure to manage the order of processing packages at their sorting center. Each package is identified by a unique code represented as an integer.

Due to a system update, the company needs to reverse the order in which the packages are processed without changing the sequence of processing for any new packages that arrive after the reversal.

The task is to implement a system that enqueues package codes, prints the original processing order, reverse the queue using a stack, and then prints the reversed processing order.

Input Format

The first line contains an integer capacity, the number of packages initially in the queue.

The second line contains capacity space-separated integers representing the package codes.

Output Format

The first line prints "Original queue:" followed by the package codes in their original order.

The second line prints "Reversed queue:" followed by the package codes in their reversed order.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 5

1 2 3 4 5

Output: Original queue: 1 2 3 4 5

Reversed queue: 5 4 3 2 1

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
int main() {
```

```
    int n;
```

```
    scanf("%d", &n);
```

```
    int queue[105];
```

```
    for (int i = 0; i < n; i++) {
```

```
        scanf("%d", &queue[i]);
```

```
    }
```

```
    // Print original queue
```

```
    printf("Original queue: ");
```

```
    for (int i = 0; i < n; i++) {
```

```
        printf("%d ", queue[i]);
```

```
    }
```

```

printf("\n");

// Use stack to reverse
int stack[105];
int top = -1;

// Push all elements to stack
for (int i = 0; i < n; i++) {
    stack[++top] = queue[i];
}

// Pop from stack   reversed order
printf("Reversed queue: ");
while (top >= 0) {
    printf("%d ", stack[top--]);
}

return 0;
}

```

Status : Correct

Marks : 10/10

4. Problem Statement

Alex is working on a customer support system that handles a queue of incoming service requests. The system needs to manage the requests efficiently by enqueue, ensuring that requests are processed in the order they arrive.

To improve processing efficiency, Alex wants to implement a feature that allows moving the last request to the front of the queue when necessary. This operation helps prioritize the most recent request for urgent handling.

Your task is to implement this feature and ensure that the queue is updated accordingly.

Input Format

The first line of input contains a single integer n , representing the number of service requests.

The second line contains n integers, each representing a service request ID, separated by spaces.

Output Format

The output displays the queue after moving the last request to the front with request IDs separated by spaces.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 4

62 85 47 98

Output: 98 62 85 47

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
int main() {
```

```
    int n;
```

```
    scanf("%d", &n);
```

```
    int arr[55];
```

```
    for (int i = 0; i < n; i++) {
```

```
        scanf("%d", &arr[i]);
```

```
    }
```

```
    // Store last element
```

```
    int last = arr[n - 1];
```

```
    // Shift elements right
```

```
    for (int i = n - 1; i > 0; i--) {
```

```
        arr[i] = arr[i - 1];
```

```
    }
```

```
    // Place last element at front
```

```
    arr[0] = last;
```

```
// Print result
for (int i = 0; i < n; i++) {
    printf("%d ", arr[i]);
}

return 0;
}
```

Status : Correct

Marks : 10/10

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2028_NeoColab_Competitive Coding II_L9

VTU_Week 5_MCQ

Attempt : 1

Total Mark : 15

Marks Obtained : 13.5

Section 1 : MCQ

1. After adding a new node at the end of a linked list, what does the current last node become?

Answer

The second last node

Status : Correct

Marks : 1/1

2. When inserting a new element at the beginning of a linked list, what happens to the current head of the list?

Answer

It becomes the second element

Status : Correct

Marks : 1/1

3. Which of the following are true about a singly linked list?

Answer

It can grow dynamically during runtime.

Status : Partially correct

Marks : 0.5/1

4. When inserting a new node into a singly linked list, which of the following are the possible cases for insertion?

Answer

All of the mentioned options

Status : Correct

Marks : 1/1

5. Given the linked list: 5 -> 10 -> 15 -> 20 -> 25 -> NULL. What will be the output of traversing the list and printing each node's data?

Answer

5 10 15 20 25

Status : Correct

Marks : 1/1

6. Which of the following is the process of finding information in a linked list?

Answer

Start from the head of the list and traverse sequentially until finding (or not finding) the node

Status : Correct

Marks : 1/1

7. Which of the following actions needs to be taken when the last item in a linked list is deleted?

Answer

Change the reference of the second last item to NULL reference

Status : Correct

Marks : 1/1

8. Which pointer helps in traversing a doubly linked list in reverse order?

Answer

prev

Status : Correct

Marks : 1/1

9. Which of the following is true about the last node in a doubly linked list?

Answer

Its next pointer is NULL

Status : Correct

Marks : 1/1

10. What will be the effect of setting the prev pointer of a node to NULL in a doubly linked list?

Answer

The node will become the new head

Status : Correct

Marks : 1/1

11. How do you delete a node from the middle of a doubly linked list?

Answer

All of the mentioned options

Status : Correct

Marks : 1/1

12. What happens if we insert a node at the beginning of a doubly linked list?

Answer

The previous pointer of the new node is NULL

Status : Correct

Marks : 1/1

13. How many pointers does a node in a doubly linked list have?

Answer

2

Status : Correct

Marks : 1/1

14. In deletion from the end of a doubly linked list, which pointer needs to be updated?

Answer

prev->next = NULL

Status : Correct

Marks : 1/1

15. Which of the following is true about a Doubly Linked List?

Answer

-

Status : -

Marks : 0/1

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2028_NeoColab_Competitive Coding II_L9

VTU_Week 5_COD

Attempt : 1

Total Mark : 50

Marks Obtained : 50

Section 1 : Coding

1. Problem Statement

Arun is tasked with implementing a program to reverse a singly linked list. The program should populate the list elements by inserting them at the beginning, traverse the list and reverse the order of its elements.

Since Arun is new to programming, you have to guide him through the task.

Input Format

The first line of input consists of an integer N, representing the linked list size.

The second line consists of N space-separated list elements.

Output Format

The first line of output prints the list elements after inserting them at the beginning.

The second line prints the reversed linked list.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 5

23 85 47 62 31

Output: 31 62 47 85 23

23 85 47 62 31

Answer

```
#include <stdio.h>
#include <stdlib.h>
```

```
// Define the structure for a node
struct Node {
    int data;
    struct Node* next;
};
```

```
// Function to insert a node at the beginning of the list
void insertAtBeginning(struct Node** head_ref, int new_data) {
    struct Node* new_node = (struct Node*)malloc(sizeof(struct Node));
    new_node->data = new_data;
    new_node->next = (*head_ref);
    (*head_ref) = new_node;
}
```

```
// Function to reverse the linked list
void reverseList(struct Node** head_ref) {
    struct Node* prev = NULL;
    struct Node* current = *head_ref;
    struct Node* next = NULL;
    while (current != NULL) {
        next = current->next; // Store next
        current->next = prev; // Reverse current node's pointer
        prev = current;      // Move pointers one position forward
        current = next;
    }
```

```

    }
    *head_ref = prev;
}

// Function to print the linked list
void printList(struct Node* node) {
    while (node != NULL) {
        printf("%d ", node->data);
        node = node->next;
    }
    printf("\n");
}

int main() {
    int n, value;
    struct Node* head = NULL;

    // Read the size of the list
    if (scanf("%d", &n) != 1) return 0;

    // Read elements and insert them at the beginning
    for (int i = 0; i < n; i++) {
        scanf("%d", &value);
        insertAtBeginning(&head, value);
    }

    // Output 1: List elements after inserting at the beginning
    printList(head);

    // Reverse the list
    reverseList(&head);

    // Output 2: The reversed linked list
    printList(head);

    return 0;
}

```

Status : Correct

Marks : 10/10

2. Problem Statement

In a computer science course project, students are required to write a program to insert values at the end of a singly linked list and remove the last node from the list. The program should traverse the list and delete the last node, adjusting pointers accordingly.

Assist students in the task.

Input Format

The first line of input consists of an integer N, representing the list size.

The second line consists of N space-separated integers, representing the list elements.

Output Format

The output prints space-separated integers, representing the list after the deletion of the last element.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 8

54 25 14 38 55 73 95 82

Output: 54 25 14 38 55 73 95

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
struct Node {  
    int data;  
    struct Node* next;  
};
```

```
// Function to add elements to the end of the list
```

```
void append(struct Node** head_ref, int new_data) {
```

```
    struct Node* new_node = (struct Node*)malloc(sizeof(struct Node));
```

```

    struct Node* last = *head_ref;
    new_node->data = new_data;
    new_node->next = NULL;

    if (*head_ref == NULL) {
        *head_ref = new_node;
        return;
    }

    while (last->next != NULL) {
        last = last->next;
    }
    last->next = new_node;
}

// Function to delete the last node
void deleteLast(struct Node* head) {
    if (head == NULL || head->next == NULL) return;

    struct Node* temp = head;
    // Move to the second-to-last node
    while (temp->next->next != NULL) {
        temp = temp->next;
    }

    free(temp->next); // Free the last node
    temp->next = NULL; // Break the link
}

void printList(struct Node* node) {
    while (node != NULL) {
        printf("%d ", node->data);
        node = node->next;
    }
}

int main() {
    int n, val;
    struct Node* head = NULL;
    if (scanf("%d", &n) != 1) return 0;

```

```
for (int i = 0; i < n; i++) {  
    scanf("%d", &val);  
    append(&head, val);  
}  
  
deleteLast(head);  
printList(head);  
  
return 0;  
}
```

Status : Correct

Marks : 10/10

3. Problem Statement

Dev is tasked with creating a program that efficiently finds the middle element of a linked list. The program should take user input to populate the linked list by inserting each element into the front of the list and then determining the middle element.

Assist Dev, as he needs to ensure that the middle element is accurately identified from the constructed singly linked list:

If it's an odd-length linked list, return the middle element. If it's an even-length linked list, return the second middle element of the two elements.

Input Format

The first line of input consists of an integer n , representing the number of elements in the linked list.

The second line consists of n space-separated integers, representing the elements of the list.

Output Format

The first line of output displays the linked list after inserting elements at the front.

The second line displays "Middle Element: " followed by the middle element of the linked list.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 5

10 20 30 40 50

Output: 50 40 30 20 10

Middle Element: 30

Answer

// You are using GCC

#include <stdio.h>

#include <stdlib.h>

```
struct Node {  
    int data;  
    struct Node* next;  
};
```

// Insert at the front (head)

```
void push(struct Node** head_ref, int new_data) {  
    struct Node* new_node = (struct Node*)malloc(sizeof(struct Node));  
    new_node->data = new_data;  
    new_node->next = (*head_ref);  
    (*head_ref) = new_node;  
}
```

// Find middle using slow and fast pointers

```
int findMiddle(struct Node* head) {  
    struct Node *slow = head, *fast = head;
```

 // Fast moves twice as fast as slow

```
    while (fast != NULL && fast->next != NULL) {  
        slow = slow->next;  
        fast = fast->next->next;  
    }
```

```
    return slow->data;
```

```
}
```

```
void printList(struct Node* node) {
```

```
    while (node != NULL) {
```

```

        printf("%d ", node->data);
        node = node->next;
    }
    printf("\n");
}

int main() {
    int n, val;
    struct Node* head = NULL;

    if (scanf("%d", &n) != 1) return 0;

    for (int i = 0; i < n; i++) {
        scanf("%d", &val);
        push(&head, val);
    }

    printList(head);

    if (head != NULL) {
        printf("Middle Element: %d", findMiddle(head));
    }

    return 0;
}

```

Status : Correct

Marks : 10/10

4. Problem Statement

Saran is tasked with implementing a doubly linked list in data structures, allowing users to append elements to the end of the list, and remove the last element. He needs to ensure efficient memory management in the implementation.

Assist him with a suitable program.

Input Format

The first line of input consists of an integer n, representing the size of the list.

The second line consists of n space-separated integers, representing the elements of the list.

Output Format

The output displays integers, representing the list after removing the last element, separated by a space.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 5

48 22 5 62 87

Output: 48 22 5 62

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
// Structure for a doubly linked list node
```

```
struct Node {
```

```
    int data;
```

```
    struct Node* prev;
```

```
    struct Node* next;
```

```
};
```

```
// Function to append at the end
```

```
void append(struct Node** head_ref, int new_data) {
```

```
    struct Node* new_node = (struct Node*)malloc(sizeof(struct Node));
```

```
    struct Node* last = *head_ref;
```

```
    new_node->data = new_data;
```

```
    new_node->next = NULL;
```

```
    if (*head_ref == NULL) {
```

```
        new_node->prev = NULL;
```

```
        *head_ref = new_node;
```

```
        return;
```

```
    }
```

```

    while (last->next != NULL) {
        last = last->next;
    }

    last->next = new_node;
    new_node->prev = last;
}

// Function to remove the last node
void deleteLast(struct Node** head_ref) {
    if (*head_ref == NULL) return;

    struct Node* last = *head_ref;

    // Traverse to the last node
    while (last->next != NULL) {
        last = last->next;
    }

    // If there is only one node
    if (last->prev == NULL) {
        *head_ref = NULL;
    } else {
        last->prev->next = NULL;
    }

    free(last); // Efficient memory management
}

void printList(struct Node* node) {
    while (node != NULL) {
        printf("%d ", node->data);
        node = node->next;
    }
}

int main() {
    int n, val;
    struct Node* head = NULL;

    if (scanf("%d", &n) != 1) return 0;

```

```
for (int i = 0; i < n; i++) {  
    scanf("%d", &val);  
    append(&head, val);  
}  
  
deleteLast(&head);  
printList(head);  
  
return 0;  
}
```

Status : Correct

Marks : 10/10

5. Problem Statement:

Ravi is building a simple application to manage a doubly linked list of integers. He wants to insert new integers into the list and delete a node at a specific position. After deleting it, he needs to display the updated list.

Write a program to ensure that the task is done by performing insertion, deletion at a given position, and then printing the modified list.

Input Format

The first line contains an integer n , representing the number of elements to be inserted into the list.

The second line contains n space-separated integers, representing the values to insert.

The third line contains an integer pos , representing the position of the node to be deleted from the list.

Output Format

If the given position is valid, print "Node at position X with value Y deleted successfully".

If the position is invalid, print "Position X not found in the list."

In both cases, print

"Data present in the list:"

followed by the updated list elements on the next line, space separated."

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 5

10 20 30 40 50

4

Output: Node at position 4 with value 40 deleted successfully.

Data present in the list:

10 20 30 50

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
struct Node {  
    int data;  
    struct Node* prev;  
    struct Node* next;  
};
```

```
// Function to add elements at the end
```

```
void append(struct Node** head_ref, int new_data) {  
    struct Node* new_node = (struct Node*)malloc(sizeof(struct Node));  
    struct Node* last = *head_ref;  
    new_node->data = new_data;  
    new_node->next = NULL;
```

```

    if (*head_ref == NULL) {
        new_node->prev = NULL;
        *head_ref = new_node;
        return;
    }

    while (last->next != NULL)
        last = last->next;

    last->next = new_node;
    new_node->prev = last;
}

// Function to delete node at a specific position
void deleteAtPosition(struct Node** head_ref, int pos, int n) {
    if (*head_ref == NULL || pos < 1 || pos > n) {
        printf("Position %d not found in the list.\n", pos);
        return;
    }

    struct Node* temp = *head_ref;

    // Traverse to the node at position 'pos'
    for (int i = 1; temp != NULL && i < pos; i++) {
        temp = temp->next;
    }

    // This case covers if pos was somehow out of bounds despite the n check
    if (temp == NULL) {
        printf("Position %d not found in the list.\n", pos);
        return;
    }

    printf("Node at position %d with value %d deleted successfully.\n", pos, temp->data);

    // If node to be deleted is head node
    if (*head_ref == temp)
        *head_ref = temp->next;

    // Change next only if node to be deleted is NOT the last node
    if (temp->next != NULL)

```

```

temp->next->prev = temp->prev;

// Change prev only if node to be deleted is NOT the first node
if (temp->prev != NULL)
    temp->prev->next = temp->next;

free(temp);
}

void printList(struct Node* node) {
    printf("Data present in the list:\n");
    while (node != NULL) {
        printf("%d ", node->data);
        node = node->next;
    }
    printf("\n");
}

int main() {
    int n, val, pos;
    struct Node* head = NULL;

    if (scanf("%d", &n) != 1) return 0;

    for (int i = 0; i < n; i++) {
        scanf("%d", &val);
        append(&head, val);
    }

    scanf("%d", &pos);

    deleteAtPosition(&head, pos, n);
    printList(head);

    return 0;
}

```

Status : Correct

Marks : 10/10

Vel Tech Group of Institutions

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2028_NeoColab_Competitive Coding II_L9

VTU_Week 5_CY

Attempt : 1

Total Mark : 40

Marks Obtained : 40

Section 1 : Coding

1. Problem Statement

Given a singly linked list and two integers M and N, implement a program to traverse the linked list such that you retain every M node and then delete the next N nodes.

Continue this process until the end of the linked list.

Input Format

The first line consists of two integers M and N, separated by a space, representing the values to retain and delete.

The second line consists of a series of integers separated by spaces, representing the values of the linked list nodes.

The last input is -1 which will terminate the inputs.

Output Format

The output prints the linked list after applying the process of retaining every M node and deleting the next N nodes represented as a series of integers separated by "->".

Sample Test Case

Input: 2 2

1 2 3 4 5 6 7 8 -1

Output: 1->2->5->6

Answer

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
struct Node {  
    int data;  
    struct Node* next;  
};
```

```
struct Node* insert(struct Node* head, int data) {  
    struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));  
    newNode->data = data;  
    newNode->next = NULL;
```

```
    if (head == NULL)  
        return newNode;
```

```
    struct Node* temp = head;  
    while (temp->next != NULL)  
        temp = temp->next;
```

```
    temp->next = newNode;  
    return head;
```

```
}
```

```
struct Node* retainDelete(struct Node* head, int M, int N) {  
    struct Node* curr = head;  
    struct Node* temp;
```



```

while (curr != NULL) {
    for (int i = 1; i < M && curr != NULL; i++)
        curr = curr->next;

    if (curr == NULL)
        break;

    temp = curr->next;
    for (int i = 0; i < N && temp != NULL; i++) {
        struct Node* del = temp;
        temp = temp->next;
        free(del);
    }

    curr->next = temp;
    curr = temp;
}

return head;
}

```

```

void printList(struct Node* head) {
    struct Node* temp = head;
    while (temp != NULL) {
        printf("%d", temp->data);
        if (temp->next != NULL)
            printf("->");
        temp = temp->next;
    }
}

```

```

int main() {
    int M, N, x;
    struct Node* head = NULL;

    scanf("%d %d", &M, &N);

    while (1) {

```

```

scanf("%d", &x);
if (x == -1)
    break;
head = insert(head, x);
}

head = retainDelete(head, M, N);
printList(head);

return 0;
}

```

Status : Correct

Marks : 10/10

2. Problem Statement

Imagine you are working on a text processing tool and need to implement a feature that allows users to insert characters at a specific position.

Implement a program that takes user inputs to create a singly linked list of characters and inserts a new character after a given index in the list.

Input Format

The first line of input consists of an integer N, representing the number of characters in the linked list.

The second line consists of a sequence of N characters, representing the linked list.

The third line consists of an integer index, representing the index(0-based) after which the new character node needs to be inserted.

The fourth line consists of a character value representing the character to be inserted after the given index.

Output Format

If the provided index is out of bounds (larger than the list size):

1. The first line prints "Invalid index".
2. The second line prints "Updated list: " followed by the unchanged linked list values.

Otherwise, prints "Updated list: " followed by the updated space separated linked list after inserting the new character after the given index.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 5
a b c d e
2
X

Output: Updated list: a b c X d e

Answer

```
// You are using GCC
#include <stdio.h>
#include <stdlib.h>
```

```
struct Node {
    char data;
    struct Node* next;
};
```

```
struct Node* insertEnd(struct Node* head, char data) {
    struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));
    newNode->data = data;
    newNode->next = NULL;
```

```
    if (head == NULL)
        return newNode;
```

```
    struct Node* temp = head;
    while (temp->next != NULL)
        temp = temp->next;
```

```
    temp->next = newNode;
```

```
    return head;
}
```

```
void printList(struct Node* head) {
    struct Node* temp = head;
    while (temp != NULL) {
        printf("%c ", temp->data);
        temp = temp->next;
    }
}
```

```
struct Node* insertAfterIndex(struct Node* head, int index, char val, int size) {
    if (index >= size) {
        printf("Invalid index\n");
        printf("Updated list: ");
        printList(head);
        return head;
    }
```

```
    struct Node* temp = head;
```

```
    for (int i = 0; i < index; i++) {
        temp = temp->next;
    }
```

```
    struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));
    newNode->data = val;
    newNode->next = temp->next;
    temp->next = newNode;
```

```
    printf("Updated list: ");
    printList(head);
```

```
    return head;
}
```

```
int main() {
    int n, index;
    char ch, val;
    struct Node* head = NULL;
```

```

scanf("%d", &n);

for (int i = 0; i < n; i++) {
    scanf(" %c", &ch);
    head = insertEnd(head, ch);
}

scanf("%d", &index);
scanf(" %c", &val);

head = insertAfterIndex(head, index, val, n);

return 0;
}

```

Status : Correct

Marks : 10/10

3. Problem Statement

Imagine Anu is tasked with finding the middle element of a doubly linked list. Given a doubly linked list where each node contains an integer value and is inserted at the end, implement a program to find the middle element of the list. If the number of nodes is even, return the middle element pair.

Input Format

The first line of input consists of an integer N, representing the number of nodes in the doubly linked list.

The second line consists of N space-separated integers, representing the values of the nodes in the doubly linked list.

Output Format

The first line of output prints the space-separated elements of the doubly linked list.

The second line prints the middle element(s) of the doubly linked list, depending on whether the number of nodes is odd or even.

Refer to the sample outputs for the formatting specifications.

Sample Test Case

Input: 5

10 20 30 40 50

Output: 10 20 30 40 50

30

Answer

```
#include <stdio.h>
#include <stdlib.h>
struct Node {
    int data;
    struct Node *next, *prev;
};

void insertEnd(struct Node** head, int data) {
    struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));
    newNode->data = data;
    newNode->next = NULL;
    if (*head == NULL) {
        newNode->prev = NULL;
        *head = newNode;
        return;
    }
    struct Node* temp = *head;
    while (temp->next != NULL) temp = temp->next;
    temp->next = newNode;
    newNode->prev = temp;
}

void display(struct Node* head) {
    while (head != NULL) {
        printf("%d ", head->data);
        head = head->next;
    }
    printf("\n");
}

void findMiddle(struct Node* head, int n) {
    struct Node* temp = head;
    if (n % 2 != 0) {
        for (int i = 0; i < n / 2; i++) temp = temp->next;
```

```

        printf("%d\n", temp->data);
    } else {
        for (int i = 0; i < (n / 2) - 1; i++) temp = temp->next;
        printf("%d %d\n", temp->data, temp->next->data);
    }
}
int main() {
    int n, val;
    if (scanf("%d", &n) != 1 || n <= 0) return 0;
    struct Node* head = NULL;
    for (int i = 0; i < n; i++) {
        scanf("%d", &val);
        insertEnd(&head, val);
    }
    display(head);
    findMiddle(head, n);
    return 0;
}

```

Status : Correct

Marks : 10/10

4. Problem Statement

Pranav wants to clockwise rotate a doubly linked list by a specified number of positions. He needs your help to implement a program to achieve this. Given a doubly linked list and an integer representing the number of positions to rotate, write a program to rotate the list clockwise.

Input Format

The first line of input consists of an integer n , representing the number of elements in the linked list.

The second line consists of n space-separated linked list elements.

The third line consists of an integer k , representing the number of positions to rotate the list.

Output Format

The output displays the elements of the doubly linked list after rotating it by k positions.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 5

1 2 3 4 5

1

Output: 5 1 2 3 4

Answer

```
#include <stdio.h>
#include <stdlib.h>
struct Node {
    int data;
    struct Node* next;
    struct Node* prev;
};
void append(struct Node** head, int data) {
    struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));
    newNode->data = data;
    newNode->next = NULL;
    if (*head == NULL) {
        newNode->prev = NULL;
        *head = newNode;
        return;
    }
    struct Node* last = *head;
    while (last->next != NULL) last = last->next;
    last->next = newNode;
    newNode->prev = last;
}
void rotate(struct Node** head, int k, int n) {
    if (k == 0 || *head == NULL) return;
    struct Node* tail = *head;
    while (tail->next != NULL) tail = tail->next;
    tail->next = *head;
    (*head)->prev = tail;
    int stepsToNewHead = n - k;
    struct Node* newTail = *head;
```



```

    for (int i = 1; i < stepsToNewHead; i++) newTail = newTail->next;
    *head = newTail->next;
    (*head)->prev = NULL;
    newTail->next = NULL;
}
void display(struct Node* head) {
    while (head != NULL) {
        printf("%d", head->data);
        if (head->next != NULL) printf(" ");
        head = head->next;
    }
}
int main() {
    int n, val, k;
    if (scanf("%d", &n) != 1) return 0;
    struct Node* head = NULL;
    for (int i = 0; i < n; i++) {
        scanf("%d", &val);
        append(&head, val);
    }
    scanf("%d", &k);
    if (n > 0) {
        rotate(&head, k % n, n);
        display(head);
    }
    return 0;
}

```

Status : Correct

Marks : 10/10

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2028_NeoColab_Competitive Coding II_L9

VTU_Week 5_PAH

Attempt : 1

Total Mark : 40

Marks Obtained : 40

Section 1 : Coding

1. Problem Statement

Akash wants to write a program that populates values in a singly linked list by inserting each element into the front of the list. After insertion, the program should delete a node immediately after a node with a specific value X.

Assist him with a suitable program.

Input Format

The first line of input consists of an integer N, representing the number of elements.

The second line consists of N space-separated integers, representing the elements of the linked list.

The third line consists of an integer X, representing the value after which the node has to be deleted.

Output Format

The output consists of two lines:

The first line shows the original linked list in the format: Original Linked List: <elements> -> NULL.

The second line shows the modified linked list after deleting the node following the node with value X: After deleting node after <X>: <modified_elements> -> NULL.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 4
15 28 36 43
36

Output: Original Linked List: 43 -> 36 -> 28 -> 15 -> NULL
After deleting node after 36: 43 -> 36 -> 15 -> NULL

Answer

```
// You are using GCC
#include <stdio.h>
#include <stdlib.h>
struct Node{
    int data;
    struct Node* next;
};
struct Node* insertFront(struct Node* head,int data){
    struct Node* newNode=(struct Node*)malloc(sizeof(struct Node));
    newNode->data=data;
    newNode->next=head;
    return newNode;
}
void printList(struct Node* head){
    struct Node* temp=head;
    while(temp!=NULL){
```

```

    printf("%d -> ",temp->data);
    temp=temp->next;
}
printf("NULL\n");
}
void deleteAfterX(struct Node* head,int x){
    struct Node* temp=head;
    while(temp!=NULL && temp->next!=NULL){
        if(temp->data==x){
            struct Node* del=temp->next;
            temp->next=del->next;
            free(del);
            return;
        }
        temp=temp->next;
    }
}
int main(){
    int n,x,i,val;
    struct Node* head=NULL;
    scanf("%d",&n);
    for(i=0;i<n;i++){
        scanf("%d",&val);
        head=insertFront(head,val);
    }
    scanf("%d",&x);
    printf("Original Linked List: ");
    printList(head);
    printf("After deleting node after %d: ",x);
    deleteAfterX(head,x);
    printList(head);
    return 0;
}

```

Status : Correct

Marks : 10/10

2. Problem Statement

Emma is managing her daily to-do list. Each task is represented by an integer ID. Due to a change in her schedule, Emma needs to update her to-do list by removing the first task, another task from the end of the list, and

a third task from a specific position within the list.

Your task is to help Emma implement a function to perform these deletions and manage her to-do list efficiently.

Input Format

The first line of input contains an integer T, representing the number of tasks currently in the list.

The second line contains T integers, each representing the ID of a task.

The third line contains an integer p, representing the position (0-based index) of the task to be removed.

Output Format

The first line of output prints the linked list after deleting the first node, with each element separated by a space.

The second line of output prints the linked list after deleting the last node, with each element separated by a space.

The third line of output prints the linked list after deleting the node at the specified position, with each element separated by a space

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 5

11 22 33 44 55

2

Output: 22 33 44 55

22 33 44

22 33

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```

struct Node{
    int data;
    struct Node* next;
};

struct Node* insertEnd(struct Node* head,int data){
    struct Node* newNode=(struct Node*)malloc(sizeof(struct Node));
    newNode->data=data;
    newNode->next=NULL;
    if(head==NULL)return newNode;
    struct Node* temp=head;
    while(temp->next!=NULL)temp=temp->next;
    temp->next=newNode;
    return head;
}

struct Node* deleteFirst(struct Node* head){
    if(head==NULL)return NULL;
    struct Node* temp=head;
    head=head->next;
    free(temp);
    return head;
}

struct Node* deleteLast(struct Node* head){
    if(head==NULL||head->next==NULL){
        if(head!=NULL)free(head);
        return NULL;
    }
    struct Node* temp=head;
    while(temp->next->next!=NULL)temp=temp->next;
    free(temp->next);
    temp->next=NULL;
    return head;
}

struct Node* deleteAtPos(struct Node* head,int pos){
    if(head==NULL)return NULL;
    if(pos==0){
        struct Node* temp=head;
        head=head->next;
        free(temp);
        return head;
    }
    struct Node* temp=head;
    for(int i=0;i<pos-1 && temp->next!=NULL;i++)temp=temp->next;

```

```

    if(temp->next!=NULL){
        struct Node* del=temp->next;
        temp->next=del->next;
        free(del);
    }
    return head;
}

void printList(struct Node* head){
    struct Node* temp=head;
    while(temp!=NULL){
        printf("%d ",temp->data);
        temp=temp->next;
    }
    printf("\n");
}

int main(){
    int T,i,val,p;
    struct Node* head=NULL;
    scanf("%d",&T);
    for(i=0;i<T;i++){
        scanf("%d",&val);
        head=insertEnd(head,val);
    }
    scanf("%d",&p);
    head=deleteFirst(head);
    printList(head);
    head=deleteLast(head);
    printList(head);
    head=deleteAtPos(head,p);
    printList(head);
    return 0;
}

```

Status : Correct

Marks : 10/10

3. Problem Statement

Emma is working with a doubly linked list and needs a program to delete a specific value from the list. Write a program that reads a list of integers, constructs a doubly linked list, and then deletes a given value if it exists. The program should print appropriate messages if the value is found and

deleted or if it is not found in the list.

Input Format

The first line contains an integer n, representing the number of elements in the list.

The second line contains n space-separated integers, representing the elements of the doubly linked list.

The third line contains an integer x, representing the value to be deleted from the list.

Output Format

The first line prints the initial doubly linked list, with elements separated by a space.

The second line prints "Value x deleted from the list." on a new line if x is found and deleted, If x is not found, print "Value x not found in the list." on a new line.

The third line prints the updated doubly linked list on a new line after deletion, with elements separated by a space,

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 3

10 20 30

20

Output: 10 20 30

Value 20 deleted from the list.

10 30

Answer

```
// You are using GCC
#include <stdio.h>
#include <stdlib.h>
struct Node{
    int data;
```



```

    struct Node* prev;
    struct Node* next;
};

struct Node* insertEnd(struct Node* head,int data){
    struct Node* newNode=(struct Node*)malloc(sizeof(struct Node));
    newNode->data=data;
    newNode->next=NULL;
    newNode->prev=NULL;
    if(head==NULL)return newNode;
    struct Node* temp=head;
    while(temp->next!=NULL)temp=temp->next;
    temp->next=newNode;
    newNode->prev=temp;
    return head;
}

struct Node* deleteValue(struct Node* head,int x,int* found){
    struct Node* temp=head;
    while(temp!=NULL){
        if(temp->data==x){
            *found=1;
            if(temp->prev!=NULL)temp->prev->next=temp->next;
            else head=temp->next;
            if(temp->next!=NULL)temp->next->prev=temp->prev;
            free(temp);
            return head;
        }
        temp=temp->next;
    }
    return head;
}

void printList(struct Node* head){
    struct Node* temp=head;
    while(temp!=NULL){
        printf("%d ",temp->data);
        temp=temp->next;
    }
    printf("\n");
}

int main(){
    int n,x,i,val,found=0;
    struct Node* head=NULL;
    scanf("%d",&n);

```

```
for(i=0;i<n;i++){
    scanf("%d",&val);
    head=insertEnd(head,val);
}
scanf("%d",&x);
printList(head);
head=deleteValue(head,x,&found);
if(found)printf("Value %d deleted from the list.\n",x);
else printf("Value %d not found in the list.\n",x);
printList(head);
return 0;
}
```

Status : Correct

Marks : 10/10

4. Problem Statement

Imagine you're managing a store's inventory list, and some products were accidentally entered multiple times. You need to remove the duplicate products from the list to ensure each product appears only once.

You have an unsorted doubly linked list of product IDs. Some of these product IDs may appear more than once, and your goal is to remove any duplicates.

Note: Insert elements at the front of the doubly linked list.

Input Format

The first line of input consists of an integer n, representing the number of elements in the list.

The second line of input consists of n space-separated integers representing the list elements.

Output Format

The output prints the final list after removing duplicate nodes, separated by a space.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 10

12 12 10 4 8 4 6 4 4 8

Output: 8 4 6 10 12

Answer

// You are using GCC

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
struct Node{
```

```
    int data;
```

```
    struct Node* prev;
```

```
    struct Node* next;
```

```
};
```

```
struct Node* insertFront(struct Node* head,int data){
```

```
    struct Node* newNode=(struct Node*)malloc(sizeof(struct Node));
```

```
    newNode->data=data;
```

```
    newNode->prev=NULL;
```

```
    newNode->next=head;
```

```
    if(head!=NULL)head->prev=newNode;
```

```
    return newNode;
```

```
}
```

```
void removeDuplicates(struct Node* head){
```

```
    struct Node* ptr1=head;
```

```
    while(ptr1!=NULL){
```

```
        struct Node* ptr2=ptr1->next;
```

```
        while(ptr2!=NULL){
```

```
            if(ptr1->data==ptr2->data){
```

```
                struct Node* temp=ptr2;
```

```
                if(ptr2->prev!=NULL)ptr2->prev->next=ptr2->next;
```

```
                if(ptr2->next!=NULL)ptr2->next->prev=ptr2->prev;
```

```
                ptr2=ptr2->next;
```

```
                free(temp);
```

```
            }else ptr2=ptr2->next;
```

```
        }
```

```
        ptr1=ptr1->next;
```

```
    }
```

```
}
```

```
void printList(struct Node* head){
```

```
    struct Node* temp=head;
    while(temp!=NULL){
        printf("%d ",temp->data);
        temp=temp->next;
    }
    printf("\n");
}
int main(){
    int n,i,val;
    struct Node* head=NULL;
    scanf("%d",&n);
    for(i=0;i<n;i++){
        scanf("%d",&val);
        head=insertFront(head,val);
    }
    removeDuplicates(head);
    printList(head);
    return 0;
}
```

Status : Correct

Marks : 10/10

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2028_NeoColab_Competitive Coding II_L9

VTU_Week 6_MCQ

Attempt : 1

Total Mark : 15

Marks Obtained : 13

Section 1 : MCQ

1. In Radix Sort, what happens if the input array contains numbers with different lengths?

Answer

The shorter numbers are padded with zeros.

Status : Correct

Marks : 1/1

2. In heap sort, when does the actual sorting take place?

Answer

During the swap and heapify phase

Status : Correct

Marks : 1/1

3. In heap sort, which of the following heap operations is performed to build a valid heap from an unsorted array?

Answer

Heapify

Status : Correct

Marks : 1/1

4. In heap sort, the element that is swapped with the final element of the heap is _____, in the case of a max-heap, sorted in descending order.

Answer

Largest element

Status : Correct

Marks : 1/1

5. Bucket sort is most efficient in the case when _____

Answer

The input is uniformly distributed

Status : Correct

Marks : 1/1

6. Which of the following is not necessarily a stable sorting algorithm?

Answer

All of the mentioned options

Status : Wrong

Marks : 0/1

7. Which step is essential to implement Bucket Sort correctly?

Answer

Dividing the range of input into intervals (buckets)

Status : Correct

Marks : 1/1

8. Which of the following is a characteristic of Radix Sort?

Answer

It is a stable sorting algorithm.

Status : Correct

Marks : 1/1

9. A database contains the following employee IDs: [1263, 2910, 8742, 3891, 2156]. Using Radix Sort, which of the following is the correct order of employee IDs after the first pass (sorting by the least significant digit)?

Answer

[2910, 3891, 8742, 1263, 2156]

Status : Correct

Marks : 1/1

10. What is the purpose of the counting sort algorithm in Radix Sort?

Answer

To sort the elements based on the current digit.

Status : Correct

Marks : 1/1

11. When does Heap Sort perform at its worst?

Answer

When the input array elements are required to be sorted in reverse order.

Status : Wrong

Marks : 0/1

12. How does Counting Sort handle duplicate elements?

Answer

It preserves their relative order

Status : Correct

Marks : 1/1

13. What is the purpose of the count array in Counting Sort?

Answer

To store the count of each unique element of the input array at their respective indices

Status : Correct

Marks : 1/1

14. What is the purpose of calculating the prefix sum in the Counting Sort algorithm?

Answer

To determine the final positions of elements in the output array

Status : Correct

Marks : 1/1

15. What is the initial value of each element in the count array in Counting Sort?

Answer

0

Status : Correct

Marks : 1/1

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2028_NeoColab_Competitive Coding II_L9

VTU_Week 6_COD

Attempt : 1

Total Mark : 50

Marks Obtained : 50

Section 1 : Coding

1. Problem Statement

Krish is participating in a coding competition and has been given an array of integers. His task is to sort the array using the Counting Sort algorithm.

Your goal is to help Krish by implementing the countSort method to sort the array in non-decreasing order.

Input Format

The first line of input consists of an integer N, representing the number of elements in the array.

The second line of input consists of N space-separated integers, representing the elements of the array.

Output Format

The output prints a single line containing N space-separated integers representing the sorted array.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 5

4 2 2 8 3

Output: 2 2 3 4 8

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
void countSort(int arr[], int n) {  
    // 1. Find the maximum element to determine the range  
    int max = arr[0];  
    for (int i = 1; i < n; i++) {  
        if (arr[i] > max)  
            max = arr[i];  
    }  
  
    // 2. Create and initialize a frequency array with 0  
    int count[101] = {0}; // Constraint: elements <= 100  
  
    // 3. Store the count of each element  
    for (int i = 0; i < n; i++) {  
        count[arr[i]]++;  
    }  
  
    // 4. Overwrite the original array with sorted elements  
    int index = 0;  
    for (int i = 0; i <= max; i++) {  
        while (count[i] > 0) {  
            arr[index++] = i;  
            count[i]--;  
        }  
    }  
}
```

```

int main() {
    int n;

    // Read the number of elements
    if (scanf("%d", &n) != 1) return 0;

    int arr[n];

    // Read the array elements
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    // Perform Counting Sort
    countSort(arr, n);

    // Print the sorted array
    for (int i = 0; i < n; i++) {
        printf("%d%s", arr[i], (i == n - 1) ? "" : " ");
    }

    return 0;
}

```

Status : Correct

Marks : 10/10

2. Problem Statement

Raj, a keen problem solver, has come across an intriguing numerical puzzle that involves sorting numbers in ascending order. Eager to crack the code, he decides to implement the Radix Sort algorithm.

Your task is to assist Raj in designing a program to solve this numerical puzzle.

Input Format

The first line contains an integer n , denoting the number of elements to be sorted.

The second line consists of n space-separated integers, representing the numbers to be sorted.

Output Format

The output prints the array after it has been sorted using the Radix Sort algorithm, separated by a space.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 5

543 986 217 765 329

Output: 217 329 543 765 986

Answer

```
// You are using GCC
#include <stdio.h>
```

```
// Function to find the maximum value in an array to determine the number of
digits
```

```
int getMax(int arr[], int n) {
    int max = arr[0];
    for (int i = 1; i < n; i++)
        if (arr[i] > max)
            max = arr[i];
    return max;
}
```

```
// Stable counting sort used to sort based on the digit at 'exp' position ( $10^i$ )
```

```
void countSort(int arr[], int n, int exp) {
    int output[n];
    int count[10] = {0};
```

```
    // 1. Store count of occurrences of digits (0-9)
```

```
    for (int i = 0; i < n; i++)
        count[(arr[i] / exp) % 10]++;
```

```
    // 2. Update count[i] to contain the actual position in the output array
```

```
    for (int i = 1; i < 10; i++)
```

```

        count[i] += count[i - 1];

// 3. Build the output array using the count positions
for (int i = n - 1; i >= 0; i--) {
    output[count[(arr[i] / exp) % 10] - 1] = arr[i];
    count[(arr[i] / exp) % 10]--;
}

// 4. Copy sorted output back to the original array
for (int i = 0; i < n; i++)
    arr[i] = output[i];
}

```

```

// Radix Sort Implementation
void radixSort(int arr[], int n) {
    int m = getMax(arr, n);

    // Process each digit (1s, 10s, 100s, etc.)
    for (int exp = 1; m / exp > 0; exp *= 10)
        countSort(arr, n, exp);
}

```

```

int main() {
    int n;

    // Input number of elements
    if (scanf("%d", &n) != 1) return 0;

    int arr[n];
    // Input array elements
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    radixSort(arr, n);

    // Print sorted array with space separation
    for (int i = 0; i < n; i++) {
        printf("%d%s", arr[i], (i == n - 1) ? "" : " ");
    }
    printf("\n");
}

```

```
    return 0;  
}
```

Status : Correct

Marks : 10/10

3. Problem Statement

Amit, an e-commerce manager, is working on an inventory management system to track the top-selling products. Given the sales data of various products, he wants to quickly determine the Kth best-selling product to make informed restocking decisions.

Help Amit implement Heap Sort to efficiently find the Kth best-selling product from the given sales data.

Input Format

The first line of input consists of an integer n, representing the number of products.

The second line consists of n space-separated integers, representing the sales data of each product.

The third line contains an integer K, representing the rank of the product to retrieve.

Output Format

The output prints the Kth best-selling product based on the given sales data.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 5
10 20 15 5 25
2

Output: 20

Answer

```
// You are using GCC
#include <stdio.h>
```

```
// Function to swap two elements
void swap(long long* a, long long* b) {
    long long temp = *a;
    *a = *b;
    *b = temp;
}
```

```
// Function to maintain the Max-Heap property
void heapify(long long arr[], int n, int i) {
    int largest = i;
    int left = 2 * i + 1;
    int right = 2 * i + 2;
```

```
    // If left child is larger than root
    if (left < n && arr[left] > arr[largest])
        largest = left;
```

```
    // If right child is larger than largest so far
    if (right < n && arr[right] > arr[largest])
        largest = right;
```

```
    // If largest is not root
    if (largest != i) {
        swap(&arr[i], &arr[largest]);
        heapify(arr, n, largest);
    }
}
```

```
long long findKthBestSelling(long long arr[], int n, int k) {
    // 1. Build Max-Heap (rearrange array)
    for (int i = n / 2 - 1; i >= 0; i--) {
        heapify(arr, n, i);
    }
```

```
    // 2. Extract elements from heap K times
    // We swap the root (max) with the last element and heapify the reduced heap
    for (int i = n - 1; i > n - 1 - k; i--) {
        // The Kth element we "remove" is the Kth largest
        if (n - i == k) {
```

```

        return arr[0];
    }
    swap(&arr[0], &arr[i]);
    heapify(arr, i, 0);
}
return -1;
}

int main() {
    int n, k;

    // Read number of products
    if (scanf("%d", &n) != 1) return 0;

    long long sales[n];
    for (int i = 0; i < n; i++) {
        scanf("%lld", &sales[i]);
    }

    // Read K
    scanf("%d", &k);

    // Find and print the Kth best-selling product
    printf("%lld\n", findKthBestSelling(sales, n, k));

    return 0;
}

```

Status : Correct

Marks : 10/10

4. Problem Statement

Ragavi, a computer scientist, is working on efficient text processing for search engines. She has a list of words in random order and needs to sort them in lexicographic order using Heap Sort to improve search performance.

Help Ragavi implement Heap Sort to arrange the words in lexicographic order efficiently.

Input Format

The first line consists of an integer N denoting the number of strings.

The next line consists of N space-separated strings.

Output Format

The output displays N space-separated sorted strings in lexicographic order.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 10

banana pineapple mango chikoo jack star guava lemon tomato orange

Output: banana chikoo guava jack lemon mango orange pineapple star tomato

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
#include <string.h>
```

```
// Function to swap two strings
```

```
void swap(char arr[][100], int i, int j) {
```

```
    char temp[100];
```

```
    strcpy(temp, arr[i]);
```

```
    strcpy(arr[i], arr[j]);
```

```
    strcpy(arr[j], temp);
```

```
}
```

```
// Function to maintain Max-Heap property for strings
```

```
void heapify(char arr[][100], int n, int i) {
```

```
    int largest = i;
```

```
    int left = 2 * i + 1;
```

```
    int right = 2 * i + 2;
```

```
// Compare strings lexicographically using strcmp
```

```
if (left < n && strcmp(arr[left], arr[largest]) > 0)
```

```
    largest = left;
```

```

    if (right < n && strcmp(arr[right], arr[largest]) > 0)
        largest = right;

    if (largest != i) {
        swap(arr, i, largest);
        heapify(arr, n, largest);
    }
}

void heapSort(char arr[][100], int n) {
    // 1. Build Max-Heap
    for (int i = n / 2 - 1; i >= 0; i--) {
        heapify(arr, n, i);
    }

    // 2. Extract elements from heap one by one
    for (int i = n - 1; i > 0; i--) {
        swap(arr, 0, i); // Move current root to end
        heapify(arr, i, 0); // Call max heapify on the reduced heap
    }
}

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;

    char words[n][100]; // Array of strings (max length 100)

    for (int i = 0; i < n; i++) {
        scanf("%s", words[i]);
    }

    heapSort(words, n);

    // Print sorted strings
    for (int i = 0; i < n; i++) {
        printf("%s%s", words[i], (i == n - 1) ? "" : " ");
    }
    printf("\n");

    return 0;
}

```

Status : Correct

Marks : 10/10

5. Problem Statement

You're organizing a gift-giving event where participants exchange gifts of varying values. To ensure fairness, you use the Bucket Sort algorithm. You gather the participants' gift values and initialize a count array.

Then, you iterate through the values, updating the count array. Next, you sort the values in descending order by iterating through the count array. Finally, you display the sorted values, ensuring fairness in the gift exchange.

Input Format

The first line contains an integer N, representing the number of gift values.

The second line contains N space-separated integers representing the gift values brought by the participants. Each gift value is an integer between 1 and 999, inclusive.

Output Format

The output displays a single line containing N space-separated integers representing the sorted gift values in descending order.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 6

156 378 453 278 281 987

Output: 987 453 378 281 278 156

Answer

```
// You are using GCC
#include <stdio.h>
```

```
int main() {
    int n, giftValue;
```

```

// Read the number of gifts
if (scanf("%d", &n) != 1) return 0;

// Initialize the count array for values 1 to 999
int count[1000] = {0};

// 1. Fill the count array based on input values
for (int i = 0; i < n; i++) {
    if (scanf("%d", &giftValue) == 1) {
        count[giftValue]++;
    }
}

// 2. Iterate through the count array in reverse to sort descending
int first = 1; // Flag to handle trailing space
for (int i = 999; i >= 1; i--) {
    // While there are gifts of this specific value
    while (count[i] > 0) {
        if (!first) {
            printf(" ");
        }
        printf("%d", i);
        count[i]--;
        first = 0;
    }
}
printf("\n");

return 0;
}

```

Status : Correct

Marks : 10/10

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2028_NeoColab_Competitive Coding II_L9

VTU_Week 6_CY

Attempt : 1

Total Mark : 40

Marks Obtained : 40

Section 1 : Coding

1. Problem Statement

Yuvana, an avid programmer, is keen on mastering various sorting algorithms and has recently delved into Counting Sort. As a practice exercise, she is given an array of N integers where each element ranges from 1 to N-1. Some numbers may appear multiple times in the array. Her task is to identify and print all numbers that appear more than once. If no such duplicates are found, she should print -1.

Help Yuvana by implementing a function that efficiently finds and prints duplicate numbers using the Counting Sort technique.

Input Format

The first line of input consists of an integer N, representing the number of elements in the array.

The second line of input consists of N space-separated integers, representing the elements of the array.

Output Format

The output prints all the duplicate numbers that appear more than once in the array, each separated by a space.

If no duplicates are found, print -1.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 5

1 3 4 2 2

Output: 2

Answer

// You are using GCC

#include <stdio.h>

```
int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;

    int arr[n];
    // Frequency array initialized to 0
    // Size n+1 covers elements ranging from 1 to n-1
    int count[21] = {0};
    int found = 0;

    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
        count[arr[i]]++;
    }

    for (int i = 1; i <= n; i++) {
        if (count[i] > 1) {
            printf("%d ", i);
        }
    }
}
```

```
        found = 1;
    }
}

if (!found) {
    printf("-1");
}

return 0;
}
```

Status : Correct

Marks : 10/10

2. Problem Statement

Anu is tasked with organizing a list of fruit names stored as strings. To ensure that the list is easy to navigate, she decides to sort the strings in ascending alphabetical order.

Being a fan of efficient sorting algorithms, Anu opts to use radix sort for this task. Help Anu with a program that can take a list of fruit names as input and output the sorted list.

Input Format

The first line of input consists of an integer n , representing the number of fruit names in the list.

The next n lines each consist of a string, representing the names of the fruits.

Output Format

The output prints n lines containing the sorted list of fruit names in ascending alphabetical order.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 4

apple
banana
orange
grape

Output: apple
banana
grape
orange

Answer

```
// You are using GCC
#include <stdio.h>
#include <string.h>
#include <stdlib.h>
void radixSort(char arr[][101], int n) {
    int maxLen = 0;
    for (int i = 0; i < n; i++) {
        int len = strlen(arr[i]);
        if (len > maxLen) maxLen = len;
    }
    char temp[n][101];
    for (int pos = maxLen - 1; pos >= 0; pos--) {
        int count[256] = {0};
        for (int i = 0; i < n; i++) {
            unsigned char c = (pos < strlen(arr[i])) ? (unsigned char)arr[i][pos] : 0;
            count[c]++;
        }
        for (int i = 1; i < 256; i++) {
            count[i] += count[i - 1];
        }
        for (int i = n - 1; i >= 0; i--) {
            unsigned char c = (pos < strlen(arr[i])) ? (unsigned char)arr[i][pos] : 0;
            strcpy(temp[--count[c]], arr[i]);
        }
        for (int i = 0; i < n; i++) {
            strcpy(arr[i], temp[i]);
        }
    }
}
int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;
```



```
char fruits[n][101];
for (int i = 0; i < n; i++) {
    scanf("%s", fruits[i]);
}
radixSort(fruits, n);
for (int i = 0; i < n; i++) {
    printf("%s\n", fruits[i]);
}
return 0;
}
```

Status : Correct

Marks : 10/10

3. Problem Statement

Riya, an HR manager, needs a quick way to identify the Kth highest salary among employees for bonus distribution. To efficiently retrieve this information, she wants to use the max-heap technique.

Help Riya implement a solution that finds the Kth highest salary from the given list of salaries.

Input Format

The first line of input consists of an integer N, representing the number of employees.

The second line consists of N space-separated integers, representing the salaries of N employees.

The third line consists of an integer K, indicating the rank of the desired Kth highest salary.

Output Format

The output prints: <k>th largest Salary: <result>

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 7

30000 25000 40000 35000 20000 28000 29500

5

Output: 5th largest Salary: 28000

Answer

// You are using GCC

#include <stdio.h>

```
void swap(int *a, int *b) {
```

```
    int temp = *a;
```

```
    *a = *b;
```

```
    *b = temp;
```

```
}
```

```
void maxHeapify(int arr[], int n, int i) {
```

```
    int largest = i;
```

```
    int left = 2 * i + 1;
```

```
    int right = 2 * i + 2;
```

```
    if (left < n && arr[left] > arr[largest])
```

```
        largest = left;
```

```
    if (right < n && arr[right] > arr[largest])
```

```
        largest = right;
```

```
    if (largest != i) {
```

```
        swap(&arr[i], &arr[largest]);
```

```
        maxHeapify(arr, n, largest);
```

```
    }
```

```
}
```

```
int main() {
```

```
    int n, k;
```

```
    if (scanf("%d", &n) != 1) return 0;
```

```
    int salaries[n];
```

```
    for (int i = 0; i < n; i++) {
```

```
        scanf("%d", &salaries[i]);
```

```
    }
```

```

scanf("%d", &k);

// Build Max-Heap
for (int i = n / 2 - 1; i >= 0; i--) {
    maxHeapify(salaries, n, i);
}

// Extract maximum K times
int result;
int currentSize = n;
for (int i = 0; i < k; i++) {
    result = salaries[0];
    salaries[0] = salaries[currentSize - 1];
    currentSize--;
    maxHeapify(salaries, currentSize, 0);
}

printf("%dth largest Salary: %d", k, result);

return 0;
}

```

Status : Correct

Marks : 10/10

4. Problem Statement

Manage a warehouse inventory with unique barcode numbers utilizing a specialized sorting algorithm designed for barcodes with exactly 16 digits. Input the number of barcode numbers (N) and their corresponding values.

Employ the Bucket Sort algorithm to distribute and sort the numbers based on their digits. Display the sorted barcode numbers, ensuring an efficient and organized warehouse inventory management system tailored for barcode systems with 16 digits.

Input Format

The first line contains an integer N, representing the number of barcode numbers to be sorted.

The following N lines contain exactly 16-digit long long integers, each representing a unique barcode number.

Output Format

The output displays a single line containing "Sorted barcode numbers:" followed by the sorted barcode numbers in ascending order. Each barcode number is displayed on a separate line.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 5

2345678901234567
1234567890123456
3456789012345678
5678901234567890
4567890123456789

Output: Sorted barcode numbers:

1234567890123456
2345678901234567
3456789012345678
4567890123456789
5678901234567890

Answer

```
// You are using GCC
#include <stdio.h>
#include <stdlib.h>
```

```
typedef struct Node {
    long long data;
    struct Node* next;
} Node;
```

```
void insert(Node** bucket, long long val) {
    Node* newNode = (Node*)malloc(sizeof(Node));
    newNode->data = val;
    newNode->next = NULL;
    if (*bucket == NULL || (*bucket)->data >= val) {
```

```

        newNode->next = *bucket;
        *bucket = newNode;
    } else {
        Node* current = *bucket;
        while (current->next != NULL && current->next->data < val) {
            current = current->next;
        }
        newNode->next = current->next;
        current->next = newNode;
    }
}

```

```

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;
    long long arr[n];
    long long min = 9999999999999999, max = 0;
    for (int i = 0; i < n; i++) {
        scanf("%lld", &arr[i]);
        if (arr[i] < min) min = arr[i];
        if (arr[i] > max) max = arr[i];
    }
    int bucketCount = n;
    Node* buckets[bucketCount];
    for (int i = 0; i < bucketCount; i++) buckets[i] = NULL;
    for (int i = 0; i < n; i++) {
        int idx = (n - 1) * (arr[i] - min) / (max - min + 1);
        insert(&buckets[idx], arr[i]);
    }
    printf("Sorted barcode numbers:\n");
    for (int i = 0; i < bucketCount; i++) {
        Node* curr = buckets[i];
        while (curr) {
            printf("%lld\n", curr->data);
            Node* temp = curr;
            curr = curr->next;
            free(temp);
        }
    }
    return 0;
}

```

Status : Correct

Marks : 10/10

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2028_NeoColab_Competitive Coding II_L9

VTU_Week 6_PAH

Attempt : 1

Total Mark : 40

Marks Obtained : 40

Section 1 : Coding

1. Problem Statement

Alex is working on a data analysis project and needs to find specific elements within a dataset. He has an array of integers where some numbers may be repeated. His goal is to identify the Kth smallest element in this array. The challenge is that the array can contain duplicate values, so finding the exact Kth smallest element requires careful consideration.

Alex has decided to use a counting sort approach to solve this problem. Given an array of integers and a number K, where K is smaller than the size of the array, he needs to find and print the Kth smallest element in the array.

Help Alex by implementing a function that efficiently finds and prints the Kth smallest element using the counting sort approach.

Input Format

The first line of input consists of an integer N, representing the number of elements in the array.

The second line of input consists of N space-separated integers, representing the elements of the array.

The third line of input consists of an integer K, representing the position of the smallest element to find.

Output Format

The output prints the Kth smallest element in the array as "Kth Smallest element is <smallest element>".

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 6

7 10 4 3 20 15

3

Output: Kth Smallest element is 7

Answer

```
// You are using GCC
#include <stdio.h>
```

```
int main() {
    int n, k;
    if (scanf("%d", &n) != 1) return 0;
```

```
    int arr[n];
    int max = 0;
```

```
    // Read elements and find the maximum value for the count array range
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
        if (arr[i] > max) max = arr[i];
    }
```



```

scanf("%d", &k);

// Frequency array (size max + 1 to accommodate the largest element)
int count[101] = {0};

// Step 1: Fill the frequency array
for (int i = 0; i < n; i++) {
    count[arr[i]]++;
}

// Step 2: Traverse the frequency array to find the Kth smallest
int cumulativeCount = 0;
for (int i = 0; i <= 100; i++) {
    if (count[i] > 0) {
        cumulativeCount += count[i];

        // If the cumulative count reaches or exceeds K, 'i' is our target
        if (cumulativeCount >= k) {
            printf("Kth Smallest element is %d", i);
            return 0;
        }
    }
}

return 0;
}

```

Status : Correct

Marks : 10/10

2. Problem Statement

Rohan is developing a text analysis tool and wants to sort a list of words based on the count of vowels in each word. He believes that sorting the words based on vowel count will help him analyze the text more effectively.

To achieve this, he plans to use radix sort. Assist Rohan with a suitable program to implement this sorting method for his project.

Input Format

The first line of input consists of an integer n, representing the number of words.

The next n lines each consist of a string, representing the words.

Output Format

The output prints n lines containing the sorted list of words based on the count of vowels in each word.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 4

another

stuck

door

you

Output: stuck

door

you

another

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
#include <string.h>
```

```
#include <ctype.h>
```

```
#include <stdlib.h>
```

```
int countVowels(char *s) {
```

```
    int count = 0;
```

```
    for (int i = 0; s[i] != '\0'; i++) {
```

```
        char c = tolower(s[i]);
```

```
        if (c == 'a' || c == 'e' || c == 'i' || c == 'o' || c == 'u') count++;
```

```
    }
```

```
    return count;
```

```
}
```

```
void stableSort(char words[][101], int n) {
```

```
    int counts[n];
```

```

    for (int i = 0; i < n; i++) counts[i] = countVowels(words[i]);
    char temp[n][101];
    int freq[101] = {0};
    for (int i = 0; i < n; i++) freq[counts[i]]++;
    for (int i = 1; i <= 100; i++) freq[i] += freq[i - 1];
    for (int i = n - 1; i >= 0; i--) {
        int vCount = counts[i];
        strcpy(temp[--freq[vCount]], words[i]);
    }
    for (int i = 0; i < n; i++) strcpy(words[i], temp[i]);
}

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;
    char words[n][101];
    for (int i = 0; i < n; i++) scanf("%s", words[i]);
    stableSort(words, n);
    for (int i = 0; i < n; i++) printf("%s\n", words[i]);
    return 0;
}

```

Status : Correct

Marks : 10/10

3. Problem Statement

In a manufacturing plant, each product is assigned an integer production code. To streamline quality control, the products have to be sorted in ascending order based on the last digit of their production codes.

Develop a program that uses Radix Sort to sort the products according to their code's last digit.

Input Format

The first line consists of an integer n , representing the number of products.

The second line consists of n space-separated integers each representing a product code.

Output Format

The output prints the sorted product codes based on the last digit in ascending

order, separated by a space.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 3

24570 61064 65391

Output: 24570 65391 61064

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
int main() {
```

```
    int n;
```

```
    if (scanf("%d", &n) != 1) return 0;
```

```
    int codes[n];
```

```
    int count[10] = {0};
```

```
    int output[n];
```

```
    // Read input and count frequencies of the last digit
```

```
    for (int i = 0; i < n; i++) {
```

```
        scanf("%d", &codes[i]);
```

```
        int lastDigit = codes[i] % 10;
```

```
        count[lastDigit]++;
```

```
    }
```

```
    // Change count[i] so that it contains actual position of this digit in output[]
```

```
    for (int i = 1; i < 10; i++) {
```

```
        count[i] += count[i - 1];
```

```
    }
```

```
    // Build the output array (traverse backwards to maintain stability)
```

```
    for (int i = n - 1; i >= 0; i--) {
```

```
        int lastDigit = codes[i] % 10;
```

```
        output[count[lastDigit] - 1] = codes[i];
```

```
        count[lastDigit]--;
```

```
    }
```

```
// Print the sorted codes
for (int i = 0; i < n; i++) {
    printf("%d ", output[i]);
}

return 0;
}
```

Status : Correct

Marks : 10/10

4. Problem Statement

Meera, a librarian, wants to organize a list of book titles based on how many words each title contains. If two titles have the same number of words, they should be arranged in lexicographical (dictionary) order.

Write a program to sort the given book titles using a bucket-based approach, where titles are first grouped by word count and then alphabetically sorted within each group.

Input Format

The first line contains an integer N, representing the number of book titles.

The next N lines: Each line contains a book title (may contain spaces and special characters).

Output Format

The output prints "Sorted book titles:"

Then print the sorted book titles, each on a new line.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 4
Harry Potter and the Philosopher's Stone

To Kill a Mockingbird
The Great Gatsby
1984

Output: Sorted book titles:

1984
The Great Gatsby
To Kill a Mockingbird
Harry Potter and the Philosopher's Stone

Answer

// You are using GCC

#include <stdio.h>

#include <string.h>

#include <stdlib.h>

```
typedef struct {  
    char title[101];  
} Book;
```

// Function to count words in a string

```
int countWords(char *str) {  
    int count = 0;  
    char temp[101];  
    strcpy(temp, str);  
    char *token = strtok(temp, " \\t\\n");  
    while (token != NULL) {  
        count++;  
        token = strtok(NULL, " \\t\\n");  
    }  
    return count;  
}
```

// Comparison function for qsort (Lexicographical)

```
int compare(const void *a, const void *b) {  
    return strcmp(((Book *)a)->title, ((Book *)b)->title);  
}
```

```
int main() {  
    int n;  
    if (scanf("%d", &n) != 1) return 0;  
    getchar(); // Consume newline
```

```

Book books[n];
int wordCounts[n];
int maxWords = 0;

for (int i = 0; i < n; i++) {
    fgets(books[i].title, 101, stdin);
    books[i].title[strcspn(books[i].title, "\n")] = 0; // Remove newline
    wordCounts[i] = countWords(books[i].title);
    if (wordCounts[i] > maxWords) maxWords = wordCounts[i];
}

printf("Sorted book titles:\n");

// Process buckets from 1 word up to maxWords
for (int w = 1; w <= maxWords; w++) {
    Book bucket[50];
    int bucketSize = 0;

    // Collect all books with current word count 'w'
    for (int i = 0; i < n; i++) {
        if (wordCounts[i] == w) {
            bucket[bucketSize++] = books[i];
        }
    }

    // Sort the current bucket alphabetically
    if (bucketSize > 0) {
        qsort(bucket, bucketSize, sizeof(Book), compare);
        for (int i = 0; i < bucketSize; i++) {
            printf("%s\n", bucket[i].title);
        }
    }
}

return 0;
}

```

Status : Correct

Marks : 10/10

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2028_NeoColab_Competitive Coding II_L9

VTU_Week 7_MCQ

Attempt : 2

Total Mark : 15

Marks Obtained : 15

Section 1 : MCQ

1. Which of the following operations can be used to traverse a Binary Search Tree (BST) in ascending order?

Answer

Inorder traversal

Status : Correct

Marks : 1/1

2. Find the pre-order traversal of the given binary search tree.

Answer

13, 2, 1, 4, 14, 18

Status : Correct

Marks : 1/1

3. Find the postorder traversal of the given binary search tree.

Answer

1, 4, 2, 18, 14, 13

Status : Correct

Marks : 1/1

4. Find the inorder traversal of the given binary search tree.

Answer

1, 2, 4, 6, 7, 9, 10, 14

Status : Correct

Marks : 1/1

5. When comparing a value to be searched with the root node in a BST?

Answer

If it's larger, go to the right subtree

Status : Correct

Marks : 1/1

6. In a binary search tree (BST), if the value to be searched is smaller than the value of the root, where should the search proceed?

Answer

Left subtree

Status : Correct

Marks : 1/1

7. In a binary search tree with nodes 18, 28, 12, 11, 16, 14, 17, what is the value of the right child of the node 12?

Answer

16

Status : Correct

Marks : 1/1

8. In a binary search tree with nodes 18, 28, 12, 11, 16, 14, 17, what is the value of the left child of the node 16?

Answer

14

Status : Correct

Marks : 1/1

9. The preorder traversal sequence of a binary search tree is 30, 20, 10, 15, 25, 23, 39, 35, 42. Which one of the following is the post-order traversal sequence of the same tree?

Answer

15, 10, 23, 25, 20, 35, 42, 39, 30

Status : Correct

Marks : 1/1

10. In a binary search tree, which of the following traversals is used for getting ascending order values?

Answer

In-order

Status : Correct

Marks : 1/1

11. Consider the given binary search tree, if the root node is deleted, then the new root can be _____.

Answer

48 or 59

Status : Correct

Marks : 1/1

12. After deleting a node in a binary search tree, how is the replacement node determined if the node to be deleted has only one child?

Answer

Replacement node is the child node itself

Status : Correct

Marks : 1/1

13. What will be the in-order traversal sequence of the following binary search tree, after deleting node 40?

Answer

8 12 20 22 30

Status : Correct

Marks : 1/1

14. Which of the following nodes would replace node 40 if it were to be deleted?

Answer

30

Status : Correct

Marks : 1/1

15. Which of the following statements is true about deleting a node with one child from a binary search tree?

Answer

The child node replaces the deleted node

Status : Correct

Marks : 1/1

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2028_NeoColab_Competitive Coding II_L9

VTU_Week 7_COD

Attempt : 1

Total Mark : 50

Marks Obtained : 50

Section 1 : Coding

1. Problem Statement

A publishing company is building a digital index system for storing book references. Each reference is represented by a single uppercase character. To ensure efficient organization, the system uses a Binary Search Tree (BST):

If the new character is smaller than the current node, it is placed in the left subtree. If the new character is greater, it is placed in the right subtree.

Once all references are stored in the BST, the system needs to perform a postorder traversal to generate a summary sequence, which is then used for indexing and cross-referencing.

Your task is to help the publishing company build the BST from the given characters and print the postorder traversal.

Example:

Input:

5

Z E W T Y

Output:

Postorder traversal: T Y W E Z

Input Format

The first line consists of an integer N, representing the number of characters.

The second line consists of N space-separated characters.

Output Format

The output displays the post-order traversal of the given inputs as space-separated.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 5

Z E W T Y

Output: Postorder traversal: T Y W E Z

Answer

```
// You are using GCC
#include <stdio.h>
#include <stdlib.h>
```

```
typedef struct Node {
    char data;
    struct Node* left;
    struct Node* right;
} Node;
```

```
Node* createNode(char data) {  
    Node* newNode = (Node*)malloc(sizeof(Node));  
    newNode->data = data;  
    newNode->left = NULL;  
    newNode->right = NULL;  
    return newNode;  
}
```

```
Node* insert(Node* root, char data) {  
    if (root == NULL) {  
        return createNode(data);  
    }  
  
    if (data < root->data) {  
        root->left = insert(root->left, data);  
    } else if (data > root->data) {  
        root->right = insert(root->right, data);  
    }  
  
    return root;  
}
```

```
void postorder(Node* root) {  
    if (root == NULL) {  
        return;  
    }  
  
    postorder(root->left);  
    postorder(root->right);  
    printf("%c ", root->data);  
}
```

```
int main() {  
    int N;  
    scanf("%d", &N);  
  
    Node* root = NULL;  
  
    for (int i = 0; i < N; i++) {  
        char ch;  
        scanf(" %c", &ch);  
        root = insert(root, ch);  
    }
```

```
}  
printf("Postorder traversal: ");  
postorder(root);  
printf("\n");  
  
return 0;  
}
```

Status : Correct

Marks : 10/10

2. Problem Statement

Thiru is learning about binary search trees and wants to practice creating and traversing them. He needs to write a program that takes a series of characters as input and constructs a binary search tree with those characters.

After constructing the tree, Thiru wants to perform an in-order traversal and print all the characters in sorted order.

Input Format

The first line contains an integer n , representing the number of characters to be inserted into the binary search tree.

The second line contains n space-separated characters c_i , representing the characters to be inserted into the binary search tree.

Output Format

The output prints a single line containing all the characters of the binary search tree in sorted order, obtained by performing an in-order traversal of the tree, separated by a space.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 4

r e f s

Output: e f r s

Answer

// You are using GCC

#include <stdio.h>

#include <stdlib.h>

```
typedef struct Node {  
    char data;  
    struct Node* left;  
    struct Node* right;  
} Node;
```

```
Node* createNode(char data) {  
    Node* newNode = (Node*)malloc(sizeof(Node));  
    newNode->data = data;  
    newNode->left = NULL;  
    newNode->right = NULL;  
    return newNode;  
}
```

```
Node* insert(Node* root, char data) {  
    if (root == NULL) {  
        return createNode(data);  
    }  
    if (data < root->data) {  
        root->left = insert(root->left, data);  
    } else if (data > root->data) {  
        root->right = insert(root->right, data);  
    }  
  
    return root;  
}
```

```
void inorder(Node* root) {  
    if (root == NULL) {  
        return;  
    }  
    inorder(root->left);  
    printf("%c", root->data);  
    inorder(root->right);  
}
```



```
    printf("%c ", root->data);  
    inorder(root->right);  
}
```

```
int main() {  
    int n;  
    scanf("%d", &n);  
  
    Node* root = NULL;  
  
    for (int i = 0; i < n; i++) {  
        char ch;  
        scanf(" %c", &ch);  
        root = insert(root, ch);  
    }  
  
    inorder(root);  
    printf("\n");  
  
    return 0;  
}
```

Status : Correct

Marks : 10/10

3. Problem Statement

Thilak is studying binary search trees (BSTs) and wants to implement a program that constructs a binary search tree from a given set of characters. He aims to perform a pre-order traversal of the constructed tree to understand its structure better.

Your task is to help Thilak by designing a program that accomplishes this.

Input Format

The first line of input consists of an integer N, representing the number of characters to be inserted into the BST.

The second line consists of N space-separated characters, representing the data elements to be inserted into the BST.

Output Format

The output prints the pre-order traversal of the constructed BST.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 5

C A E B D

Output: C A B E D

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
typedef struct Node {  
    char data;  
    struct Node* left;  
    struct Node* right;  
} Node;
```

```
Node* createNode(char data) {  
    Node* newNode = (Node*)malloc(sizeof(Node));  
    newNode->data = data;  
    newNode->left = NULL;  
    newNode->right = NULL;  
    return newNode;  
}
```

```
Node* insert(Node* root, char data) {  
    if (root == NULL) {  
        return createNode(data);  
    }
```

```
    if (data < root->data) {  
        root->left = insert(root->left, data);  
    } else if (data > root->data) {  
        root->right = insert(root->right, data);  
    }
```

```

    }
    return root;
}

void preorder(Node* root) {
    if (root == NULL) {
        return;
    }

    printf("%c ", root->data);
    preorder(root->left);
    preorder(root->right);
}

int main() {
    int N;
    scanf("%d", &N);

    Node* root = NULL;

    for (int i = 0; i < N; i++) {
        char ch;
        scanf(" %c", &ch);
        root = insert(root, ch);
    }

    preorder(root);
    printf("\n");

    return 0;
}

```

Status : Correct

Marks : 10/10

4. Problem Statement

Jake is learning about binary search trees(BST) and their operations. He wants to implement a program that can delete a node from a BST based on the given key value and print the remaining nodes in an in-order traversal.

Assist Jake in the program.

Input Format

The first line of input consists of an integer n, representing the number of elements in BST.

The second line consists of n space-separated integers, representing the elements of the tree.

The third line consists of an integer x, representing the key value of the node to be deleted.

Output Format

The first line of output prints "Before deletion: " followed by the in-order traversal of the initial BST.

The second line prints "After deletion: " followed by the in-order traversal after the deletion of the key value.

If the key value is not present in the BST, print the original tree as it is.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 5
8 6 4 3 1
4

Output: Before deletion: 1 3 4 6 8
After deletion: 1 3 6 8

Answer

```
// You are using GCC
#include <stdio.h>
#include <stdlib.h>
```

```
typedef struct Node {
    int data;
```

```
    struct Node* left;  
    struct Node* right;  
} Node;
```

```
Node* createNode(int data) {  
    Node* newNode = (Node*)malloc(sizeof(Node));  
    newNode->data = data;  
    newNode->left = NULL;  
    newNode->right = NULL;  
    return newNode;  
}
```

```
Node* insert(Node* root, int data) {  
    if (root == NULL) {  
        return createNode(data);  
    }  
  
    if (data < root->data) {  
        root->left = insert(root->left, data);  
    } else if (data > root->data) {  
        root->right = insert(root->right, data);  
    }  
  
    return root;  
}
```

```
Node* findMin(Node* root) {  
    Node* current = root;  
    while (current && current->left != NULL) {  
        current = current->left;  
    }  
    return current;  
}
```

```
Node* deleteNode(Node* root, int key) {  
    if (root == NULL) {  
        return root;  
    }  
  
    if (key < root->data) {  
        root->left = deleteNode(root->left, key);  
    } else if (key > root->data) {
```

```

    root->right = deleteNode(root->right, key);
} else {
    // Node with only one child or no child
    if (root->left == NULL) {
        Node* temp = root->right;
        free(root);
        return temp;
    } else if (root->right == NULL) {
        Node* temp = root->left;
        free(root);
        return temp;
    }

    // Node with two children: Get inorder successor
    Node* temp = findMin(root->right);
    root->data = temp->data;
    root->right = deleteNode(root->right, temp->data);
}
return root;
}

```

```

void inorder(Node* root) {
    if (root == NULL) {
        return;
    }

```

```

    inorder(root->left);
    printf("%d ", root->data);
    inorder(root->right);
}

```

```

int main() {
    int n;
    scanf("%d", &n);

    Node* root = NULL;

    for (int i = 0; i < n; i++) {
        int val;
        scanf("%d", &val);
        root = insert(root, val);
    }
}

```

```
int x;
scanf("%d", &x);

printf("Before deletion: ");
inorder(root);
printf("\n");

root = deleteNode(root, x);

printf("After deletion: ");
inorder(root);
printf("\n");

return 0;
}
```

Status : Correct

Marks : 10/10

5. Problem Statement

You are required to implement basic operations on a Binary Search Tree (BST), like insertion and searching.

Insertion: Given a list of integers, construct a Binary Search Tree by repeatedly inserting each integer into the tree according to the rules of a BST.

Searching: Given an integer, search for its presence in the constructed Binary Search Tree. Print whether the integer is found or not.

Write a program to calculate this efficiently.

Input Format

The first line of input consists of an integer n , representing the number of nodes in the binary search tree.

The second line consists of the values of the nodes, separated by space as integers.

The third line consists of an integer representing, the value that is to be searched.

Output Format

The output prints, "Value <value> is found in the tree." if the given value is present, otherwise it prints: "Value <value> is not found in the tree."

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 7

8 3 10 1 6 14 23

6

Output: Value 6 is found in the tree.

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
typedef struct Node {  
    int data;  
    struct Node* left;  
    struct Node* right;  
} Node;
```

```
Node* createNode(int data) {  
    Node* newNode = (Node*)malloc(sizeof(Node));  
    newNode->data = data;  
    newNode->left = NULL;  
    newNode->right = NULL;  
    return newNode;  
}
```

```
Node* insert(Node* root, int data) {  
    if (root == NULL) {  
        return createNode(data);  
    }  
  
    if (data < root->data) {
```



```

        root->left = insert(root->left, data);
    } else if (data > root->data) {
        root->right = insert(root->right, data);
    }

    return root;
}

```

```

int search(Node* root, int key) {
    if (root == NULL) {
        return 0;
    }

    if (root->data == key) {
        return 1;
    } else if (key < root->data) {
        return search(root->left, key);
    } else {
        return search(root->right, key);
    }
}

```

```

int main() {
    int n;
    scanf("%d", &n);

    Node* root = NULL;

    for (int i = 0; i < n; i++) {
        int val;
        scanf("%d", &val);
        root = insert(root, val);
    }

```

```

    int key;
    scanf("%d", &key);

```

```

    if (search(root, key)) {
        printf("Value %d is found in the tree.\n", key);
    } else {
        printf("Value %d is not found in the tree.\n", key);
    }
}

```

```
} return 0;
```

Status : Correct

Marks : 10/10

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2028_NeoColab_Competitive Coding II_L9

VTU_Week 7_CY

Attempt : 1

Total Mark : 40

Marks Obtained : 40

Section 1 : Coding

1. Problem Statement

Given a BST, write an efficient function to delete a given key in it.

There are three cases that can be considered:

Deleting a node with no children: remove the node from the tree. Deleting a node with two children: Deleting a node with one child

Input Format

The first line of the input consists of an integer n representing, the number of nodes.

The second line of the input consists of integers space-separated representing, input node values.

The third line of the input consists of an integer q representing, the node value to

be deleted.

Output Format

The output prints the node values before and after the deletion of the key value in the format:

"Before deletion of key value:"

"After deleting key value:"

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 5

10 30 5 6 7

30

Output: Before deletion of key value:

5 6 7 10 30

After deleting key value:

5 6 7 10

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
typedef struct Node {  
    int data;  
    struct Node* left;  
    struct Node* right;  
} Node;
```

```
Node* createNode(int data) {  
    Node* newNode = (Node*)malloc(sizeof(Node));  
    newNode->data = data;  
    newNode->left = NULL;  
    newNode->right = NULL;  
    return newNode;  
}
```

```

Node* insert(Node* root, int data) {
    if (root == NULL) {
        return createNode(data);
    }

    if (data < root->data) {
        root->left = insert(root->left, data);
    } else if (data > root->data) {
        root->right = insert(root->right, data);
    }

    return root;
}

```

```

Node* findMin(Node* root) {
    Node* current = root;
    while (current && current->left != NULL) {
        current = current->left;
    }
    return current;
}

```

```

Node* deleteNode(Node* root, int key) {
    if (root == NULL) {
        return root;
    }

    if (key < root->data) {
        root->left = deleteNode(root->left, key);
    } else if (key > root->data) {
        root->right = deleteNode(root->right, key);
    } else {
        // Node with only one child or no child
        if (root->left == NULL) {
            Node* temp = root->right;
            free(root);
            return temp;
        } else if (root->right == NULL) {
            Node* temp = root->left;
            free(root);
            return temp;
        }
    }
}

```

```

    }
    // Node with two children: Get inorder successor
    Node* temp = findMin(root->right);
    root->data = temp->data;
    root->right = deleteNode(root->right, temp->data);
}
return root;
}

```

```

void inorder(Node* root) {
    if (root == NULL) {
        return;
    }
    inorder(root->left);
    printf("%d ", root->data);
    inorder(root->right);
}

```

```

int main() {
    int n;
    scanf("%d", &n);

    Node* root = NULL;

    for (int i = 0; i < n; i++) {
        int val;
        scanf("%d", &val);
        root = insert(root, val);
    }

    int q;
    scanf("%d", &q);

    printf("Before deletion of key value:\n");
    inorder(root);
    printf("\n");

    root = deleteNode(root, q);

    printf("After deleting key value:\n");
}

```

```
inorder(root);  
printf("\n");  
  
return 0;  
}
```

Status : Correct

Marks : 10/10

2. Problem Statement

Revi is studying binary search trees (BSTs) and wants to explore all the unique BSTs that can be constructed using distinct integers from 1 to n. He decides to write a program that generates all such BSTs and prints their preorder traversals.

Write a program to ensure that the preorder traversal of all possible BSTs constructed from numbers 1 to n is printed, each on a new line.

Example

Input

3

Output

1 2 3

1 3 2

2 1 3

3 1 2

3 2 1

Explanation

Input Format

The input consists of a single integer n, representing the number of nodes.

Output Format

Each line contains the preorder traversal (space-separated) of one possible BST using values from 1 to n.

Each traversal is printed on a new line.

There should be no blank lines between the outputs, and the order of output may vary depending on the internal recursive tree generation.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 3

Output: 1 2 3

1 3 2

2 1 3

3 1 2

3 2 1

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
typedef struct Node {  
    int data;  
    struct Node* left;  
    struct Node* right;  
} Node;
```

```
Node* createNode(int data) {  
    Node* newNode = (Node*)malloc(sizeof(Node));  
    newNode->data = data;  
    newNode->left = NULL;  
    newNode->right = NULL;  
    return newNode;  
}
```



```
void preorder(Node* root) {  
    if (root == NULL) {  
        return;  
    }  
    printf("%d ", root->data);  
    preorder(root->left);  
    preorder(root->right);  
}
```

```
Node* cloneTree(Node* root) {  
    if (root == NULL) {  
        return NULL;  
    }  
    Node* newNode = createNode(root->data);  
    newNode->left = cloneTree(root->left);  
    newNode->right = cloneTree(root->right);  
    return newNode;  
}
```

```
Node** generateTrees(int start, int end, int* count) {  
    if (start > end) {  
        Node** trees = (Node**)malloc(sizeof(Node*));  
        trees[0] = NULL;  
        *count = 1;  
        return trees;  
    }
```

```
    Node** allTrees = (Node**)malloc(sizeof(Node*) * 1000);  
    *count = 0;
```

```
    for (int i = start; i <= end; i++) {  
        int leftCount, rightCount;  
        Node** leftTrees = generateTrees(start, i - 1, &leftCount);  
        Node** rightTrees = generateTrees(i + 1, end, &rightCount);
```

```
        for (int l = 0; l < leftCount; l++) {  
            for (int r = 0; r < rightCount; r++) {  
                Node* root = createNode(i);  
                root->left = cloneTree(leftTrees[l]);  
                root->right = cloneTree(rightTrees[r]);  
                allTrees[( *count )++] = root;  
            }  
        }
```

```

    }
    free(leftTrees);
    free(rightTrees);
}

return allTrees;
}

int main() {
    int n;
    scanf("%d", &n);

    int count;
    Node** trees = generateTrees(1, n, &count);

    for (int i = 0; i < count; i++) {
        preorder(trees[i]);
        printf("\n");
        free(trees[i]);
    }

    free(trees);

    return 0;
}

```

Status : Correct

Marks : 10/10

3. Problem Statement

Jai is fascinated by binary trees and wants to create and manipulate them. He needs your help to implement basic operations like insertion and deletion in a binary search tree.

Given a sequence of characters representing the values to be inserted into a binary search tree, followed by a character V to be deleted from the tree, your task is to construct the binary search tree and delete the specified character from it.

Your program should perform the following operations:

Construct a binary search tree from the given sequence of characters. Delete the specified character V from the constructed tree. Print the elements of the tree in sorted order after deletion.

Input Format

The first line contains an integer N, denoting the number of characters in the sequence.

The second line contains N space-separated characters, representing the sequence of characters to be inserted into the binary search tree.

The third line contains a single character V (which is guaranteed to be present in the tree), representing the value to be deleted from the binary search tree.

Output Format

The output prints a single line containing the elements of the binary search tree in sorted order after deleting the specified character V.

If the specified value is not available in the tree, print the given input values in order traversal.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 5
F B G A D
B

Output: A D F G

Answer

```
// You are using GCC
#include <stdio.h>
#include <stdlib.h>
```

```
typedef struct Node {
    char data;
    struct Node* left;
```

```
    struct Node* right;
} Node;
```

```
Node* createNode(char data) {
    Node* newNode = (Node*)malloc(sizeof(Node));
    newNode->data = data;
    newNode->left = NULL;
    newNode->right = NULL;
    return newNode;
}
```

```
Node* insert(Node* root, char data) {
    if (root == NULL) {
        return createNode(data);
    }
    if (data < root->data) {
        root->left = insert(root->left, data);
    } else if (data > root->data) {
        root->right = insert(root->right, data);
    }

    return root;
}
```

```
Node* findMin(Node* root) {
    Node* current = root;
    while (current && current->left != NULL) {
        current = current->left;
    }
    return current;
}
```

```
Node* deleteNode(Node* root, char key) {
    if (root == NULL) {
        return root;
    }

    if (key < root->data) {
        root->left = deleteNode(root->left, key);
    } else if (key > root->data) {
        root->right = deleteNode(root->right, key);
    }
}
```

```

    } else {
        // Node with only one child or no child
        if (root->left == NULL) {
            Node* temp = root->right;
            free(root);
            return temp;
        } else if (root->right == NULL) {
            Node* temp = root->left;
            free(root);
            return temp;
        }

        // Node with two children: Get inorder successor
        Node* temp = findMin(root->right);
        root->data = temp->data;
        root->right = deleteNode(root->right, temp->data);
    }
    return root;
}

```

```

void inorder(Node* root) {
    if (root == NULL) {
        return;
    }

```

```

    inorder(root->left);
    printf("%c ", root->data);
    inorder(root->right);
}

```

```

int main() {
    int N;
    scanf("%d", &N);

    Node* root = NULL;

    for (int i = 0; i < N; i++) {
        char ch;
        scanf(" %c", &ch);
        root = insert(root, ch);
    }
}

```

```
char V;  
scanf("%c", &V);  
  
root = deleteNode(root, V);  
  
inorder(root);  
printf("\n");  
  
return 0;  
}
```

Status : Correct

Marks : 10/10

4. Problem Statement:

Arun is exploring operations on binary search trees (BST). He wants to write a program that takes an unsorted array of distinct integers and constructs a BST by inserting elements in the given order (not height-balanced).

After constructing the BST, he wants to delete all nodes at the deepest level of the tree and perform a post-order traversal to print the remaining nodes.

Example

Input:

7

1 7 4 9 2 11 12

Output:

Before deleting: 2 4 12 11 9 7 1

After deleting: 2 4 11 9 7 1

Explanation:

The BST is constructed by inserting elements in the given order: 1, 7, 4, 9, 2, 11, 12. This creates a regular BST (not height-balanced) where each element is inserted according to BST properties.

The tree is traversed in post-order before deleting the deepest level, showing: 2, 4, 12, 11, 9, 7, 1.

The deepest level nodes (in this case, node 12) are then deleted.

The tree is traversed in post-order after deleting the deepest level, showing: 2, 4, 11, 9, 7, 1.

Input Format

The first line contains an integer n , representing the number of nodes to be inserted in the BST.

The second line contains n space-separated integers, representing the elements to be inserted into the BST in the given order.

Output Format

The first line prints "Before deleting: " followed by the space-separated elements of the tree in post-order traversal before deleting the deepest level, ending with a space and newline.

The second line prints "After deleting: " followed by the space-separated elements of the tree in post-order traversal after deleting the deepest level, ending with a space and newline.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 7

1 7 4 9 2 11 12

Output: Before deleting: 2 4 12 11 9 7 1

After deleting: 2 4 11 9 7 1

Answer

```
// You are using GCC
#include <stdio.h>
```

```
#include <stdlib.h>
#include <limits.h>
```

```
typedef struct Node {
    int data;
    struct Node* left;
    struct Node* right;
} Node;
```

```
int maxHeight = 0;
```

```
Node* createNode(int data) {
    Node* newNode = (Node*)malloc(sizeof(Node));
    newNode->data = data;
    newNode->left = NULL;
    newNode->right = NULL;
    return newNode;
}
```

```
Node* insert(Node* root, int data) {
    if (root == NULL) {
        return createNode(data);
    }

    if (data < root->data) {
        root->left = insert(root->left, data);
    } else if (data > root->data) {
        root->right = insert(root->right, data);
    }

    return root;
}
```

```
int getHeight(Node* root) {
    if (root == NULL) {
        return 0;
    }
    int leftHeight = getHeight(root->left);
    int rightHeight = getHeight(root->right);
    return (leftHeight > rightHeight ? leftHeight : rightHeight) + 1;
}
```



```

Node* deleteDeepestNodes(Node* root, int currentHeight, int targetHeight) {
    if (root == NULL) {
        return NULL;
    }

    if (currentHeight == targetHeight) {
        free(root);
        return NULL;
    }

    root->left = deleteDeepestNodes(root->left, currentHeight + 1, targetHeight);
    root->right = deleteDeepestNodes(root->right, currentHeight + 1, targetHeight);

    return root;
}

```

```

void postorder(Node* root) {
    if (root == NULL) {
        return;
    }

    postorder(root->left);
    postorder(root->right);
    printf("%d ", root->data);
}

```

```

int main() {
    int n;
    scanf("%d", &n);

    Node* root = NULL;

    for (int i = 0; i < n; i++) {
        int val;
        scanf("%d", &val);
        root = insert(root, val);
    }

    int height = getHeight(root);

    printf("Before deleting: ");
    postorder(root);
}

```

```
printf("\n");  
root = deleteDeepestNodes(root, 1, height);  
  
printf("After deleting: ");  
postorder(root);  
printf("\n");  
  
return 0;  
}
```

Status : Correct

Marks : 10/10

Vel Tech Group of Institutions

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2028_NeoColab_Competitive Coding II_L9

VTU_Week 7_PAH

Attempt : 1

Total Mark : 40

Marks Obtained : 40

Section 1 : Coding

1. Problem Statement

Dhruv is working on a project where he needs to implement a Binary Search Tree (BST) data structure and perform various operations on it.

He wants to create a program that allows him to build a BST, traverse it in different orders (inorder, preorder, postorder), and exit the program when needed.

Help Dhruv by designing a program that fulfils his requirements.

Input Format

The first input consists of the choice.

If the choice is 1, enter the number of elements N and the elements inserted into

the tree, separated by a space in a new line.

If the choice is 2, print the in-order traversal.

If the choice is 3, print the pre-order traversal.

If the choice is 4, print the post-order traversal.

If the choice is 5, exit.

Output Format

The output prints the results based on the choice.

For choice 1, print "BST with N nodes is ready to use," where N is the number of nodes inserted.

For choice 2, print the inorder traversal of the BST after "BST Traversal in INORDER"

For choice 3, print the pre-order traversal of the BST after "BST Traversal in PREORDER"

For choice 4, print the post-order traversal of the BST after "BST Traversal in POSTORDER"

For choice 5, the program exits.

If the choice is greater than 5, print "Wrong choice".

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 1

5

12 78 96 34 55

2

3

4

5

Output: BST with 5 nodes is ready to use

BST Traversal in INORDER

12 34 55 78 96

BST Traversal in PREORDER

12 78 34 55 96

BST Traversal in POSTORDER

55 34 96 78 12

Answer

// You are using GCC

#include <stdio.h>

#include <stdlib.h>

```
typedef struct Node {  
    int data;  
    struct Node* left;  
    struct Node* right;  
} Node;
```

```
Node* createNode(int data) {  
    Node* newNode = (Node*)malloc(sizeof(Node));  
    newNode->data = data;  
    newNode->left = NULL;  
    newNode->right = NULL;  
    return newNode;  
}
```

```
Node* insert(Node* root, int data) {  
    if (root == NULL) {  
        return createNode(data);  
    }  
  
    if (data < root->data) {  
        root->left = insert(root->left, data);  
    } else if (data > root->data) {  
        root->right = insert(root->right, data);  
    }  
  
    return root;  
}
```

```
void inorder(Node* root) {
```

```
    if (root == NULL) {  
        return;  
    }  
    inorder(root->left);  
    printf("%d ", root->data);  
    inorder(root->right);  
}
```

```
void preorder(Node* root) {  
    if (root == NULL) {  
        return;  
    }  
    printf("%d ", root->data);  
    preorder(root->left);  
    preorder(root->right);  
}
```

```
void postorder(Node* root) {  
    if (root == NULL) {  
        return;  
    }  
    postorder(root->left);  
    postorder(root->right);  
    printf("%d ", root->data);  
}
```

```
void freeTree(Node* root) {  
    if (root == NULL) {  
        return;  
    }  
    freeTree(root->left);  
    freeTree(root->right);  
    free(root);  
}
```

```
int main() {  
    Node* root = NULL;  
    int choice;
```

```
    while (1) {  
        scanf("%d", &choice);
```

```

if (choice == 1) {
    int n;
    scanf("%d", &n);
    root = NULL;
    for (int i = 0; i < n; i++) {
        int val;
        scanf("%d", &val);
        root = insert(root, val);
    }
    printf("BST with %d nodes is ready to use\n", n);
}
else if (choice == 2) {
    printf("BST Traversal in INORDER\n");
    inorder(root);
    printf("\n");
}
else if (choice == 3) {
    printf("BST Traversal in PREORDER\n");
    preorder(root);
    printf("\n");
}
else if (choice == 4) {
    printf("BST Traversal in POSTORDER\n");
    postorder(root);
    printf("\n");
}
else if (choice == 5) {
    freeTree(root);
    break;
}
else {
    printf("Wrong choice\n");
}
}

return 0;
}

```

Status : Correct

Marks : 10/10

2. Problem Statement

Singh is studying binary search trees (BSTs) in his computer science class. He is particularly interested in finding the inorder predecessor of a given node in a BST.

The inorder predecessor of a node in a BST is the node with the largest value that is smaller than the given node's value.

Given a binary search tree and a node value, Singh needs your help to write a program that finds and outputs the in-order predecessor of the given node.

Input Format

The first line of input consists of an integer n , representing the number of keys to be inserted into the Binary Search Tree (BST).

The second line of input consists of n integers, representing the value of each key to be inserted into the BST, separated by a space.

The third line of input consists of an integer value key , representing the value of the node for which the inorder predecessor is to be found.

Output Format

The output displays "Inorder predecessor of X is Y ", where Y is the value of the inorder predecessor of the node and X is the key.

If the node with the given key is not found in the BST, the output displays "Key not found".

If the given key has no inorder predecessor in the BST, the output displays "No inorder predecessor found".

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 3
20 70 90
90

Output: Inorder predecessor of 90 is 70

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
typedef struct Node {  
    int data;  
    struct Node* left;  
    struct Node* right;  
} Node;
```

```
Node* createNode(int data) {  
    Node* newNode = (Node*)malloc(sizeof(Node));  
    newNode->data = data;  
    newNode->left = NULL;  
    newNode->right = NULL;  
    return newNode;  
}
```

```
Node* insert(Node* root, int data) {  
    if (root == NULL) {  
        return createNode(data);  
    }  
  
    if (data < root->data) {  
        root->left = insert(root->left, data);  
    } else if (data > root->data) {  
        root->right = insert(root->right, data);  
    }  
  
    return root;  
}
```

```
Node* search(Node* root, int key) {  
    if (root == NULL || root->data == key) {  
        return root;  
    }  
  
    if (key < root->data) {  
        return search(root->left, key);  
    }  
}
```

```

    }
    return search(root->right, key);
}

int findInorderPredecessor(Node* root, int key) {
    Node* current = search(root, key);
    if (current == NULL) {
        return -2; // Key not found
    }

    // Case 1: If left subtree exists, predecessor is the rightmost node in left
    subtree
    if (current->left != NULL) {
        Node* temp = current->left;
        while (temp->right != NULL) {
            temp = temp->right;
        }
        return temp->data;
    }

    // Case 2: No left subtree, go up the tree
    Node* predecessor = NULL;
    Node* ancestor = root;

    while (ancestor != current) {
        if (current->data > ancestor->data) {
            predecessor = ancestor;
            ancestor = ancestor->right;
        } else {
            ancestor = ancestor->left;
        }
    }

    if (predecessor != NULL) {
        return predecessor->data;
    }
    return -1; // No predecessor found
}

int main() {
    int n;
    scanf("%d", &n);

```

```

Node* root = NULL;

for (int i = 0; i < n; i++) {
    int val;
    scanf("%d", &val);
    root = insert(root, val);
}

int key;
scanf("%d", &key);

int result = findInorderPredecessor(root, key);

if (result == -2) {
    printf("Key not found\n");
} else if (result == -1) {
    printf("No inorder predecessor found\n");
} else {
    printf("Inorder predecessor of %d is %d\n", key, result);
}

return 0;
}

```

Status : Correct

Marks : 10/10

3. Problem Statement

Toy loves trees, especially alphabet trees! He wants to build his alphabet tree by inserting characters one by one. Given a sequence of characters, Toy needs to construct a binary search tree where each node contains a character. After constructing the tree, Toy wants to search for a specific character in the tree to see if it exists or not.

Write a program to help Toy construct his alphabet tree and perform searches.

Input Format

The first line contains an integer, N, denoting the number of characters Toy

wants to insert into the tree.

The second line contains N space-separated characters, representing the characters Toy wants to insert into the tree.

The third line contains a single character, V, representing the character Toy wants to search for in the tree.

Output Format

If the character V is found in the constructed alphabet tree, print "Character - [V] is Found".

If the character V is not found in the constructed alphabet tree, print "Character - [V] is Not Found".

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 5

F A B C D

B

Output: Character - B is Found

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
typedef struct Node {  
    char data;  
    struct Node* left;  
    struct Node* right;  
} Node;
```

```
Node* createNode(char data) {  
    Node* newNode = (Node*)malloc(sizeof(Node));  
    newNode->data = data;  
    newNode->left = NULL;  
    newNode->right = NULL;
```

```
    return newNode;
}

Node* insert(Node* root, char data) {
    if (root == NULL) {
        return createNode(data);
    }

    if (data < root->data) {
        root->left = insert(root->left, data);
    } else if (data > root->data) {
        root->right = insert(root->right, data);
    }

    return root;
}
```

```
int search(Node* root, char key) {
    if (root == NULL) {
        return 0;
    }

    if (root->data == key) {
        return 1;
    } else if (key < root->data) {
        return search(root->left, key);
    } else {
        return search(root->right, key);
    }
}
```

```
int main() {
    int N;
    scanf("%d", &N);

    Node* root = NULL;

    for (int i = 0; i < N; i++) {
        char ch;
        scanf("%c", &ch);
        root = insert(root, ch);
    }
```

```

char V;
scanf("%c", &V);

if (search(root, V)) {
    printf("Character - %c is Found\n", V);
} else {
    printf("Character - %c is Not Found\n", V);
}

return 0;
}

```

Status : Correct

Marks : 10/10

4. Problem Statement

Jay is studying binary search trees (BSTs) as part of his computer science curriculum. He is particularly interested in understanding operations like insertion and deletion in a BST.

Jay wants to write a program that takes a sequence of characters as input, constructs a BST using those characters, deletes the minimum element from the BST, and then outputs the elements of the tree in sorted order.

Help Jay by designing a program that achieves this.

Input Format

The first line of input contains an integer n , denoting the number of characters to be inserted into the BST.

The second line contains n space-separated characters, each character representing a node to be inserted into the BST.

Output Format

The output displays a single line displaying the space-separated characters of the BST in sorted order after deleting the minimum element in the format:

"Inorder traversal after deleting the minimum element:"

"[char1] [char2] ...[char_n]"

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 6

K J M V H F

Output: Inorder traversal after deleting the minimum element:

H J K M V

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
typedef struct Node {  
    char data;  
    struct Node* left;  
    struct Node* right;  
} Node;
```

```
Node* createNode(char data) {  
    Node* newNode = (Node*)malloc(sizeof(Node));  
    newNode->data = data;  
    newNode->left = NULL;  
    newNode->right = NULL;  
    return newNode;  
}
```

```
Node* insert(Node* root, char data) {  
    if (root == NULL) {  
        return createNode(data);  
    }  
  
    if (data < root->data) {  
        root->left = insert(root->left, data);  
    } else if (data > root->data) {  
        root->right = insert(root->right, data);  
    }  
}
```

```
    return root;
}
```

```
Node* deleteMin(Node* root) {
    if (root == NULL) {
        return NULL;
    }

    if (root->left == NULL) {
        Node* temp = root->right;
        free(root);
        return temp;
    }

    root->left = deleteMin(root->left);
    return root;
}
```

```
void inorder(Node* root) {
    if (root == NULL) {
        return;
    }
```

```
    inorder(root->left);
    printf("%c ", root->data);
    inorder(root->right);
}
```

```
int main() {
    int n;
    scanf("%d", &n);
```

```
    Node* root = NULL;
```

```
    for (int i = 0; i < n; i++) {
        char ch;
        scanf(" %c", &ch);
        root = insert(root, ch);
    }
```

```
    root = deleteMin(root);
```



```
printf("Inorder traversal after deleting the minimum element:\n");  
inorder(root);  
printf("\n");  
  
return 0;  
}
```

Status : Correct

Marks : 10/10

Vel Tech Group of Institutions

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2028_NeoColab_Competitive Coding II_L9

VTU_Week 8_MCQ

Attempt : 1

Total Mark : 15

Marks Obtained : 13

Section 1 : MCQ

1. Which data structure is used in Breadth First Search?

Answer

Queue

Status : Correct

Marks : 1/1

2. Which of the following problems can be solved by using both BFS & DFS?

Answer

Finding the shortest path between two vertices in an unweighted graph

Status : Correct

Marks : 1/1

3. In BFS, what happens when a node is visited for the second time?

Answer

It is skipped and the traversal continues to the next unvisited node

Status : Correct

Marks : 1/1

4. Given two vertices in a graph s and t, which of the two traversals (BFS and DFS) can be used to find if there is path from s to t?

Answer

Both BFS and DFS

Status : Correct

Marks : 1/1

5. What will be the result of depth-first traversal in the following tree?

Answer

1 2 4 5 3

Status : Correct

Marks : 1/1

6. The Breadth First Search (BFS) algorithm has been implemented using the queue data structure. Which one of the following is a possible order of visiting the nodes in the graph below?

Answer

POQNMR

Status : Correct

Marks : 1/1

7. In the _____ traversal, we process all of a vertex's descendants before we move to an adjacent vertex.

Answer

Depth First

Status : Correct

Marks : 1/1

8. Which of the following conditions is sufficient to detect a cycle in a directed graph?

Answer

There is an edge from the currently visited node to an ancestor of the currently visited node in the DFS forest.

Status : Correct

Marks : 1/1

9. Regarding the implementation of Breadth First Search using queues, what is the maximum distance between two nodes present in the queue? (considering each edge length 1)

Answer

At most 1

Status : Correct

Marks : 1/1

10. Given the following adjacency matrix of a graph(G) determine the number of components in the G.

[0 1 1 0 0 0],

[1 0 1 0 0 0],

[1 1 0 0 0 0],

[0 0 0 0 1 0],

[0 0 0 1 0 0],

[0 0 0 0 0 0].

Answer

3

Status : Correct

Marks : 1/1

11. Which of the following statements is true about DFS?

Answer

DFS can be used to detect cycles in a graph.

Status : Correct

Marks : 1/1

12. Consider the following graph.

Apply breadth-first search traversal of the above graph. Which of the following traversal is possible if the start vertex is G? (Assume lexicographic ordering)

Answer

GAHIJBDEKC

Status : Correct

Marks : 1/1

13. Which of the following is a possible result of depth-first traversal of the given graph(consider 1 to be the source element)?

Answer

1 4 5 3 2

Status : Wrong

Marks : 0/1

14. In an adjacency list representation of a graph, what does each node in the list represent?

Answer

Edges

Status : Wrong

Marks : 0/1

15. The number of elements in the adjacency matrix of a graph having 7

vertices is _____.

Answer

49

Status : Correct

Marks : 1/1

Vel Tech Group of Institutions

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2028_NeoColab_Competitive Coding II_L9

VTU_Week 8_COD

Attempt : 1

Total Mark : 50

Marks Obtained : 48.5

Section 1 : Coding

1. Problem Statement

Sanjay is writing a program to manage an undirected graph using an adjacency list representation. Users will initially input the vertices to create the edges. The program will display the adjacency list of the graph.

Assist Sanjay in creating the program.

Input Format

The input consists of 5 nodes.

Edge information: ab ae bc bd be cd de.

Output Format

The output prints the adjacency list representation of the graph.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 0 1 2 3 4

Output: vertex 0: head -> 4 -> 1
vertex 1: head -> 4 -> 3 -> 2 -> 0
vertex 2: head -> 3 -> 1
vertex 3: head -> 4 -> 2 -> 1
vertex 4: head -> 3 -> 1 -> 0

Answer

```
// You are using GCC
#include <stdio.h>
#include <stdlib.h>

#define MAX 11

struct Node {
    int data;
    struct Node* next;
};

struct Node* adj[MAX];

// Insert at head
void addEdge(int u, int v) {
    struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));
    newNode->data = v;
    newNode->next = adj[u];
    adj[u] = newNode;
}

int main() {
    int a, b, c, d, e;

    // Initialize adjacency list
    for(int i = 0; i < MAX; i++)
        adj[i] = NULL;
```



```

// Input
scanf("%d %d %d %d %d", &a, &b, &c, &d, &e);

// Given edges
addEdge(a, b); addEdge(b, a);
addEdge(a, e); addEdge(e, a);
addEdge(b, c); addEdge(c, b);
addEdge(b, d); addEdge(d, b);
addEdge(b, e); addEdge(e, b);
addEdge(c, d); addEdge(d, c);
addEdge(d, e); addEdge(e, d);

// Print adjacency list for vertices 0 to 4
for(int i = 0; i < 5; i++) {
    printf("vertex %d: head", i);
    struct Node* temp = adj[i];
    while(temp != NULL) {
        printf(" -> %d", temp->data);
        temp = temp->next;
    }
    printf("\n");
}

return 0;
}

```

Status : Correct

Marks : 10/10

2. Problem Statement

Sophia is designing a transportation network for a city with multiple interconnected districts. Each district is represented as a vertex, and every road connecting two districts is represented as a directed edge in a graph.

To evaluate connectivity, Sophia needs a program that performs a Breadth-First Search (BFS) traversal of the city's network starting from a specified source district, and outputs the order in which districts are visited.

Write a program to ensure that Sophia can perform the BFS traversal correctly and view the order of visited districts.

Input Format

The first line of input consists of two space-separated integers, v , and e , representing the total number of cities and the number of connections between cities.

The next e lines each contain two space-separated integers a and b , representing a directed edge from city a to city b .

The last line consists of a single integer s , representing the source city from which BFS traversal should start.

Output Format

The output prints the order in which the cities are visited during the BFS traversal, separated by a space.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 6 7

0 1

0 2

1 3

1 4

2 4

3 5

4 5

0

Output: 0 1 2 3 4 5

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
#include <stdbool.h>
```

```
#define MAX 100
```

```
int graph[MAX][MAX];
```

```
int visited[MAX];
```

```
int queue[MAX];  
int front = 0, rear = 0;
```

```
void enqueue(int vertex) {  
    queue[rear++] = vertex;  
}
```

```
int dequeue() {  
    return queue[front++];  
}
```

```
bool isEmpty() {  
    return front == rear;  
}
```

```
void bfs(int start, int v) {  
    enqueue(start);  
    visited[start] = 1;
```

```
    while (!isEmpty()) {  
        int current = dequeue();  
        printf("%d ", current);
```

```
        for (int i = 0; i < v; i++) {  
            if (graph[current][i] == 1 && !visited[i]) {  
                visited[i] = 1;  
                enqueue(i);  
            }  
        }  
    }  
}
```

```
int main() {  
    int v, e;  
    scanf("%d %d", &v, &e);
```

```
    // Initialize graph  
    for (int i = 0; i < v; i++) {  
        visited[i] = 0;  
        for (int j = 0; j < v; j++) {  
            graph[i][j] = 0;  
        }  
    }
```

```

    }

    // Read edges
    for (int i = 0; i < e; i++) {
        int a, b;
        scanf("%d %d", &a, &b);
        graph[a][b] = 1;
    }

    // Read source vertex
    int s;
    scanf("%d", &s);

    // Perform BFS
    bfs(s, v);

    return 0;
}

```

Status : Correct

Marks : 10/10

3. Problem Statement

Liam is developing a social media analytics tool for tracking user interactions. He needs to identify the connected components of users in the network by traversing a directed graph. Each user is represented as a vertex, and the connections between users are represented as directed edges. Liam's task is to implement a Depth-First Search (DFS) algorithm to perform a traversal of the network, starting from a given user's profile.

Can you help Liam by implementing the DFS algorithm to identify all connected users starting from a specified user profile?

Example

Input:

6 6

0 1

0 2

1 2
2 0
2 3
3 3
2

Output:

2 0 1 3

Explanation:

Starting from vertex 2, it visits and marks vertex 2 as visited.

Vertex 2 is printed.

Checking the neighbors of vertex 2: 0 and 3.

Vertex 0 is visited, so DFS is called with vertex 0.

Vertex 0 is visited and printed.

Vertex 1 is visited and printed.

Vertex 3 is visited and printed.

The final output is "2 0 1 3".

Input Format

The first line contains two space-separated integers n and m , representing the total number of users (vertices) and the number of connections (edges) in the network, respectively.

The next m lines contain two space-separated integers u and v , representing a directed connection from user u to user v .

The last line contains a single integer s , representing the ID of the source user profile from which the DFS traversal should start.

Output Format

The output displays a single line containing the user IDs in the order they were visited during the DFS traversal, separated by a space.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 6 6

0 1

0 2

1 2

2 0

2 3

3 3

2

Output: 2 0 1 3

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
#include <stdbool.h>
```

```
#define MAX 100
```

```
int graph[MAX][MAX];
```

```
int visited[MAX];
```

```
int n, m;
```

```
void dfs(int vertex) {  
    visited[vertex] = 1;  
    printf("%d ", vertex);
```

```
    for (int i = 0; i < n; i++) {  
        if (graph[vertex][i] == 1 && !visited[i]) {  
            dfs(i);  
        }  
    }  
}
```

```
int main() {  
    scanf("%d %d", &n, &m);
```

```

// Initialize graph
for (int i = 0; i < n; i++) {
    visited[i] = 0;
    for (int j = 0; j < n; j++) {
        graph[i][j] = 0;
    }
}

// Read edges
for (int i = 0; i < m; i++) {
    int u, v;
    scanf("%d %d", &u, &v);
    graph[u][v] = 1;
}

// Read source vertex
int s;
scanf("%d", &s);

// Perform DFS
dfs(s);

return 0;
}

```

Status : Correct

Marks : 10/10

4. Problem Statement

Given a graph with V vertices numbered from 0 to $V-1$ and a list of edges, determine whether the graph is bipartite or not.

Note: A bipartite graph is a type of graph where the set of vertices can be divided into two disjoint sets, say U and V , such that every edge connects a vertex in U to a vertex in V , there are no edges between vertices within the same set.

Example:

Input: $V = 4$, edges $[][] = [[0, 1], [0, 2], [1, 2], [2, 3]]$

Output: No, the given graph is not Bipartite

Explanation

The graph is not bipartite because no matter how we try to color the nodes using two colors, there exists a cycle of odd length (like 1–2–0–1), which leads to a situation where two adjacent nodes end up with the same color. This violates the bipartite condition, which requires that no two connected nodes share the same color.

Input: $V = 4$, edges $[[0, 1], [1, 2], [2, 3]]$

Output: Yes, the given graph is Bipartite

Explanation:

The given graph can be colored in two colors so, it is a bipartite graph.

Input Format

The first line contains two space-separated integers V and E , representing the number of vertices and edges respectively.

The next E lines each contain two space-separated integers u and v , indicating an undirected edge between vertices u and v .

Output Format

The output print "Yes, the given graph is Bipartite" if the graph is bipartite.

Otherwise, print "No, the given graph is not Bipartite".

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 4 3

0 1

1 2

2 3

Output: Yes, the given graph is Bipartite

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
#include <stdbool.h>
```

```
#define MAX 10
```

```
int graph[MAX][MAX];
```

```
int color[MAX];
```

```
int V, E;
```

```
bool isBipartite() {
```

```
    // Initialize colors as -1 (uncolored)
```

```
    for (int i = 0; i < V; i++) {  
        color[i] = -1;
```

```
    }
```

```
    // Handle disconnected graph - check each component
```

```
    for (int start = 0; start < V; start++) {
```

```
        if (color[start] == -1) {
```

```
            // Start BFS from this vertex
```

```
            int queue[MAX];
```

```
            int front = 0, rear = 0;
```

```
            color[start] = 0;
```

```
            queue[rear++] = start;
```

```
            while (front < rear) {
```

```
                int u = queue[front++];
```

```
                for (int v = 0; v < V; v++) {
```

```
                    if (graph[u][v] == 1) {
```

```
                        if (color[v] == -1) {
```

```
                            // Assign opposite color
```

```
                            color[v] = 1 - color[u];
```

```
                            queue[rear++] = v;
```

```
                        }
```

```
                    else if (color[v] == color[u]) {
```

```
                        return false;
```

```
                    }
```

```

    }
    }
    }
    }
    return true;
}

int main() {
    scanf("%d %d", &V, &E);

    // Initialize graph
    for (int i = 0; i < V; i++) {
        for (int j = 0; j < V; j++) {
            graph[i][j] = 0;
        }
    }

    // Read edges
    for (int i = 0; i < E; i++) {
        int u, v;
        scanf("%d %d", &u, &v);
        graph[u][v] = 1;
        graph[v][u] = 1; // Undirected graph
    }

    if (isBipartite()) {
        printf("Yes, the given graph is Bipartite\n");
    } else {
        printf("No, the given graph is not Bipartite\n");
    }

    return 0;
}

```

Status : Correct

Marks : 10/10

5. Problem Statement

David is working on a project that involves traversing graphs for analyzing network connectivity. He is tasked with performing a Depth-First Search

(DFS) traversal on a connected, undirected graph to explore all its vertices starting from a given node.

He needs help writing a program that can process multiple test cases where each test case describes a graph with its vertices and edges, and the program should output the DFS traversal order for each graph.

Can you assist David by writing a program that performs the DFS traversal for each test case?

Example

Input:

2

5 4

0 1 0 2 0 3 2 4

4 3

0 1 1 2 0 3

Output:

0 1 2 4 3

0 1 2 3

Explanation:

Visit all the vertices using a depth-first search algorithm using 0 as a source node.

Input Format

The first line of input contains an integer T denoting the number of test cases.

Each test case consists of two lines.

- The first line of each test case contains two integers N and E which denote the number of vertices and number of edges respectively.
- The second line of each test case contains E space-separated pairs u and v denoting that there is an edge from u to v.

Output Format

For each test case, the output displays the graph vertices in the depth-first traversal order.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 2

5 4

0 1 0 2 0 3 2 4

4 3

0 1 1 2 0 3

Output: 0 1 2 4 3

0 1 2 3

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <stdbool.h>
```

```
#define MAX 10000
```

```
// Adjacency list representation
```

```
typedef struct Node {
```

```
    int vertex;
```

```
    struct Node* next;
```

```
} Node;
```

```
Node* adj[MAX];
```

```
int visited[MAX];
```

```
int stack[MAX];
```

```
int top = -1;
```

```
void push(int vertex) {
```

```
    stack[++top] = vertex;
```

```
}
```

```
int pop() {  
    return stack[top--];  
}
```

```
bool isEmpty() {  
    return top == -1;  
}
```

```
void addEdge(int u, int v) {  
    Node* newNode = (Node*)malloc(sizeof(Node));  
    newNode->vertex = v;  
    newNode->next = adj[u];  
    adj[u] = newNode;  
  
    newNode = (Node*)malloc(sizeof(Node));  
    newNode->vertex = u;  
    newNode->next = adj[v];  
    adj[v] = newNode;  
}
```

```
void dfs(int start, int N) {  
    // Initialize visited array  
    for (int i = 0; i < N; i++) {  
        visited[i] = 0;  
    }  
}
```

```
push(start);  
visited[start] = 1;
```

```
while (!isEmpty()) {  
    int current = pop();  
    printf("%d ", current);
```

```
    // Sort neighbors in ascending order for consistent output  
    int neighbors[MAX];  
    int count = 0;
```

```
    Node* temp = adj[current];  
    while (temp != NULL) {  
        neighbors[count++] = temp->vertex;  
        temp = temp->next;  
    }
```

```

// Sort neighbors in descending order (since stack gives reverse order)
for (int i = 0; i < count - 1; i++) {
    for (int j = i + 1; j < count; j++) {
        if (neighbors[i] < neighbors[j]) {
            int t = neighbors[i];
            neighbors[i] = neighbors[j];
            neighbors[j] = t;
        }
    }
}

for (int i = 0; i < count; i++) {
    int neighbor = neighbors[i];
    if (!visited[neighbor]) {
        visited[neighbor] = 1;
        push(neighbor);
    }
}
}
}

```

```

void freeGraph(int N) {
    for (int i = 0; i < N; i++) {
        Node* temp = adj[i];
        while (temp != NULL) {
            Node* toFree = temp;
            temp = temp->next;
            free(toFree);
        }
        adj[i] = NULL;
    }
}

```

```

int main() {
    int T;
    scanf("%d", &T);

    for (int t = 0; t < T; t++) {
        int N, E;
        scanf("%d %d", &N, &E);
    }
}

```

```
// Initialize adjacency list
for (int i = 0; i < N; i++) {
    adj[i] = NULL;
}

// Read edges
for (int i = 0; i < E; i++) {
    int u, v;
    scanf("%d %d", &u, &v);
    addEdge(u, v);
}

// Perform DFS from vertex 0
dfs(0, N);
printf("\n");

// Free memory
freeGraph(N);
}

return 0;
}
```

Status : Partially correct

Marks : 8.5/10

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2028_NeoColab_Competitive Coding II_L9

VTU_Week 8_CY

Attempt : 1

Total Mark : 40

Marks Obtained : 40

Section 1 : Coding

1. Problem Statement

Given an $M \times N$ matrix of characters, find the length of the longest path in the matrix starting from a given character.

In alphabetical order, all characters in the longest path should be increasing and consecutive. We are allowed to search the string in all eight possible directions, i.e., North, West, South, East, North-East, North-West, South-East, and South-West.

Use depth-first search (DFS) to solve this problem.

For example, consider the following matrix of characters:

The length of the longest path with consecutive characters starting from

character C is 6.

The longest path is:

Input Format

The first line of input is an integer n representing the row size.

The second line of input is an integer m representing the column size.

The following lines of input are the characters of the matrix.

The last line of the input is the character for which the length of the longest path is to be found.

Output Format

The output prints "The length of the longest path with consecutive characters starting from character <startChar> is <length>"

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 5

5

d e h x b

a o g p e

d d c f d

e b e a s

c d y e n

c

Output: The length of the longest path with consecutive characters starting from character c is 6

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
#include <stdbool.h>
```

```
#define MAX 10
```

```
char matrix[MAX][MAX];
```

```
int n, m;
```

```
int maxLength = 0;
```

```
// All 8 possible directions
```

```
int dx[] = {-1, -1, -1, 0, 0, 1, 1, 1};
```

```
int dy[] = {-1, 0, 1, -1, 1, -1, 0, 1};
```

```
void dfs(int row, int col, char expectedChar, int length) {
```

```
    // Update maximum length found
```

```
    if (length > maxLength) {
```

```
        maxLength = length;
```

```
    }
```

```
    // Explore all 8 directions
```

```
    for (int i = 0; i < 8; i++) {
```

```
        int newRow = row + dx[i];
```

```
        int newCol = col + dy[i];
```

```
        // Check if the new position is within bounds
```

```
        if (newRow >= 0 && newRow < n && newCol >= 0 && newCol < m) {
```

```
            // Check if the next character is exactly the next expected character
```

```
            if (matrix[newRow][newCol] == expectedChar) {
```

```
                dfs(newRow, newCol, expectedChar + 1, length + 1);
```

```
            }
```

```
        }
```

```
    }
```

```
}
```

```
int main() {
```

```
    scanf("%d %d", &n, &m);
```

```
    // Read the matrix
```

```
    for (int i = 0; i < n; i++) {
```

```
        for (int j = 0; j < m; j++) {
```

```
            scanf(" %c", &matrix[i][j]);
```

```
        }
```

```
    }
```

```
    char startChar;
```

```

scanf("%c", &startChar);
maxLength = 0;

// Start DFS from every occurrence of startChar
for (int i = 0; i < n; i++) {
    for (int j = 0; j < m; j++) {
        if (matrix[i][j] == startChar) {
            dfs(i, j, startChar + 1, 1);
        }
    }
}

printf("The length of the longest path with consecutive characters starting
from character %c is %d\n", startChar, maxLength);

return 0;
}

```

Status : Correct

Marks : 10/10

2. Problem Statement

Priya is developing a learning module to simulate breadth-first traversal of directed graphs. Given multiple test cases where each describes a directed graph with vertices and edges, write a program to perform BFS starting from vertex 0 and print the traversal order for each test case.

Example

Input:

1

5 4

0 1 0 2 0 3 2 4

Output:

0 1 2 3 4

Explanation:

For the given graph, the BFS is 0 1 2 3 4.

Input Format

The first line of input contains an integer T, denoting the number of test cases.

Then T test cases follow. Each test case consists of two lines.

The descriptions of the test cases are as follows:

1. The first line of each test case contains two space-separated integers, N and E, which denote the number of vertices and edges, respectively.
2. The second line of each test case contains E space-separated pairs u and v, denoting that there is an edge from u to v.

Output Format

For each test case, the output prints the BFS order of the graph vertices.

Each vertex should appear once in the order it is visited, separated by a space.

Refer to the sample input and output for formatting specifications.

Sample Test Case

Input: 2

5 4

0 1 0 2 0 3 2 4

3 2

0 1 0 2

Output: 0 1 2 3 4

0 1 2

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
#include <stdbool.h>
```

```
#define MAX 10
```

```
int graph[MAX][MAX];  
int visited[MAX];  
int queue[MAX];  
int front, rear;
```

```
void enqueue(int vertex) {  
    queue[rear++] = vertex;  
}
```

```
int dequeue() {  
    return queue[front++];  
}
```

```
bool isEmpty() {  
    return front == rear;  
}
```

```
void bfs(int start, int N) {  
    // Initialize  
    front = 0;  
    rear = 0;  
    for (int i = 0; i < N; i++) {  
        visited[i] = 0;  
    }
```

```
    enqueue(start);  
    visited[start] = 1;
```

```
    while (!isEmpty()) {  
        int current = dequeue();  
        printf("%d ", current);
```

```
        // Visit neighbors in increasing order  
        for (int i = 0; i < N; i++) {  
            if (graph[current][i] == 1 && !visited[i]) {  
                visited[i] = 1;  
                enqueue(i);  
            }  
        }  
    }
```

```

    }
}

int main() {
    int T;
    scanf("%d", &T);

    for (int t = 0; t < T; t++) {
        int N, E;
        scanf("%d %d", &N, &E);

        // Initialize graph
        for (int i = 0; i < N; i++) {
            for (int j = 0; j < N; j++) {
                graph[i][j] = 0;
            }
        }

        // Read edges
        for (int i = 0; i < E; i++) {
            int u, v;
            scanf("%d %d", &u, &v);
            graph[u][v] = 1;
        }

        // Perform BFS starting from vertex 0
        bfs(0, N);
        printf("\n");
    }

    return 0;
}

```

Status : Correct

Marks : 10/10

3. Problem Statement

Aayush is building a network verifier that must detect the presence of cycles in an undirected graph. Given a graph with a set number of vertices and edges, write a program to determine whether the graph contains at least one cycle using depth-first traversal.

Example

Input

$N = 4, E = 4$

Output

Yes

Explanation

The diagram clearly shows a cycle 0 to 2 to 1 to 0

Input

$N = 4, E = 3$

Output

No

Explanation

There is no cycle in the given graph.

Input Format

The first line of input consists of two integers, V and E , representing the number of vertices and edges separated by space, respectively.

The next E lines contain two space-separated integers u and v , describing an edge between vertices u and v .

Output Format

The output prints "Graph contains a cycle" if a cycle exists, otherwise prints "Graph does not contain a cycle".

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 5 5

1 0

0 2

2 1

0 3

3 4

Output: Graph contains a cycle

Answer

// You are using GCC

#include <stdio.h>

#include <stdbool.h>

#define MAX 10

int graph[MAX][MAX];

int visited[MAX];

int V, E;

bool cycleFound = false;

void dfs(int vertex, int parent) {
 visited[vertex] = 1;

for (int i = 0; i < V; i++) {

if (graph[vertex][i] == 1) {

if (!visited[i]) {

dfs(i, vertex);

}

else if (i != parent) {

cycleFound = true;

}

}

}

}

int main() {

scanf("%d %d", &V, &E);

// Initialize graph

for (int i = 0; i < V; i++) {

visited[i] = 0;

for (int j = 0; j < V; j++) {


```

graph[i][j] = 0;
}
}

// Read edges
for (int i = 0; i < E; i++) {
    int u, v;
    scanf("%d %d", &u, &v);
    graph[u][v] = 1;
    graph[v][u] = 1; // Undirected graph
}

// Check for cycle in each component
for (int i = 0; i < V; i++) {
    if (!visited[i]) {
        dfs(i, -1);
        if (cycleFound) break;
    }
}

if (cycleFound) {
    printf("Graph contains a cycle\n");
} else {
    printf("Graph does not contain a cycle\n");
}

return 0;
}

```

Status : Correct

Marks : 10/10

4. Problem Statement

Emma is developing a graph traversal feature for a social networking platform. Each user is represented by a lowercase English letter (a to z), and connections between users are represented as directed edges.

Emma wants to perform a Breadth-First Search (BFS) starting from a given user, visiting friends in lexicographical order whenever multiple connections are available.

Write a program to help Emma implement this lexicographically sorted BFS traversal.

Input:

N = 10, M = 10, S = 'a',

edges[[2] = { { 'a', 'y' }, { 'a', 'z' }, { 'a', 'p' }, { 'p', 'c' }, { 'p', 'b' }, { 'y', 'm' }, { 'y', 'l' },
{ 'z', 'h' }, { 'z', 'g' }, { 'z', 'i' } }

Output:

a p y z b c l m g h i

Input Format

The first line contains an integer N, representing the number of nodes (users) in the graph.

The second line contains an integer M, representing the number of edges (friendships) in the graph.

The next M lines each contain two space-separated lowercase characters u and v, indicating a directed edge from user u to user v.

The last line contains an integer S, the starting node (user) for the BFS traversal.

Output Format

The output prints the users in the order they are visited during the BFS traversal, separated by a space.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 10

10

a y

a z

a p
p c
p b
y m
y l
z h
z g
z i
a

Output: a p y z b c l m g h i

Answer

// You are using GCC

```
#include <stdio.h>
```

```
#include <stdbool.h>
```

```
#include <stdlib.h>
```

```
#define MAX 26
```

```
int graph[MAX][MAX];
```

```
int visited[MAX];
```

```
int queue[MAX];
```

```
int front, rear;
```

```
void enqueue(int vertex) {
```

```
    queue[rear++] = vertex;
```

```
}
```

```
int dequeue() {
```

```
    return queue[front++];
```

```
}
```

```
bool isEmpty() {
```

```
    return front == rear;
```

```
}
```

```
void bfs(int start) {
```

```
    front = 0;
```

```
    rear = 0;
```

```
    for (int i = 0; i < MAX; i++) {
```

```
        visited[i] = 0;
```

```

    }
    enqueue(start);
    visited[start] = 1;

    while (!isEmpty()) {
        int current = dequeue();
        printf("%c ", current + 'a');

        // Collect neighbors and sort them lexicographically
        int neighbors[MAX];
        int count = 0;

        for (int i = 0; i < MAX; i++) {
            if (graph[current][i] == 1 && !visited[i]) {
                neighbors[count++] = i;
            }
        }

        // Sort neighbors in ascending order
        for (int i = 0; i < count - 1; i++) {
            for (int j = i + 1; j < count; j++) {
                if (neighbors[i] > neighbors[j]) {
                    int temp = neighbors[i];
                    neighbors[i] = neighbors[j];
                    neighbors[j] = temp;
                }
            }
        }

        // Enqueue sorted neighbors
        for (int i = 0; i < count; i++) {
            if (!visited[neighbors[i]]) {
                visited[neighbors[i]] = 1;
                enqueue(neighbors[i]);
            }
        }
    }
}

int main() {
    int N, M;

```

```
scanf("%d", &N);
scanf("%d", &M);

// Initialize graph
for (int i = 0; i < MAX; i++) {
    for (int j = 0; j < MAX; j++) {
        graph[i][j] = 0;
    }
}

// Read edges
for (int i = 0; i < M; i++) {
    char u, v;
    scanf(" %c %c", &u, &v);
    graph[u - 'a'][v - 'a'] = 1;
}

// Read starting node
char startChar;
scanf(" %c", &startChar);
int start = startChar - 'a';

// Perform BFS
bfs(start);

return 0;
}
```

Status : Correct

Marks : 10/10

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Batch: 2028

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2028_NeoColab_Competitive Coding II_L9

VTU_Week 8_PAH

Attempt : 1

Total Mark : 40

Marks Obtained : 37.5

Section 1 : Coding

1. Problem Statement

Maya is developing a route planning system for a large theme park. Each attraction in the park is represented as a vertex in a directed graph, and each pathway connecting attractions is a directed edge. To help visitors explore all possible paths between two attractions, Maya needs a function that finds all possible routes from a given starting attraction to a destination attraction using Depth-First Search (DFS) traversal.

Write a program to help Maya list all such possible routes from source to destination.

Input Format

The first line contains two space-separated integers V and E, representing the number of attractions and roads in the city.

Each of the next E lines contains two integers, u and v, separated by a space, representing a road from landmark u to attractions v.

The last line contains two integers, s, and d, separated by a space, representing the source attraction and the destination attraction. for which routes need to be found.

Output Format

The output prints all the possible routes from the source attraction to the destination attraction, each route on a separate line.

Each route should be represented as a space-separated sequence of attractions visited in the order of traversal.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 4 6

0 1

0 2

0 3

2 0

2 1

1 3

2 3

Output: 2 0 1 3

2 0 3

2 1 3

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
#include <stdbool.h>
```

```
#define MAX 100
```

```
int graph[MAX][MAX];
```

```
int path[MAX];
```

```
int visited[MAX];  
int V, E;  
int source, dest;
```

```
void printPath(int length) {  
    for (int i = 0; i <= length; i++) {  
        printf("%d", path[i]);  
        if (i < length) {  
            printf(" ");  
        }  
    }  
    printf("\n");  
}
```

```
void dfs(int current, int length) {  
    path[length] = current;  
  
    if (current == dest) {  
        printPath(length);  
        return;  
    }
```

```
    visited[current] = 1;
```

```
    for (int i = 0; i < V; i++) {  
        if (graph[current][i] == 1 && !visited[i]) {  
            dfs(i, length + 1);  
        }  
    }
```

```
    visited[current] = 0; // Backtrack  
}
```

```
int main() {  
    scanf("%d %d", &V, &E);
```

```
    // Initialize graph  
    for (int i = 0; i < V; i++) {  
        visited[i] = 0;  
        for (int j = 0; j < V; j++) {  
            graph[i][j] = 0;  
        }  
    }
```



```

    }

    // Read edges
    for (int i = 0; i < E; i++) {
        int u, v;
        scanf("%d %d", &u, &v);
        graph[u][v] = 1;
    }

    // Read source and destination
    scanf("%d %d", &source, &dest);

    // Find all paths from source to destination
    dfs(source, 0);

    return 0;
}

```

Status : Correct

Marks : 10/10

2. Problem Statement

Daniel is working on a navigation system for a city, where each intersection is represented as a vertex, and the roads between them are edges. He needs to calculate the shortest path between two intersections using the Breadth-First Search (BFS) algorithm.

Given the adjacency matrix of the city's road network, Daniel needs to find the shortest path from a source intersection to a destination intersection. If there are multiple shortest paths, the path with the smaller indexed intersections should be preferred.

Can you assist Daniel in implementing the solution for this problem?

Example

Input:

5

0 1 1 0 0

1 0 1 0 0

1 1 0 1 0

0 0 1 0 1

0 0 0 1 0

0 4

Output:

3

0 2 3 4

Explanation:

The graph has 5 vertices (0 to 4).

The adjacency matrix represents an undirected graph where 1 indicates an edge between two vertices.

The starting vertex is 0, and the ending vertex is 4.

The BFS starts from vertex 0, exploring its neighbors 1 and 2.

BFS continues to vertex 2, discovering vertex 3.

From vertex 3, it reaches the destination vertex 4.

The shortest path found is 0 2 3 4, with a length of 3.

The program prints the path length (3) and the path itself (0 2 3 4).

Input Format

The first line contains an integer N, the number of intersections (vertices) in the city.

The next N lines each contain N integers, representing the adjacency matrix. If there is a road between intersection i and intersection j, the matrix entry will be 1, otherwise 0.

The last line contains two integers, source and goal, representing the source and destination intersections (indexed from 0).

Output Format

The first line should print the length of the shortest path from the source to the goal intersection.

The second line should print the path from the source to the goal, including both the source and the goal.

If the source or goal index is invalid, print "Not found".

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 5

```
0 1 1 0 0
1 0 1 0 0
1 1 0 1 0
0 0 1 0 1
0 0 0 1 0
0 4
```

Output: 3

```
0 2 3 4
```

Answer

```
// You are using GCC
#include <stdio.h>
#include <stdbool.h>
#include <limits.h>
```

```
#define MAX 10
```

```
int graph[MAX][MAX];
int parent[MAX];
int visited[MAX];
int dist[MAX];
int N;
```

```
void bfs(int start, int goal) {
    int queue[MAX];
    int front = 0, rear = 0;
```

```
// Initialize
```

```
for (int i = 0; i < N; i++) {  
    visited[i] = 0;  
    parent[i] = -1;  
    dist[i] = -1;  
}
```

```
queue[rear++] = start;  
visited[start] = 1;  
dist[start] = 0;
```

```
while (front < rear) {  
    int current = queue[front++];  
    if (current == goal) {  
        return;  
    }  
}
```

```
// Explore neighbors in order (smaller index first)
```

```
for (int i = 0; i < N; i++) {  
    if (graph[current][i] == 1 && !visited[i]) {  
        visited[i] = 1;  
        parent[i] = current;  
        dist[i] = dist[current] + 1;  
        queue[rear++] = i;  
    }  
}
```

```
void printPath(int start, int goal) {  
    if (parent[goal] == -1 && start != goal) {  
        printf("Not found\n");  
        return;  
    }  
}
```

```
int path[MAX];  
int count = 0;  
int current = goal;
```

```
while (current != -1) {  
    path[count++] = current;
```

```

        current = parent[current];
    }

    // Print length
    printf("%d\n", dist[goal]);

    // Print path in reverse order
    for (int i = count - 1; i >= 0; i--) {
        printf("%d", path[i]);
        if (i > 0) {
            printf(" ");
        }
    }
    printf("\n");
}

int main() {
    scanf("%d", &N);

    // Read adjacency matrix
    for (int i = 0; i < N; i++) {
        for (int j = 0; j < N; j++) {
            scanf("%d", &graph[i][j]);
        }
    }

    int source, goal;
    scanf("%d %d", &source, &goal);

    // Validate indices
    if (source < 0 || source >= N || goal < 0 || goal >= N) {
        printf("Not found\n");
        return 0;
    }

    bfs(source, goal);

    if (source == goal) {
        printf("0\n%d\n", source);
    } else if (parent[goal] == -1) {
        printf("Not found\n");
    } else {

```

```
    printPath(source, goal);  
}  
  
    return 0;  
}
```

Status : Correct

Marks : 10/10

3. Problem Statement

Sarah is working on a traffic management system for a city's road network. She needs to find the shortest cycle in the road network, where each intersection is represented as a vertex and each road between intersections is a directed edge with a specific travel time (weight). The goal is to identify the shortest cycle in the road network, where a cycle is defined as a path that starts and ends at the same intersection, and the length of the cycle is the total travel time of all roads in the cycle.

Can you assist Sarah by writing a program to find the length of the shortest cycle in the city's road network?

Example1

Input:

4

4

0 1 2

1 2 3

2 0 4

3 1 5

Output:

Length of the shortest cycle: 9

Explanation:

After exploring all starting vertices, the code checks for cycles by examining the distances. If an edge leads back to the starting vertex, it means there is a cycle. In this case, edge 1 (vertex 1) leads back to the starting vertex 0. The code calculates the length of the cycle by adding the distances of the vertices along the cycle: $2 (0 \rightarrow 1) + 3 (1 \rightarrow 2) + 4 (2 \rightarrow 0) = 9$.

Therefore, the output is "Length of the shortest cycle: 9".

Example2

Input:

3

2

0 1 2

1 2 3

Output:

No cycle was found in the graph.

Explanation:

After exploring all starting vertices, the code checks for cycles by examining the distances. If an edge leads back to the starting vertex, it means there is a cycle. However, in this case, there are no edges that lead back to the starting vertex. Hence, there is no cycle in the graph.

Therefore, the output is "No cycle found in the graph."

Input Format

The first line contains an integer v , representing the number of intersections (vertices) in the city's road network.

The second line contains an integer e , representing the number of roads (edges) in the city.

The next e lines each contain three integers: source, destination, and weight, representing a directed road from intersection source to intersection destination with a travel time (weight).

Output Format

If a cycle exists, the output prints "Length of the shortest cycle: X", where X is the length of the shortest cycle.

If no cycle exists, then the output prints "No cycle found in the graph."

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 4

4

0 1 2

1 2 3

2 0 4

3 1 5

Output: Length of the shortest cycle: 9

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
#include <limits.h>
```

```
#define MAX 10
```

```
#define INF 99999
```

```
int graph[MAX][MAX];
```

```
int V, E;
```

```
void initializeGraph() {
```

```
    for (int i = 0; i < V; i++) {
```

```
        for (int j = 0; j < V; j++) {
```

```
            graph[i][j] = INF;
```

```
        }
```

```
    }
```

```
}
```

```
int findShortestCycle() {
```

```
    int shortestCycle = INF;
```



```

// Try each vertex as potential starting point
for (int start = 0; start < V; start++) {
    int dist[MAX];
    int visited[MAX];

    // Initialize distances
    for (int i = 0; i < V; i++) {
        dist[i] = INF;
        visited[i] = 0;
    }

    dist[start] = 0;

    // Simple DFS-based cycle detection with distance tracking
    for (int count = 0; count < V - 1; count++) {
        for (int u = 0; u < V; u++) {
            for (int v = 0; v < V; v++) {
                if (graph[u][v] != INF && dist[u] != INF) {
                    if (dist[u] + graph[u][v] < dist[v]) {
                        dist[v] = dist[u] + graph[u][v];
                    }
                }
            }
        }
    }

    // Check for cycles starting from 'start'
    for (int v = 0; v < V; v++) {
        if (graph[v][start] != INF && dist[v] != INF) {
            int cycleLength = dist[v] + graph[v][start];
            if (cycleLength < shortestCycle) {
                shortestCycle = cycleLength;
            }
        }
    }
}

return shortestCycle;
}

int main() {

```

```

scanf("%d", &V);
scanf("%d", &E);

initializeGraph();

// Read edges
for (int i = 0; i < E; i++) {
    int u, v, w;
    scanf("%d %d %d", &u, &v, &w);
    graph[u][v] = w;
}

int shortestCycle = findShortestCycle();

if (shortestCycle != INF) {
    printf("Length of the shortest cycle: %d\n", shortestCycle);
} else {
    printf("No cycle found in the graph.\n");
}

return 0;
}

```

Status : Correct

Marks : 10/10

4. Problem Statement

Ravi is developing a network simulator that checks if a given undirected graph contains a cycle. His task is to build a program that determines whether any loop exists in the graph and also counts the number of nodes visited during the cycle detection process.

Write a program to ensure that the task is done by printing whether a cycle exists (1 or 0) and reporting the total number of recursive visits during traversal.

Input Format

The first line of the input consists of an integer V, representing the number of vertices.

The second line of the input consists of an integer E, representing the number of edges.

The next E lines each contain two space-separated integers u and v, representing an edge between vertices u and v.

Output Format

The first line prints 1 if the graph contains a cycle, else it prints 0

The second line prints "Number of nodes visited: X" where X is the number of nodes visited during the cycle detection function.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 3

3

0 1

1 2

2 0

Output: 1

Number of nodes visited: 3

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
#include <stdbool.h>
```

```
#define MAX 10
```

```
int graph[MAX][MAX];
```

```
int visited[MAX];
```

```
int V, E;
```

```
bool cycleFound = false;
```

```
int visitedCount = 0;
```

```
void dfs(int vertex, int parent) {
```

```
    visited[vertex] = 1;
```

```
    visitedCount++;
```

```

    for (int i = 0; i < V; i++) {
        if (graph[vertex][i] == 1) {
            if (!visited[i]) {
                dfs(i, vertex);
            }
            else if (i != parent) {
                cycleFound = true;
            }
        }
    }
}

```

```

int main() {
    scanf("%d", &V);
    scanf("%d", &E);

```

```

    // Initialize graph
    for (int i = 0; i < V; i++) {
        visited[i] = 0;
        for (int j = 0; j < V; j++) {
            graph[i][j] = 0;
        }
    }

```

```

    // Read edges
    for (int i = 0; i < E; i++) {
        int u, v;
        scanf("%d %d", &u, &v);
        graph[u][v] = 1;
        graph[v][u] = 1;
    }

```

```

    // Check for cycle in each component
    for (int i = 0; i < V; i++) {
        if (!visited[i]) {
            dfs(i, -1);
        }
    }

```

```

    // Output results
    if (cycleFound) {

```

```
        printf("1\n");
    } else {
        printf("0\n");
    }
    printf("Number of nodes visited: %d\n", visitedCount);

    return 0;
}
```

Status : Partially correct

Marks : 7.5/10

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2028_NeoColab_Competitive Coding II_L9

VTU_Week 9_MCQ

Attempt : 1

Total Mark : 15

Marks Obtained : 13

Section 1 : MCQ

1. In Backtracking, what is the purpose of marking vertices as visited during graph traversal?

Answer

To avoid revisiting the same vertex

Status : Correct

Marks : 1/1

2. In a Sudoku solver, when does the backtracking algorithm backtrack?

Answer

When no valid number can be placed in a cell

Status : Correct

Marks : 1/1

3. Which of the following best describes the backtracking approach to generating permutations of a string?

Answer

Fix one character, recursively permute the rest, and backtrack

Status : Correct

Marks : 1/1

4. In the Hamiltonian Cycle problem using backtracking, what must be true for a path to be valid?

Answer

All vertices are included once, and the last connects to the first

Status : Correct

Marks : 1/1

5. In backtracking solutions for maze problems, when do we backtrack from a cell?

Answer

When all directions from the current cell are either visited or blocked

Status : Correct

Marks : 1/1

6. Which of the following constraints are typically used in Sudoku backtracking?

Answer

All the mentioned options

Status : Correct

Marks : 1/1

7. In Backtracking, what is the base case for checking connectivity in an undirected graph?

Answer

All vertices are visited

Status : Correct

Marks : 1/1

8. Which of the following is essential for backtracking to function correctly?

Answer

A recursive call with a rollback (backtrack step)

Status : Correct

Marks : 1/1

9. What is the main idea behind Backtracking to check if a graph is connected?

Answer

Trying all possible paths from a starting vertex and marking visited vertices

Status : Correct

Marks : 1/1

10. Which of the following problems cannot be solved using backtracking?

Answer

Finding shortest path in weighted graph

Status : Wrong

Marks : 0/1

11. In general, backtracking can be used to solve?

Answer

Combinatorial problems

Status : Correct

Marks : 1/1

12. What is a base case in a backtracking algorithm?

Answer

A condition when a solution is found or invalid

Status : Correct

Marks : 1/1

13. Which data structure is commonly used in backtracking to store partial solutions?

Answer

Stack or List

Status : Correct

Marks : 1/1

14. What is the primary technique used to solve the Rat in a Maze problem?

Answer

Backtracking

Status : Correct

Marks : 1/1

15. Backtracking may lead to a solution that is _____.

Answer

optimal

Status : Wrong

Marks : 0/1

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2028_NeoColab_Competitive Coding II_L9

VTU_Week 9_COD

Attempt : 1

Total Mark : 50

Marks Obtained : 43.5

Section 1 : Coding

1. Problem Statement

Given an array of integers and a sum B, find all unique combinations in the array where the sum is equal to B. The same number may be chosen from the array any number of times to make B.

1. All numbers will be positive integers.
2. Elements in a combination ($a_1, a_2, \dots, a_k$) must be in non-descending order. (ie, $a_1 \leq a_2 \leq \dots \leq a_k$).
3. The combinations themselves must be sorted in ascending order.

Examples:

Input:

`arr[] = {7,2,6,5}`

`B = 16`

Output:

(2 2 2 2 2 2 2 2)

(2 2 2 2 2 6)

(2 2 2 5 5)

(2 2 5 7)

(2 2 6 6)

(2 7 7)

(5 5 6)

Input:

`arr[] = {6,5,7,1,8,2,9,9,7,7,9}`

`B = 6`

Output:

(1 1 1 1 1 1)

(1 1 1 1 2)

(1 1 2 2)

(1 5)

(2 2 2)

(6)

Input Format

The first line of input consists of space-separated positive integers representing the elements of the array.

The second line consists of a single positive integer representing the target sum B.

Output Format

The output prints "Combinations that sum to [B] are:" followed by the combinations in parentheses, each combination separated by a space.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 7 2 6 5

16

Output: Combinations that sum to 16 are:

(2 2 2 2 2 2 2) (2 2 2 2 2 6) (2 2 2 5 5) (2 2 5 7) (2 2 6 6) (2 7 7) (5 5 6)

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
int arr[100];
```

```
int n;
```

```
int target;
```

```
int combination[100];
```

```
int found = 0;
```

```
// Function to sort the array
```

```
void sort(int arr[], int n) {
```

```
    for (int i = 0; i < n - 1; i++) {
```

```
        for (int j = i + 1; j < n; j++) {
```

```
            if (arr[i] > arr[j]) {
```

```
                int temp = arr[i];
```

```
                arr[i] = arr[j];
```

```
                arr[j] = temp;
```

```
            }
```

```
        }
```

```
    }
```

```
}
```

```
// Function to remove duplicates from sorted array
```

```
int removeDuplicates(int arr[], int n) {
```

```
    if (n == 0) return 0;
```

```
    int j = 0;
```

```

    for (int i = 1; i < n; i++) {
        if (arr[i] != arr[j]) {
            j++;
            arr[j] = arr[i];
        }
    }
    return j + 1;
}

```

```

// Recursive function to find combinations
void findCombinations(int start, int remaining, int depth) {
    if (remaining == 0) {
        // Print the combination
        printf(" ");
        for (int i = 0; i < depth; i++) {
            printf("%d ", combination[i]);
        }
        printf("\n");
        found = 1;
        return;
    }

```

```

    for (int i = start; i < n; i++) {
        if (arr[i] > remaining) {
            break;
        }

        combination[depth] = arr[i];
        // Allow using the same element again (i, not i+1)
        findCombinations(i, remaining - arr[i], depth + 1);
    }
}

```

```

int main() {
    // Read the array
    n = 0;
    while (scanf("%d", &arr[n]) == 1) {
        n++;
        char c = getchar();
        if (c == '\n') break;
    }
}

```

```

// Read the target sum
scanf("%d", &target);

// Sort the array
sort(arr, n);

// Remove duplicates
n = removeDuplicates(arr, n);

printf("Combinations that sum to %d are:\n", target);
findCombinations(0, target, 0);

if (!found) {
    printf("\n");
}

return 0;
}

```

Status : Correct

Marks : 10/10

2. Problem Statement

Imagine you're helping a friend plan gift packages with a total budget of N. Your task is to find all unique ways to spend the entire budget on different combinations of gift prices, ensuring no combination is repeated by rearranging the prices.

Note:

By unique, it means that no matter that no other composition can be expressed as a permutation of the generated composition. For eg. [1,2,1] and [1, 1, 2] are not unique. You need to print all combinations in non-decreasing order, for eg. [1, 2, 1] or [1,1,2] will be printed as [1,1,2]. The order of printing all the combinations should be in lexicographical order.

Use backtracking to solve the given problem.

Input Format

The input consists of a single integer, n, which represents the number to be

broken down.

Output Format

The output consists of a list of lists, where each inner list represents a possible breakdown of the given number.

Each breakdown is represented as a list of integers that add up to the given number.

The breakdowns are printed in lexicographic order.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 1

Output: [[1]]

Answer

```
// You are using GCC
#include <stdio.h>
```

```
int combination[25];
int result[100][25];
int resultLengths[100];
int resultCount = 0;
```

```
void findCombinations(int remaining, int start, int depth) {
    if (remaining == 0) {
        // Store the combination
        for (int i = 0; i < depth; i++) {
            result[resultCount][i] = combination[i];
        }
        resultLengths[resultCount] = depth;
        resultCount++;
        return;
    }
    for (int i = start; i <= remaining; i++) {
        combination[depth] = i;
```

```

        findCombinations(remaining - i, i, depth + 1);
    }
}

int main() {
    int n;
    scanf("%d", &n);

    findCombinations(n, 1, 0);

    // Print the result in the required format
    printf("[");
    for (int i = 0; i < resultCount; i++) {
        printf("[");
        for (int j = 0; j < resultLengths[i]; j++) {
            printf("%d", result[i][j]);
            if (j < resultLengths[i] - 1) {
                printf(", ");
            }
        }
        printf("]");
        if (i < resultCount - 1) {
            printf(", ");
        }
    }
    printf("]\n");

    return 0;
}

```

Status : Partially correct

Marks : 3.5/10

3. Problem Statement

Rohan is designing a security system that generates all possible rearrangements of a given access code made up of distinct characters. He inputs a string representing the access code into the system. The system must display every possible permutation of the characters without repetition.

Write a program to ensure that all permutations of the given string are printed in separate lines.

Input Format

The input consists of a single line containing a string S, consisting of lowercase or upper-case alphabets.

Output Format

The output prints each permutation of the string on a new line.

The permutations should be printed in the order generated by swapping characters starting from the first position

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: ABC

Output: ABC

ACB

BAC

BCA

CBA

CAB

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
#include <string.h>
```

```
void swap(char *a, char *b) {  
    char temp = *a;  
    *a = *b;  
    *b = temp;  
}
```

```
void permute(char *str, int left, int right) {  
    if (left == right) {  
        printf("%s\n", str);  
    }
```

```

    return;
}

for (int i = left; i <= right; i++) {
    swap(&str[left], &str[i]);
    permute(str, left + 1, right);
    swap(&str[left], &str[i]); // backtrack
}
}

```

```

int main() {
    char str[6];
    scanf("%s", str);

    int n = strlen(str);
    permute(str, 0, n - 1);

    return 0;
}

```

Status : Correct

Marks : 10/10

4. Problem Statement

Given a number K and string str of digits denoting a positive integer, build the largest number possible by performing swap operations on the digits of str at most K times.

Input Format

The first line of input consists of a positive integer K.

The second line of input consists of a string of digits of positive integers.

Output Format

The output displays the largest number possible by performing swap operations on the digits of str at most K times.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 1

254

Output: 524

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
#include <string.h>
```

```
#include <limits.h>
```

```
char maxStr[31];
```

```
int k;
```

```
void swap(char *a, char *b) {
```

```
    char temp = *a;
```

```
    *a = *b;
```

```
    *b = temp;
```

```
}
```

```
void findMaxUtil(char str[], int index, int swapsRemaining) {
```

```
    if (swapsRemaining == 0 || index == strlen(str) - 1) {
```

```
        if (strcmp(str, maxStr) > 0) {
```

```
            strcpy(maxStr, str);
```

```
        }
```

```
        return;
```

```
    }
```

```
    // Find the maximum digit from current position to end
```

```
    char maxDigit = str[index];
```

```
    for (int i = index + 1; i < strlen(str); i++) {
```

```
        if (str[i] > maxDigit) {
```

```
            maxDigit = str[i];
```

```
        }
```

```
    }
```

```
    // If current digit is already the maximum, no need to swap
```

```
    if (maxDigit == str[index]) {
```

```
        findMaxUtil(str, index + 1, swapsRemaining);
```

```
        return;
```

```

    }
    // Try swapping with each occurrence of maxDigit from index+1 to end
    for (int i = index + 1; i < strlen(str); i++) {
        if (str[i] == maxDigit) {
            swap(&str[index], &str[i]);
            findMaxUtil(str, index + 1, swapsRemaining - 1);
            swap(&str[index], &str[i]); // backtrack
        }
    }
}

int main() {
    scanf("%d", &k);

    char str[31];
    scanf("%s", str);

    strcpy(maxStr, str);
    findMaxUtil(str, 0, k);

    printf("%s\n", maxStr);

    return 0;
}

```

Status : Correct

Marks : 10/10

5. Problem Statement

Mythili has a list of item prices and a customer's gift card balance. She needs to find a combination of two or more items whose total price exactly matches the gift card balance. If such a combination exists, she should print the item prices in the order they appear. If multiple valid combinations exist, she must print the first one found. If no valid combination is possible, print "No combination of items"

Write a function that takes the list of item prices, the number of items, and the gift card balance as input, and prints the result based on the above rules.

Input Format

The first line of input consists of an integer, representing the number of items in the store.

The second line consists of an array of integers separated by a space, representing the prices of the items.

The last line consists of an integer representing the customer's gift card balance.

Output Format

If there exists a combination of items whose prices sum up to the gift card balance, then print this combination.

If no such combination exists, print "No combination of items".

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 3
10 20 30
50

Output: 20 30

Answer

```
// You are using GCC
#include <stdio.h>
#include <stdbool.h>
```

```
int prices[5];
int n;
int balance;
int combination[5];
int found = 0;
```

```
void findCombination(int start, int currentSum, int depth) {
    if (found) return;

    if (currentSum == balance && depth >= 2) {
```

```

        // Print the combination
        for (int i = 0; i < depth; i++) {
            if (i > 0) printf(" ");
            printf("%d", combination[i]);
        }
        printf("\n");
        found = 1;
        return;
    }

    if (currentSum > balance) return;

    for (int i = start; i < n; i++) {
        combination[depth] = prices[i];
        findCombination(i + 1, currentSum + prices[i], depth + 1);
        if (found) return;
    }
}

int main() {
    scanf("%d", &n);

    for (int i = 0; i < n; i++) {
        scanf("%d", &prices[i]);
    }

    scanf("%d", &balance);

    findCombination(0, 0, 0);

    if (!found) {
        printf("No combination of items\n");
    }

    return 0;
}

```

Status : Correct

Marks : 10/10

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2028_NeoColab_Competitive Coding II_L9

VTU_Week 9_CY

Attempt : 1

Total Mark : 40

Marks Obtained : 40

Section 1 : Coding

1. Problem Statement

Emily is working on arranging queens on a chessboard where one queen must already be fixed at a given position. She wants to find all possible configurations of placing N queens such that no two queens attack each other, with the condition that the queen at the specified position cannot be moved.

Write a program to ensure that the chessboard solutions are displayed correctly, or state "No solution" if none exist.

Input Format

The first line contains an integer N representing the size of the chessboard.

The second line contains an integer i, representing the row index of the fixed queen.

The third line contains an integer j , representing the column index of the fixed queen.

Output Format

If solutions exist: Print each valid board configuration, where "Q" represents a queen and "." represents an empty cell.

Each row of the board should be space-separated.

Each configuration should be followed by a blank line.

If no configuration is possible, print "No solution".

Refer to the sample output for the formatting specification.

Sample Test Case

Input: 4

0

1

Output: . Q . .

. . . Q

Q . . .

. . Q .

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
#include <stdbool.h>
```



```
#include <stdlib.h>
```

```
int N;  
int fixedRow, fixedCol;  
int board[10][10];  
int solutions = 0;
```

```
// Function to check if a queen can be placed at (row, col)
```

```
bool isSafe(int row, int col) {  
    // Check if this is the fixed queen position  
    if (row == fixedRow && col == fixedCol) {  
        return true;  
    }
```

```
    // Check if fixed queen is in same column  
    if (col == fixedCol) {  
        return false;  
    }
```

```
    // Check if fixed queen is in same row  
    if (row == fixedRow) {  
        return false;  
    }
```

```
    // Check if fixed queen is on same diagonal  
    if (abs(row - fixedRow) == abs(col - fixedCol)) {  
        return false;  
    }
```

```
    // Check this column on upper side  
    for (int i = 0; i < row; i++) {  
        if (board[i][col] == 1) {  
            return false;  
        }  
    }
```

```
    // Check upper left diagonal  
    for (int i = row, j = col; i >= 0 && j >= 0; i--, j--) {  
        if (board[i][j] == 1) {  
            return false;  
        }  
    }
```

```

// Check upper right diagonal
for (int i = row, j = col; i >= 0 && j < N; i--, j++) {
    if (board[i][j] == 1) {
        return false;
    }
}

return true;
}

```

// Function to solve N-Queens problem

```

void solveNQueens(int row) {
    if (row == N) {
        // Print the solution
        for (int i = 0; i < N; i++) {
            for (int j = 0; j < N; j++) {
                if (board[i][j] == 1) {
                    printf("Q ");
                } else {
                    printf(". ");
                }
            }
            printf("\n");
        }
        printf("\n");
        solutions++;
        return;
    }
}

```

// If this row contains the fixed queen, skip to next row

```

if (row == fixedRow) {
    solveNQueens(row + 1);
    return;
}

```

```

for (int col = 0; col < N; col++) {
    // Skip the fixed queen's column
    if (col == fixedCol) {
        continue;
    }
}

```

```

        if (isSafe(row, col)) {
            board[row][col] = 1;
            solveNQueens(row + 1);
            board[row][col] = 0; // backtrack
        }
    }
}

```

```

int main() {
    scanf("%d", &N);
    scanf("%d", &fixedRow);
    scanf("%d", &fixedCol);

    // Initialize board
    for (int i = 0; i < N; i++) {
        for (int j = 0; j < N; j++) {
            board[i][j] = 0;
        }
    }

    // Place the fixed queen
    board[fixedRow][fixedCol] = 1;

    // Start solving from row 0
    solveNQueens(0);

    if (solutions == 0) {
        printf("No solution\n");
    }

    return 0;
}

```

Status : Correct

Marks : 10/10

2. Problem Statement

Consider a rat placed at (0, 0) in a square matrix of order $N \times N$. It has to reach the destination at $(N - 1, N - 1)$. Find all possible paths that the rat can take to reach from the source to the destination. The directions in which the rat can move are 'U'(up), 'D'(down), 'L' (left), and 'R' (right). Value

0 at a cell in the matrix represents that it is blocked and the rat cannot move to it while value 1 at a cell in the matrix represents that the rat can travel through it. Return the list of paths in lexicographically increasing order.

Note: In a path, no cell can be visited more than once. If the source cell is 0, the rat cannot move to any other cell.

Example 1

Input:

$N = 4$

$m[][] =$

$\{\{1, 0, 0, 0\},$

$\{1, 1, 0, 1\},$

$\{1, 1, 0, 0\},$

$\{0, 1, 1, 1\}\}$

Output:

DDRDRR DRDDRR

Explanation:

The rat can reach the destination at (3, 3) from (0, 0) by two paths - DRDDRR and DDRDRR, when printed in sorted order we get DDRDRR DRDDRR.

Example 2

Input:

$N = 2$

$m[][] =$

$\{\{1, 0\},$

$\{1, 0\}\}$

Output:

-1

Explanation:

No path exists and the destination cell is blocked.

Input Format

The first line of input contains an integer n, representing the size of the maze.

The next n lines each contain n space-separated integers, representing the maze.

Output Format

The output consists of

If at least one path is found, print all valid paths as space-separated strings.

If no path is found, prints -1.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 4

1 0 0 0

1 1 0 1

1 1 0 0

0 1 1 1

Output: DDRDRR DRDDRR

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
#include <string.h>
```

```
#include <stdlib.h>
```

```
int N;
```

```
int maze[5][5];
```

```
int visited[5][5];
```

```

char paths[100][100];
int pathCount = 0;

// Possible moves in lexicographical order: D, L, R, U
int dx[] = {1, 0, 0, -1};
int dy[] = {0, -1, 1, 0};
char dir[] = {'D', 'L', 'R', 'U'};

int compareStrings(const void *a, const void *b) {
    return strcmp((char *)a, (char *)b);
}

void findPaths(int x, int y, char path[], int index) {
    // If destination is reached
    if (x == N - 1 && y == N - 1) {
        path[index] = '\0';
        strcpy(paths[pathCount], path);
        pathCount++;
        return;
    }

    // Explore all 4 directions in lexicographical order
    for (int i = 0; i < 4; i++) {
        int newX = x + dx[i];
        int newY = y + dy[i];

        // Check if the move is valid
        if (newX >= 0 && newX < N && newY >= 0 && newY < N &&
            !visited[newX][newY] && maze[newX][newY] == 1) {

            visited[newX][newY] = 1;
            path[index] = dir[i];
            findPaths(newX, newY, path, index + 1);
            visited[newX][newY] = 0; // Backtrack
        }
    }
}

int main() {
    scanf("%d", &N);

    for (int i = 0; i < N; i++) {

```

```

    for (int j = 0; j < N; j++) {
        scanf("%d", &maze[i][j]);
        visited[i][j] = 0;
    }
}

// Check if source or destination is blocked
if (maze[0][0] == 0 || maze[N - 1][N - 1] == 0) {
    printf("-1\n");
    return 0;
}

char path[100];
visited[0][0] = 1;
findPaths(0, 0, path, 0);

if (pathCount == 0) {
    printf("-1\n");
} else {
    // Sort paths lexicographically
    qsort(paths, pathCount, sizeof(paths[0]), compareStrings);

    for (int i = 0; i < pathCount; i++) {
        printf("%s ", paths[i]);
    }
    printf("\n");
}

return 0;
}

```

Status : Correct

Marks : 10/10

3. Problem Statement

Given a partially filled 9x9 2D array, the goal is to assign digits (from 1 to 9) to the empty cells so that every row, column, and sub-grid of size 3x3 contains exactly one instance of the digits from 1 to 9.

Write a program for the same.

Example

Input:

```
3 0 6 5 0 8 4 0 0
5 2 0 0 0 0 0 0 0
0 8 7 0 0 0 0 3 1
0 0 3 0 1 0 0 8 0
9 0 0 8 6 3 0 0 5
0 5 0 0 9 0 6 0 0
1 3 0 0 0 0 2 5 0
0 0 0 0 0 0 0 7 4
0 0 5 2 0 6 3 0 0
```

Output:

```
3 1 6 5 7 8 4 9 2
5 2 9 1 3 4 7 6 8
4 8 7 6 2 9 5 3 1
2 6 3 4 1 5 9 8 7
9 7 4 8 6 3 1 2 5
8 5 1 7 9 2 6 4 3
1 3 8 9 4 7 2 5 6
6 9 2 3 5 1 8 7 4
7 4 5 2 8 6 3 1 9
```

Explanation

To solve the given Sudoku puzzle, we apply the fundamental rules of Sudoku: each row, column, and 3x3 subgrid must contain the digits 1 through 9 without any repetition. Starting with the given partially filled grid, we systematically fill in the empty cells. For each empty cell, we check the

numbers already present in the corresponding row, column, and 3x3 subgrid to determine which number can legally go in that position. This process involves logical deduction and elimination. In the case of the provided puzzle, the empty cells are filled one by one, ensuring that each row, column, and subgrid adheres to the Sudoku rules. For example, in Row 1, the missing numbers (1, 7, 9, 2) are placed in the available spots based on their placement in the columns and subgrids. This continues for each row, progressively narrowing down the possibilities for the empty cells until the entire grid is complete. The final solution, after applying this reasoning, looks like the output grid:

Input Format

The input consists of a 9x9 matrix in 9 rows and 9 columns separated by space.

Output Format

The output displays the 9x9 matrix with the updated values.

If the conditions are not satisfied, then display "No Solution exists".

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 3 0 6 5 0 8 4 0 0

5 2 0 0 0 0 0 0 0

0 8 7 0 0 0 0 3 1

0 0 3 0 1 0 0 8 0

9 0 0 8 6 3 0 0 5

0 5 0 0 9 0 6 0 0

1 3 0 0 0 0 2 5 0

0 0 0 0 0 0 0 7 4

0 0 5 2 0 6 3 0 0

Output: 3 1 6 5 7 8 4 9 2

5 2 9 1 3 4 7 6 8

4 8 7 6 2 9 5 3 1

2 6 3 4 1 5 9 8 7

9 7 4 8 6 3 1 2 5

8 5 1 7 9 2 6 4 3

```
1 3 8 9 4 7 2 5 6
6 9 2 3 5 1 8 7 4
7 4 5 2 8 6 3 1 9
```

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
#include <stdbool.h>
```

```
#define N 9
```

```
int grid[N][N];
```

```
// Function to check if a number can be placed at grid[row][col]
```

```
bool isSafe(int row, int col, int num) {
```

```
    // Check row
```

```
    for (int x = 0; x < N; x++) {
```

```
        if (grid[row][x] == num) {
```

```
            return false;
```

```
        }
```

```
    }
```

```
    // Check column
```

```
    for (int x = 0; x < N; x++) {
```

```
        if (grid[x][col] == num) {
```

```
            return false;
```

```
        }
```

```
    }
```

```
    // Check 3x3 subgrid
```

```
    int startRow = row - row % 3;
```

```
    int startCol = col - col % 3;
```

```
    for (int i = 0; i < 3; i++) {
```

```
        for (int j = 0; j < 3; j++) {
```

```
            if (grid[i + startRow][j + startCol] == num) {
```

```
                return false;
```

```
            }
```

```
        }
```

```
    }
```

```
    return true;
```

```
}
```

```

// Function to find an empty cell
bool findEmptyCell(int *row, int *col) {
    for (*row = 0; *row < N; (*row)++) {
        for (*col = 0; *col < N; (*col)++) {
            if (grid[*row][*col] == 0) {
                return true;
            }
        }
    }
    return false;
}

```

```

// Function to solve Sudoku using backtracking
bool solveSudoku() {
    int row, col;

```

```

    // If no empty cell found, puzzle is solved
    if (!findEmptyCell(&row, &col)) {
        return true;
    }

```

```

    // Try digits 1 to 9
    for (int num = 1; num <= 9; num++) {
        if (isSafe(row, col, num)) {
            grid[row][col] = num;

            if (solveSudoku()) {
                return true;
            }

```

```

            grid[row][col] = 0; // Backtrack
        }
    }

```

```

    return false;
}

```

```

// Function to print the grid
void printGrid() {
    for (int i = 0; i < N; i++) {
        for (int j = 0; j < N; j++) {

```

```

        printf("%d ", grid[i][j]);
    }
    printf("\n");
}

int main() {
    // Read the Sudoku grid
    for (int i = 0; i < N; i++) {
        for (int j = 0; j < N; j++) {
            scanf("%d", &grid[i][j]);
        }
    }

    if (solveSudoku()) {
        printGrid();
    } else {
        printf("No Solution exists\n");
    }

    return 0;
}

```

Status : Correct

Marks : 10/10

4. Problem Statement

Given an undirected graph and a number m , determine if the graph can be colored with at most m colors such that no two adjacent vertices of the graph are colored with the same color.

Note: Here, the coloring of a graph means the assignment of colors to all vertices.

Example

Input:

```

0 1 1 1
1 0 1 0

```

1 1 0 1

1 0 1 0

3

Output:

Solution Exists:

1 2 3 2

Explanation:

A minimum of 3 colors is required for the above graph.

Input Format

The input consists of four lines each containing four space-separated integers (1 if connected, 0 if not), representing the adjacency matrix of a graph where each element represents the connection between vertices.

The last line consists of an integer m, representing the number of colors available.

Output Format

If a solution exists:

1. The first line prints "Solution Exists:".
2. The second line prints the assigned colors for each vertex, separated by a space.

If no solution exists, the output prints "Solution does not exist".

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 0 1 1 1

```
1 0 1 0
1 1 0 1
1 0 1 0
3
```

Output: Solution Exists:

```
1 2 3 2
```

Answer

// You are using GCC

```
#include <stdio.h>
```

```
#include <stdbool.h>
```

```
#define V 4
```

```
int graph[V][V];
```

```
int colors[V];
```

```
int m;
```

// Function to check if assigning color to vertex v is safe

```
bool isSafe(int v, int color) {
```

```
    for (int i = 0; i < V; i++) {
```

```
        if (graph[v][i] && colors[i] == color) {
```

```
            return false;
```

```
        }
```

```
    }
```

```
    return true;
```

```
}
```

// Recursive function to solve graph coloring

```
bool graphColoringUtil(int v) {
```

```
    // If all vertices are assigned a color
```

```
    if (v == V) {
```

```
        return true;
```

```
    }
```

// Try different colors for vertex v

```
    for (int c = 1; c <= m; c++) {
```

```
        if (isSafe(v, c)) {
```

```
            colors[v] = c;
```

```
            if (graphColoringUtil(v + 1)) {
```

```
                return true;
```

```

    }

    colors[v] = 0; // Backtrack
}
}

return false;
}

int main() {
    // Read the adjacency matrix
    for (int i = 0; i < V; i++) {
        for (int j = 0; j < V; j++) {
            scanf("%d", &graph[i][j]);
        }
    }

    // Read number of colors
    scanf("%d", &m);

    // Initialize colors array
    for (int i = 0; i < V; i++) {
        colors[i] = 0;
    }

    // Solve graph coloring
    if (graphColoringUtil(0)) {
        printf("Solution Exists:\n");
        for (int i = 0; i < V; i++) {
            printf("%d ", colors[i]);
        }
        printf("\n");
    } else {
        printf("Solution does not exist\n");
    }

    return 0;
}

```

Status : Correct

Marks : 10/10

Vel Tech Group of Institutions

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Scan to verify results



2028_NeoColab_Competitive Coding II_L9

VTU_Week 9_PAH

Attempt : 1

Total Mark : 40

Marks Obtained : 31

Section 1 : Coding

1. Problem Statement

In chess, each step of a knight comprises stepping by two squares horizontally and one square vertically, or by one square horizontally and two squares vertically. A knight making one step from location (0,0) of an infinite chess board would finish up at one of the following eight locations: (1,2), (-1,2), (1,-2), (-1,-2), (2,1), (-2,1), (2,-1), (-2,-1).

Beginning from location (0,0), what is the minimum number of steps needed for a knight to get to some other arbitrary location (x,y)?

Input Format

The first line of input contains two integers x and y, representing the location of a knight on a chess board.

Output Format

The output displays a single integer m, representing the minimum number of steps needed for a knight to move from (0,0) to (x,y).

Sample Test Case

Input: 1 2

Output: 1

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
#include <stdbool.h>
```

```
#include <stdlib.h>
```

```
#define MAX 10
```

```
typedef struct {
```

```
    int x;
```

```
    int y;
```

```
    int dist;
```

```
} Cell;
```

```
int knightMoves(int targetX, int targetY) {
```

```
    // Possible knight moves
```

```
    int movesX[] = {1, 1, -1, -1, 2, 2, -2, -2};
```

```
    int movesY[] = {2, -2, 2, -2, 1, -1, 1, -1};
```

```
    bool visited[MAX][MAX] = {false};
```

```
    Cell queue[100];
```

```
    int front = 0, rear = 0;
```

```
    // Start from (0,0)
```

```
    queue[rear].x = 0;
```

```
    queue[rear].y = 0;
```

```
    queue[rear].dist = 0;
```

```
    rear++;
```

```
    visited[0][0] = true;
```

```
    while (front < rear) {
```

```
        Cell current = queue[front];
```

```
        front++;
```

```

    if (current.x == targetX && current.y == targetY) {
        return current.dist;
    }

    for (int i = 0; i < 8; i++) {
        int newX = current.x + movesX[i];
        int newY = current.y + movesY[i];

        if (newX >= 0 && newX <= 8 && newY >= 0 && newY <= 8 && !visited[newX]
[newY]) {
            visited[newX][newY] = true;
            queue[rear].x = newX;
            queue[rear].y = newY;
            queue[rear].dist = current.dist + 1;
            rear++;
        }
    }
}

return -1; // Should not happen for valid inputs
}

int main() {
    int x, y;
    scanf("%d %d", &x, &y);

    int steps = knightMoves(x, y);
    printf("%d\n", steps);

    return 0;
}

```

Status : Correct

Marks : 10/10

2. Problem Statement

You are given a chessboard of size $N \times N$. The objective is to place N queens on the board such that no two queens threaten each other (i.e., no two queens share the same row, column, or diagonal).

Additionally, the board may already have K queens pre-placed at valid positions. You must:

Verify that the initial placements are valid (i.e., not attacking each other).

Try to place the remaining $N - K$ queens.

Print one valid arrangement if it exists.

Input Format

The first line of input contains two integers N and K, separated by a space:

N: Size of the chessboard ($1 \leq N \leq 20$)

K: Number of pre-placed queens ($0 \leq K \leq N$)

The next K lines each contain two integers r and c separated by a space:

These represent the row and column indices (0-based) of the pre-placed queens.

Output Format

If a valid solution exists, print the board as N lines, each containing N integers (0 or 1), where 1 represents a queen and 0 is an empty square.

If no solution is possible, print: "No solution found."

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 4

1

0 1

Output: 0 1 0 0

0 0 0 1

```
1 0 0 0
0 0 1 0
```

Answer

```
// You are using GCC
#include <stdio.h>
#include <stdbool.h>
#include <stdlib.h>

#define MAX 20

int N, K;
int board[MAX][MAX];
int preRow[MAX], preCol[MAX];
bool solutionFound = false;

// Function to check if placing a queen at (row, col) is safe
bool isSafe(int row, int col) {
    // Check column
    for (int i = 0; i < N; i++) {
        if (board[i][col] == 1 && i != row) {
            return false;
        }
    }

    // Check row
    for (int j = 0; j < N; j++) {
        if (board[row][j] == 1 && j != col) {
            return false;
        }
    }

    // Check diagonals
    for (int i = 0; i < N; i++) {
        for (int j = 0; j < N; j++) {
            if (board[i][j] == 1 && (i != row || j != col)) {
                if (abs(i - row) == abs(j - col)) {
                    return false;
                }
            }
        }
    }
}
```

```
    return true;
}
```

```
// Function to check if pre-placed queens are valid
```

```
bool arePrePlacedValid() {
    for (int i = 0; i < K; i++) {
        int r = preRow[i];
        int c = preCol[i];

        // Check with previously placed queens
        for (int j = 0; j < i; j++) {
            if (r == preRow[j] || c == preCol[j]) {
                return false;
            }
            if (abs(r - preRow[j]) == abs(c - preCol[j])) {
                return false;
            }
        }
    }
    return true;
}
```

```
// Backtracking function to place queens
```

```
void solve(int row) {
    if (solutionFound) return;

    if (row == N) {
        solutionFound = true;
        return;
    }
}
```

```
// Check if this row already has a pre-placed queen
```

```
int preColForRow = -1;
for (int i = 0; i < K; i++) {
    if (preRow[i] == row) {
        preColForRow = preCol[i];
        break;
    }
}

if (preColForRow != -1) {
```

```

// This row has a pre-placed queen
if (isSafe(row, preColForRow)) {
    solve(row + 1);
}
} else {
    // Try placing queen in each column
    for (int col = 0; col < N; col++) {
        // Check if this column is already occupied by a pre-placed queen
        bool colOccupied = false;
        for (int i = 0; i < K; i++) {
            if (preCol[i] == col) {
                colOccupied = true;
                break;
            }
        }
        if (colOccupied) continue;

        if (isSafe(row, col)) {
            board[row][col] = 1;
            solve(row + 1);
            if (!solutionFound) {
                board[row][col] = 0;
            }
        }
    }
}
}

int main() {
    scanf("%d %d", &N, &K);

    // Initialize board
    for (int i = 0; i < N; i++) {
        for (int j = 0; j < N; j++) {
            board[i][j] = 0;
        }
    }

    // Read pre-placed queens
    for (int i = 0; i < K; i++) {
        scanf("%d %d", &preRow[i], &preCol[i]);
        board[preRow[i]][preCol[i]] = 1;
    }
}

```

```

    }

    // Check if pre-placed queens are valid
    if (!arePrePlacedValid()) {
        printf("No solution found.\n");
        return 0;
    }

    // Solve the problem
    solve(0);

    if (solutionFound) {
        for (int i = 0; i < N; i++) {
            for (int j = 0; j < N; j++) {
                printf("%d ", board[i][j]);
            }
            printf("\n");
        }
    } else {
        printf("No solution found.\n");
    }

    return 0;
}

```

Status : Correct

Marks : 10/10

3. Problem Statement

Bob has a task to find a Hamiltonian cycle in a given undirected graph represented as an adjacency matrix. A Hamiltonian cycle is a cycle that visits every vertex exactly once and returns to the starting vertex. Bob needs to write a program that will determine if such a cycle exists and, if so, print the cycle. If no cycle is found, the program should output "No solution".

Your task is to help Bob in implementing the same.

Input Format

The first line of input consists of a single integer, V, indicating the number of

vertices in the graph.

The following V lines each contain V space separated integers, representing the adjacency matrix of the graph, where a '1' means an edge exists between two vertices, and a '0' means it does not.

Output Format

If a Hamiltonian cycle is found, print "Solution found" followed by the cycle as a series of vertices starting and ending at the starting vertex, space-separated on a new line.

If no Hamiltonian cycle is found, print "No solution".

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 8

0 1 0 1 1 0 0 0

1 0 1 0 0 1 0 0

0 1 0 1 0 0 1 0

1 0 1 0 0 0 0 1

1 0 0 0 0 1 0 1

0 1 0 0 1 0 1 0

0 0 1 0 0 1 0 1

0 0 0 1 1 0 1 0

Output: Solution found

0 1 2 3 7 6 5 4 0

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
#include <stdbool.h>
```

```
#define MAX 10
```

```
int V;
```

```
int graph[MAX][MAX];
```

```
int path[MAX];
```

```
bool visited[MAX];
```



```
// Function to check if vertex v can be added at position pos
```

```
bool isSafe(int v, int pos) {
```

```
    // Check if there's an edge from the last added vertex to v
```

```
    if (graph[path[pos - 1]][v] == 0) {
```

```
        return false;
```

```
    }
```

```
    // Check if vertex v has already been visited
```

```
    for (int i = 0; i < pos; i++) {
```

```
        if (path[i] == v) {
```

```
            return false;
```

```
        }
```

```
    }
```

```
    return true;
```

```
}
```

```
// Recursive function to find Hamiltonian cycle
```

```
bool hamiltonianCycleUtil(int pos) {
```

```
    // If all vertices are included in the cycle
```

```
    if (pos == V) {
```

```
        // Check if there's an edge from the last vertex to the first vertex
```

```
        if (graph[path[pos - 1]][path[0]] == 1) {
```

```
            return true;
```

```
        }
```

```
    } return false;
```

```
}
```

```
// Try different vertices as the next candidate
```

```
for (int v = 0; v < V; v++) {
```

```
    if (isSafe(v, pos)) {
```

```
        path[pos] = v;
```

```
        if (hamiltonianCycleUtil(pos + 1)) {
```

```
            return true;
```

```
        }
```

```
    // Backtrack
```

```
    path[pos] = -1;
```

```
}
```

```
}
```

```

    return false;
}

// Function to find and print Hamiltonian cycle
void findHamiltonianCycle() {
    // Initialize path array
    for (int i = 0; i < V; i++) {
        path[i] = -1;
    }

    // Start from vertex 0
    path[0] = 0;

    if (hamiltonianCycleUtil(1)) {
        printf("Solution found\n");
        for (int i = 0; i < V; i++) {
            printf("%d ", path[i]);
        }
        printf("%d \n", path[0]);
    } else {
        printf("No solution\n");
    }
}

int main() {
    scanf("%d", &V);

    // Read adjacency matrix
    for (int i = 0; i < V; i++) {
        for (int j = 0; j < V; j++) {
            scanf("%d", &graph[i][j]);
        }
    }

    findHamiltonianCycle();

    return 0;
}

```

Status : Correct

Marks : 10/10

4. Problem Statment

You are tasked with placing N queens on an $N \times N$ chessboard such that no two queens can attack each other. Specifically:

No two queens can be on the same row.

No two queens can be on the same column.

No two queens can be on the same diagonal.

You need to find all possible valid placements of the queens and print them.

Example Scenario: $N = 4$

For $n = 4$, we need to find all the valid placements for 4 queens on a 4×4 chessboard.

Return only the number of distinct solutions to the n -queens puzzle.

Each solution contains a distinct board configuration of the n -queens' placement, where 'Q' and '.' both indicate a queen and an empty space, respectively.

Example

Input:

4

Output:

2

..Q.

Q...

...Q

.Q..

.Q..

...Q

Q...

..Q.

Explanation:

There exist two distinct solutions to the 4-queens puzzle as shown.

Input Format

The input consists of a single integer n , representing the size of the chessboard.

Output Format

The first line contains an integer representing the number of distinct solutions to the N-Queens problem for the given value of n .

For each solution:

The board configuration is printed with n lines.

Each line contains n characters:

'Q' represents a queen.

'.' represents an empty space.

The configuration is printed in top-to-bottom row order, i.e., the first line corresponds to the top row of the board.

Solutions are printed in the order they are found by the backtracking algorithm, which places queens column by column from left to right, and in each column, tries rows from top to bottom (row 0 to row $n-1$).

A blank line is printed after each solution to separate different board configurations.

Refer to the sample output format for the formatting specifications.

Sample Test Case

Input: 4

Output: 2

..Q.

Q...

...Q

.Q..

.Q..

...Q

Q...

..Q.

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
#include <stdbool.h>
```

```
#define MAX 10
```

```
#define MAX_SOLUTIONS 100
```

```
int n;
```

```
int board[MAX][MAX];
```

```
int solutions[MAX_SOLUTIONS][MAX][MAX];
```

```
int solCount = 0;
```

```
bool isSafe(int row, int col) {
```

```
    for (int i = 0; i < row; i++) {  
        if (board[i][col] == 1) return false;
```

```
    }
```

```
    for (int i = row, j = col; i >= 0 && j >= 0; i--, j--) {  
        if (board[i][j] == 1) return false;
```

```
    }
```

```
    for (int i = row, j = col; i >= 0 && j < n; i--, j++) {  
        if (board[i][j] == 1) return false;
```

```
    }
```

```
    return true;
```

```
}
```

```
void saveBoard() {
```

```
    for (int i = 0; i < n; i++) {
```

```
        for (int j = 0; j < n; j++) {
```

```

        solutions[solCount][i][j] = board[i][j];
    }
}
solCount++;
}

```

```

void solveNQueens(int row) {
    if (row == n) {
        saveBoard();
        return;
    }
    for (int col = 0; col < n; col++) {
        if (isSafe(row, col)) {
            board[row][col] = 1;
            solveNQueens(row + 1);
            board[row][col] = 0;
        }
    }
}

```

```

void printBoard(int idx) {
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            printf("%c", solutions[idx][i][j] ? 'Q' : '.');
        }
        printf("\n");
    }
    printf("\n");
}

```

```

int main() {
    scanf("%d", &n);

    // Initialize board
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            board[i][j] = 0;
        }
    }

    solveNQueens(0);
}

```

```
// Print number of solutions
printf("%d\n", solCount);

// Print solutions in the order they were found
for (int i = 0; i < solCount; i++) {
    printBoard(i);
}

return 0;
}
```

Status : Partially correct

Marks : 1/10

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2028_NeoColab_Competitive Coding II_L9

VTU_Week 10_MCQ

Attempt : 1

Total Mark : 15

Marks Obtained : 14

Section 1 : MCQ

1. Which of the following sequences of array elements forms a heap?

Answer

{23, 17, 14, 7, 13, 10, 1, 5, 6, 12}

Status : Correct

Marks : 1/1

2. Consider a max heap, represented by the array: 23, 17, 14, 6, 13, 10, 1, 5. If a value 20 is inserted into this heap, then what will be the new max heap?

Answer

23, 20, 14, 17, 13, 10, 1, 5, 6

Status : Correct

Marks : 1/1

3. Given a binary max heap. The elements are sorted in an array as 25, 14, 16, 13, 10, 8, 12. What is the content of the array after two delete operations?

Answer

14, 13, 12, 8, 10

Status : Correct

Marks : 1/1

4. In a Min Heap, what is the condition that must be satisfied between a parent and its children?

Answer

The parent must have a lower value than its children

Status : Correct

Marks : 1/1

5. When inserting an element with a value of 8 into a Min Heap, where would it be placed if the current heap contains the elements [10, 12, 15, 20, 17]?

Answer

As the root element

Status : Correct

Marks : 1/1

6. If you're inserting an element with a value of 5 into a Min Heap, and the current heap contains the elements [11, 16, 10, 13, 18], what will the Min Heap look like after the insertion?

Answer

5, 11, 10, 13, 16, 18

Status : Correct

Marks : 1/1

7. When deleting the root element from a Min Heap, which of the following replaces it as the new root?

Answer

the last element from the tree

Status : Correct

Marks : 1/1

8. Tries are mainly used for:

Answer

All of the mentioned options

Status : Correct

Marks : 1/1

9. In a trie, how is the end of a valid word usually represented?

Answer

By using an end-of-word (terminal) marker at a node

Status : Correct

Marks : 1/1

10. Which of the following is true about tries?

Answer

All of the mentioned options

Status : Correct

Marks : 1/1

11. What is stored in a segment tree node for a Range Minimum Query (RMQ)?

Answer

Minimum element in range

Status : Correct

Marks : 1/1

12. In a Range Max Query tree, what is the internal node value?

Answer

Maximum of child values

Status : Correct

Marks : 1/1

13. Which segment tree operation is used for range max queries?

Answer

max() merge

Status : Correct

Marks : 1/1

14. What data structure is often used in segment tree nodes to count distinct elements?

Answer

Set

Status : Correct

Marks : 1/1

15. In range distinct count queries, what does each segment tree node contain?

Answer

Sorted array

Status : Wrong

Marks : 0/1

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2028_NeoColab_Competitive Coding II_L9

VTU_Week 10_COD

Attempt : 1

Total Mark : 50

Marks Obtained : 50

Section 1 : Coding

1. Problem Statement

Jovika is learning about heaps and wants to implement a program that constructs a max heap and inserts elements into it.

She wants to write a program that implements max heap insertion. The program defines functions to swap elements and perform maximum heapification. She wants to extend this program to take a list of integers, build a max heap, and insert a new element into the heap.

Help her in implementing the program.

Input Format

The first line of input consists of an integer N, representing the number of elements to be inserted into the max heap.

The second line consists of N space-separated integers, representing the elements to be inserted into the max heap.

The third line consists of an integer representing the new element to be inserted into the max heap.

Output Format

The output prints the space-separated integers, representing the elements in the max heap after inserting the new element in the order of insertion.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 5

3 1 4 2 5

6

Output: 6 3 5 2 1 4

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
void swap(int *a, int *b) {  
    int temp = *a;  
    *a = *b;  
    *b = temp;  
}
```

```
void maxHeapify(int heap[], int n, int i) {  
    int largest = i;  
    int left = 2 * i + 1;  
    int right = 2 * i + 2;
```

```
    if (left < n && heap[left] > heap[largest])  
        largest = left;
```

```
    if (right < n && heap[right] > heap[largest])  
        largest = right;
```

```

        if (largest != i) {
            swap(&heap[i], &heap[largest]);
            maxHeapify(heap, n, largest);
        }
    }
}

```

```

void buildMaxHeap(int heap[], int n) {
    for (int i = n / 2 - 1; i >= 0; i--) {
        maxHeapify(heap, n, i);
    }
}

```

```

void insertIntoMaxHeap(int heap[], int *n, int newElement) {
    // Add the new element at the end
    heap[*n] = newElement;
    (*n)++;

    // Heapify up (bubble up the new element)
    int i = *n - 1;
    while (i > 0 && heap[(i - 1) / 2] < heap[i]) {
        swap(&heap[(i - 1) / 2], &heap[i]);
        i = (i - 1) / 2;
    }
}

```

```

int main() {
    int N;
    scanf("%d", &N);

    int heap[20]; // Array to store heap elements
    for (int i = 0; i < N; i++) {
        scanf("%d", &heap[i]);
    }

    int newElement;
    scanf("%d", &newElement);

    // Build max heap from the given elements
    buildMaxHeap(heap, N);

    // Insert the new element

```

```
insertIntoMaxHeap(heap, &N, newElement);  
// Print the heap elements  
for (int i = 0; i < N; i++) {  
    printf("%d ", heap[i]);  
}  
printf("\n");  
  
return 0;  
}
```

Status : Correct

Marks : 10/10

2. Problem Statement

Bob is fascinated by numbers and loves arranging them in various ways. His recent exploration led him to Max Heaps.

Bob likes odd numbers more than even numbers. After inserting elements into a Max Heap, he wants to remove all even numbers from it while keeping the Max Heap property intact.

Your task is to help Bob build a Max Heap from the given elements, display it, delete all the even numbers from the resultant array and then rearrange the elements in the form of Max Heap.

Input Format

The first line of input consists of an integer N, representing the number of elements initially to be inserted into the Max Heap.

The second line consists of N space-separated integers, representing the elements to be inserted into the heap.

Output Format

The first line of output prints the space-separated elements of the Max Heap after building it from the given N elements.

The second line prints the space-separated elements of the Max Heap after removing even elements.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 3

3 2 1

Output: 3 2 1

3 1

Answer

// You are using GCC

#include <stdio.h>

```
void swap(int *a, int *b) {  
    int temp = *a;  
    *a = *b;  
    *b = temp;  
}
```

```
void maxHeapify(int heap[], int n, int i) {  
    int largest = i;  
    int left = 2 * i + 1;  
    int right = 2 * i + 2;
```

```
    if (left < n && heap[left] > heap[largest])  
        largest = left;
```

```
    if (right < n && heap[right] > heap[largest])  
        largest = right;
```

```
    if (largest != i) {  
        swap(&heap[i], &heap[largest]);  
        maxHeapify(heap, n, largest);  
    }
```

```
}
```

```
void buildMaxHeap(int heap[], int n) {  
    for (int i = n / 2 - 1; i >= 0; i--) {  
        maxHeapify(heap, n, i);
```



```

    }
}

void removeEvenNumbers(int heap[], int *n) {
    int temp[20];
    int newSize = 0;

    // Collect all odd numbers
    for (int i = 0; i < *n; i++) {
        if (heap[i] % 2 != 0) {
            temp[newSize++] = heap[i];
        }
    }

    // Copy back to heap array
    for (int i = 0; i < newSize; i++) {
        heap[i] = temp[i];
    }
    *n = newSize;

    // Rebuild max heap
    buildMaxHeap(heap, *n);
}

```

```

int main() {
    int N;
    scanf("%d", &N);

    int heap[20];
    for (int i = 0; i < N; i++) {
        scanf("%d", &heap[i]);
    }

    // Build max heap
    buildMaxHeap(heap, N);

    // Print the max heap
    for (int i = 0; i < N; i++) {
        printf("%d ", heap[i]);
    }
    printf("\n");
}

```

```

// Remove even numbers and rebuild heap
removeEvenNumbers(heap, &N);

// Print the heap after removing even numbers
for (int i = 0; i < N; i++) {
    printf("%d ", heap[i]);
}
printf("\n");

return 0;
}

```

Status : Correct

Marks : 10/10

3. Problem Statement

Iniya is studying data structures and algorithms, and she is currently learning about heaps.

She wants to practice implementing Min-Heap and perform a deletion operation on it. A min-heap is a binary tree in which the value of each node is less than or equal to the values of its children.

Write a program to assist Iniya in building a min-heap and delete the root element.

Input Format

The first line of input consists of an integer N, representing the number of elements in the initial Min-Heap.

The second line consists of N space-separated integers, where each integer represents an element in the Min-Heap.

Output Format

The output prints the space-separated elements of the Min-Heap after performing the deletion operation of the root.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 7

80 40 30 10 50 33 66

Output: 30 40 33 80 50 66

Answer

// You are using GCC

#include <stdio.h>

```
void swap(int *a, int *b) {  
    int temp = *a;  
    *a = *b;  
    *b = temp;  
}
```

```
void minHeapify(int heap[], int n, int i) {  
    int smallest = i;  
    int left = 2 * i + 1;  
    int right = 2 * i + 2;  
  
    if (left < n && heap[left] < heap[smallest])  
        smallest = left;  
  
    if (right < n && heap[right] < heap[smallest])  
        smallest = right;  
  
    if (smallest != i) {  
        swap(&heap[i], &heap[smallest]);  
        minHeapify(heap, n, smallest);  
    }  
}
```

```
void buildMinHeap(int heap[], int n) {  
    for (int i = n / 2 - 1; i >= 0; i--) {  
        minHeapify(heap, n, i);  
    }  
}
```

```
void deleteRoot(int heap[], int *n) {  
    if (*n <= 0) return;
```

```

// Replace root with last element
heap[0] = heap[*n - 1];
(*n)--;

// Heapify the root
minHeapify(heap, *n, 0);
}

int main() {
    int N;
    scanf("%d", &N);

    int heap[20];
    for (int i = 0; i < N; i++) {
        scanf("%d", &heap[i]);
    }

    // Build min heap
    buildMinHeap(heap, N);

    // Delete the root
    deleteRoot(heap, &N);

    // Print the heap after deletion
    for (int i = 0; i < N; i++) {
        printf("%d ", heap[i]);
    }
    printf("\n");

    return 0;
}

```

Status : Correct

Marks : 10/10

4. Problem Statement

A trie (pronounced as "try") or prefix tree is a tree data structure used to store and retrieve keys in a dataset of strings efficiently. This data structure has various applications, such as autocomplete and spellchecker.

Implement the Trie class:

Trie() Initializes the trie object.
void insert(String word): Inserts the string word into the trie.
boolean search(String word) Returns true if the string word is in the trie (i.e., was inserted before), and false otherwise.
boolean startsWith(String prefix) Returns true if a previously inserted string word has the prefix, and false otherwise.

Example

Input:

4

apple

cat

orange

mat

cat

ora

Output:

The word is present in the trie.

There is a word with the given prefix in the trie.

Input Format

The first line of the input consists of the number of words to insert, N.

The next N lines of input consist of the words in each line.

The next line of input consists of the word to search.

The last line of the input consists of the prefix to search.

Output Format

When searching for a word using the search method, the program will output one of the following messages:

- If the word is present in the trie, the output prints "The word is present in the trie."
- If the word is not present in the trie, the output prints "The word is not present in the trie."

When checking for a prefix using the startsWith method, the program will output one of the following messages:

- If there is a word in the trie with the given prefix, the output prints "There is a word with the given prefix in the trie."
- If there is no word in the trie with the given prefix, the output prints "There is no word with the given prefix in the trie."

Sample Test Case

Input: 4

apple

cat

orange

mat

catlyn

ora

Output: The word is not present in the trie.

There is a word with the given prefix in the trie.

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <string.h>
```

```
#include <stdbool.h>
```

```
#define ALPHABET_SIZE 26
```

```
// Trie node structure
```

```
struct TrieNode {
```

```
    struct TrieNode* children[ALPHABET_SIZE];
```

```
    bool isEndOfWord;
```

```
};
```

```

// Create a new trie node
struct TrieNode* createNode() {
    struct TrieNode* node = (struct TrieNode*)malloc(sizeof(struct TrieNode));
    node->isEndOfWord = false;
    for (int i = 0; i < ALPHABET_SIZE; i++) {
        node->children[i] = NULL;
    }
    return node;
}

```

```

// Insert a word into the trie
void insert(struct TrieNode* root, const char* word) {
    struct TrieNode* current = root;
    int len = strlen(word);

    for (int i = 0; i < len; i++) {
        int index = word[i] - 'a';
        if (current->children[index] == NULL) {
            current->children[index] = createNode();
        }
        current = current->children[index];
    }
    current->isEndOfWord = true;
}

```

```

// Search for a word in the trie
bool search(struct TrieNode* root, const char* word) {
    struct TrieNode* current = root;
    int len = strlen(word);

    for (int i = 0; i < len; i++) {
        int index = word[i] - 'a';
        if (current->children[index] == NULL) {
            return false;
        }
        current = current->children[index];
    }
    return current->isEndOfWord;
}

```

```

// Check if there is any word with the given prefix

```

```

bool startsWith(struct TrieNode* root, const char* prefix) {
    struct TrieNode* current = root;
    int len = strlen(prefix);

    for (int i = 0; i < len; i++) {
        int index = prefix[i] - 'a';
        if (current->children[index] == NULL) {
            return false;
        }
        current = current->children[index];
    }
    return true;
}

```

```

int main() {
    struct TrieNode* root = createNode();

    int N;
    scanf("%d", &N);

    // Read the words to insert
    char words[N][100];
    for (int i = 0; i < N; i++) {
        scanf("%s", words[i]);
        insert(root, words[i]);
    }

    // Read the word to search
    char searchWord[100];
    scanf("%s", searchWord);

    // Read the prefix to search
    char prefix[100];
    scanf("%s", prefix);

    // Search for the word
    if (search(root, searchWord)) {
        printf("The word is present in the trie.\n");
    } else {
        printf("The word is not present in the trie.\n");
    }
}

```



```

// Check for prefix
if (startsWith(root, prefix)) {
    printf("There is a word with the given prefix in the trie.\n");
} else {
    printf("There is no word with the given prefix in the trie.\n");
}

return 0;
}

```

Status : Correct

Marks : 10/10

5. Problem Statement

Given an array of integers representing the combat power of N players.

You need to efficiently perform the following operations:

Query (Q L R): Find the maximum combat power between ranks L and R (inclusive). Update (U i val): Update the power of the player at index i to a new value val.

To ensure efficiency, implement these operations using a Segment Tree.

Input Format

The first line contains two space-separated integers, N and Q, representing the number of players and the number of operations.

The second line contains N space-separated integers representing the initial combat power scores of all players, where index 0 is the top-ranked player.

The next Q lines describe the operations:

- "Q L R" Query represents the maximum combat power from ranks L to R (inclusive).
- "U i val" Updates the power score of the player at rank i to val.

All indices are 0-based.

Output Format

For each query Q L R, print the maximum combat power in that range on a new line.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 6 5

4 7 2 9 5 1

Q 1 4

U 3 6

Q 1 4

U 0 10

Q 0 2

Output: 9

7

10

Answer

// You are using GCC

#include <stdio.h>

int segmentTree[100]; // Segment tree array (size 4*N is sufficient for N<=20)

// Function to build the segment tree

void build(int arr[], int node, int start, int end) {

if (start == end) {

segmentTree[node] = arr[start];

return;

}

int mid = (start + end) / 2;

build(arr, 2 * node + 1, start, mid);

build(arr, 2 * node + 2, mid + 1, end);

// For max query

segmentTree[node] = (segmentTree[2 * node + 1] > segmentTree[2 * node + 2]) ?

segmentTree[2 * node + 1] : segmentTree[2 * node + 2];

}

// Function to update a value at index i

```

void update(int node, int start, int end, int idx, int val) {
    if (start == end) {
        segmentTree[node] = val;
        return;
    }

    int mid = (start + end) / 2;
    if (idx <= mid) {
        update(2 * node + 1, start, mid, idx, val);
    } else {
        update(2 * node + 2, mid + 1, end, idx, val);
    }

    segmentTree[node] = (segmentTree[2 * node + 1] > segmentTree[2 * node +
2]) ?
        segmentTree[2 * node + 1] : segmentTree[2 * node + 2];
}

```

```

// Function to query maximum in range [l, r]
int query(int node, int start, int end, int l, int r) {
    if (r < start || l > end) {
        return -1; // Return -1 for out of range (since values are positive)
    }

    if (l <= start && end <= r) {
        return segmentTree[node];
    }

    int mid = (start + end) / 2;
    int leftMax = query(2 * node + 1, start, mid, l, r);
    int rightMax = query(2 * node + 2, mid + 1, end, l, r);

    if (leftMax == -1) return rightMax;
    if (rightMax == -1) return leftMax;
    return (leftMax > rightMax) ? leftMax : rightMax;
}

```

```

int main() {
    int N, Q;
    scanf("%d %d", &N, &Q);

    int arr[N];

```

```

for (int i = 0; i < N; i++) {
    scanf("%d", &arr[i]);
}

// Build segment tree
build(arr, 0, 0, N - 1);

// Process operations
for (int i = 0; i < Q; i++) {
    char op;
    scanf(" %c", &op);

    if (op == 'Q') {
        int L, R;
        scanf("%d %d", &L, &R);
        int maxVal = query(0, 0, N - 1, L, R);
        printf("%d\n", maxVal);
    }
    else if (op == 'U') {
        int idx, val;
        scanf("%d %d", &idx, &val);
        update(0, 0, N - 1, idx, val);
    }
}

return 0;
}

```

Status : Correct

Marks : 10/10

Vel Tech Group of Institutions

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Batch: 2028

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2028_NeoColab_Competitive Coding II_L9

VTU_Week 10_CY

Attempt : 1

Total Mark : 40

Marks Obtained : 40

Section 1 : Coding

1. Problem Statement

Sai has been studying various algorithms related to heaps. Recently, he came across a problem where he needed to filter elements from a given array based on certain criteria related to heap properties.

He wants to build a max heap from a given array of integers, delete elements that fall outside the specified range, and then rearrange the elements in the form of a Max Heap.

Write a program to help Sai in implementing the program.

Input Format

The first line of input consists of an integer N, representing the number of elements in the array.

The second line consists of N space-separated integers, representing the elements of the array.

The third line consists of two integers, a and b, separated by a space, representing the range to filter the elements.

Output Format

The output prints the space-separated integers, representing the elements that remain in the max heap after removing elements outside the given range.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 5
5 7 8 3 12
1 6

Output: 5 3

Answer

```
// You are using GCC
#include <stdio.h>
```

```
void swap(int *a, int *b) {
    int temp = *a;
    *a = *b;
    *b = temp;
}
```

```
void maxHeapify(int heap[], int n, int i) {
    int largest = i;
    int left = 2 * i + 1;
    int right = 2 * i + 2;
```

```
    if (left < n && heap[left] > heap[largest])
        largest = left;
```

```
    if (right < n && heap[right] > heap[largest])
        largest = right;
```

```

    if (largest != i) {
        swap(&heap[i], &heap[largest]);
        maxHeapify(heap, n, largest);
    }
}

```

```

void buildMaxHeap(int heap[], int n) {
    for (int i = n / 2 - 1; i >= 0; i--) {
        maxHeapify(heap, n, i);
    }
}

```

```

void filterAndRearrange(int arr[], int *n, int a, int b) {
    int temp[20];
    int newSize = 0;

    // Filter elements within range [a, b]
    for (int i = 0; i < *n; i++) {
        if (arr[i] >= a && arr[i] <= b) {
            temp[newSize++] = arr[i];
        }
    }

    // Copy back to original array
    for (int i = 0; i < newSize; i++) {
        arr[i] = temp[i];
    }
    *n = newSize;

    // Rebuild max heap
    buildMaxHeap(arr, *n);
}

```

```

int main() {
    int N;
    scanf("%d", &N);

    int arr[20];
    for (int i = 0; i < N; i++) {
        scanf("%d", &arr[i]);
    }
}

```

```

int a, b;
scanf("%d %d", &a, &b);

// Build max heap
buildMaxHeap(arr, N);

// Filter elements and rebuild heap
filterAndRearrange(arr, &N, a, b);

// Print the result
for (int i = 0; i < N; i++) {
    printf("%d ", arr[i]);
}
printf("\n");

return 0;
}

```

Status : Correct

Marks : 10/10

2. Problem Statement

Maxwell, an enthusiastic computer science student, is exploring the fascinating world of data structures. His task is to create a program that constructs a min heap from a list of integers and identifies negative numbers within it.

Help Maxwell in completing the program.

Input Format

The first line of input consists of an integer N, representing the number of integers in the array.

The second line consists of N space-separated integers, representing the elements of the list.

Output Format

The first line of output prints the min-heap as a space-separated list of integers.

The second line prints the negative numbers, separated by a space.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 7

15 -5 20 -19 23 42 29

Output: -19 -5 20 15 23 42 29

-19 -5

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
void swap(int *a, int *b) {  
    int temp = *a;  
    *a = *b;  
    *b = temp;  
}
```

```
void minHeapify(int heap[], int n, int i) {  
    int smallest = i;  
    int left = 2 * i + 1;  
    int right = 2 * i + 2;
```

```
    if (left < n && heap[left] < heap[smallest])  
        smallest = left;
```

```
    if (right < n && heap[right] < heap[smallest])  
        smallest = right;
```

```
    if (smallest != i) {  
        swap(&heap[i], &heap[smallest]);  
        minHeapify(heap, n, smallest);  
    }
```

```
}
```

```
void buildMinHeap(int heap[], int n) {  
    for (int i = n / 2 - 1; i >= 0; i--) {
```

```

        minHeapify(heap, n, i);
    }
}

```

```

int main() {
    int N;
    scanf("%d", &N);

    int arr[20];
    for (int i = 0; i < N; i++) {
        scanf("%d", &arr[i]);
    }

    // Build min heap
    buildMinHeap(arr, N);

    // Print the min heap
    for (int i = 0; i < N; i++) {
        printf("%d ", arr[i]);
    }
    printf("\n");

    // Print negative numbers
    for (int i = 0; i < N; i++) {
        if (arr[i] < 0) {
            printf("%d ", arr[i]);
        }
    }
    printf("\n");

    return 0;
}

```

Status : Correct

Marks : 10/10

3. Problem Statement

You are given an array of n integers. You need to perform q queries of two types:

Q | r – Find the minimum value in the subarray from index l to r

(inclusive). U idx val — Update the element at index idx to val.

Implement an efficient solution using a Segment Tree to handle both query and update operations.

Input Format

The first line contains two space-separated integers n and q, representing the size of the array and the number of queries

The second line contains n space-separated integers representing the elements of the array.

Next q lines: Each line contains a query in one of the following formats:

- Q l r
- U idx val

Output Format

For each query of type Q, print the minimum value in the specified range on a new line.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 3 2

4 2 5

Q 0 2

Q 1 1

Output: 2

2

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
int segmentTree[100]; // Segment tree array (size 4*n is sufficient for n<=20)
```

```
// Function to build the segment tree for minimum queries
```

```

void build(int arr[], int node, int start, int end) {
    if (start == end) {
        segmentTree[node] = arr[start];
        return;
    }

    int mid = (start + end) / 2;
    build(arr, 2 * node + 1, start, mid);
    build(arr, 2 * node + 2, mid + 1, end);

    // For min query
    segmentTree[node] = (segmentTree[2 * node + 1] < segmentTree[2 * node +
2]) ?
        segmentTree[2 * node + 1] : segmentTree[2 * node + 2];
}

// Function to update a value at index idx
void update(int node, int start, int end, int idx, int val) {
    if (start == end) {
        segmentTree[node] = val;
        return;
    }

    int mid = (start + end) / 2;
    if (idx <= mid) {
        update(2 * node + 1, start, mid, idx, val);
    } else {
        update(2 * node + 2, mid + 1, end, idx, val);
    }

    segmentTree[node] = (segmentTree[2 * node + 1] < segmentTree[2 * node +
2]) ?
        segmentTree[2 * node + 1] : segmentTree[2 * node + 2];
}

// Function to query minimum in range [l, r]
int query(int node, int start, int end, int l, int r) {
    if (r < start || l > end) {
        return 1000000; // Return a large number for out of range
    }

    if (l <= start && end <= r) {

```

```

        return segmentTree[node];
    }

    int mid = (start + end) / 2;
    int leftMin = query(2 * node + 1, start, mid, l, r);
    int rightMin = query(2 * node + 2, mid + 1, end, l, r);

    return (leftMin < rightMin) ? leftMin : rightMin;
}

```

```

int main() {
    int n, q;
    scanf("%d %d", &n, &q);

    int arr[20];
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    // Build segment tree
    build(arr, 0, 0, n - 1);

    // Process queries
    for (int i = 0; i < q; i++) {
        char op;
        scanf(" %c", &op);

        if (op == 'Q') {
            int l, r;
            scanf("%d %d", &l, &r);
            int minVal = query(0, 0, n - 1, l, r);
            printf("%d\n", minVal);
        }
        else if (op == 'U') {
            int idx, val;
            scanf("%d %d", &idx, &val);
            update(0, 0, n - 1, idx, val);
        }
    }

    return 0;
}

```

Status : Correct

Marks : 10/10

4. Problem Statement

Meera is working on a text processing task where she wants to identify the longest common prefix among a given set of words. She decides to use a Trie structure to efficiently compute this prefix.

Write a program to help Meera insert the given words into a Trie and then determine the longest common prefix among them.

Input Format

The first line contains an integer n, representing the number of words.

The next n lines each contain a single word.

Output Format

The output prints "The longest common prefix is: <prefix>" if a common prefix exists.

Otherwise, print "There is no common prefix"

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 4
aeroplane
aerospace
aerial
aesthetic

Output: The longest common prefix is: ae

Answer

```
// You are using GCC
#include <stdio.h>
#include <stdlib.h>
```

```

#include <string.h>
#include <stdbool.h>

#define ALPHABET_SIZE 26

// Trie node structure
struct TrieNode {
    struct TrieNode* children[ALPHABET_SIZE];
    int childCount; // Count of non-null children
    bool isEndOfWord;
};

// Create a new trie node
struct TrieNode* createNode() {
    struct TrieNode* node = (struct TrieNode*)malloc(sizeof(struct TrieNode));
    node->childCount = 0;
    node->isEndOfWord = false;
    for (int i = 0; i < ALPHABET_SIZE; i++) {
        node->children[i] = NULL;
    }
    return node;
}

// Insert a word into the trie
void insert(struct TrieNode* root, const char* word) {
    struct TrieNode* current = root;
    int len = strlen(word);
    for (int i = 0; i < len; i++) {
        int index = word[i] - 'a';
        if (current->children[index] == NULL) {
            current->children[index] = createNode();
            current->childCount++;
        }
        current = current->children[index];
    }
    current->isEndOfWord = true;
}

// Find the longest common prefix
void findLongestCommonPrefix(struct TrieNode* root, char* prefix, int n) {
    struct TrieNode* current = root;

```

```

int index = 0;

while (current->childCount == 1 && !current->isEndOfWord) {
    // Find the single child
    for (int i = 0; i < ALPHABET_SIZE; i++) {
        if (current->children[i] != NULL) {
            prefix[index++] = 'a' + i;
            current = current->children[i];
            break;
        }
    }
}
prefix[index] = '\0';
}

int main() {
    struct TrieNode* root = createNode();

    int n;
    scanf("%d", &n);

    char words[10][101];
    for (int i = 0; i < n; i++) {
        scanf("%s", words[i]);
        insert(root, words[i]);
    }

    char prefix[101];
    findLongestCommonPrefix(root, prefix, n);

    if (strlen(prefix) > 0) {
        printf("The longest common prefix is: %s\n", prefix);
    } else {
        printf("There is no common prefix\n");
    }

    return 0;
}

```

Status : Correct

Marks : 10/10

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2028_NeoColab_Competitive Coding II_L9

VTU_Week 10_PAH

Attempt : 1

Total Mark : 40

Marks Obtained : 40

Section 1 : Coding

1. Problem Statement

Aarav, a computer science student, is working on a heap-based data processing system. He needs to find the second-largest element from a collection of integers represented as a max-heap. Aarav first constructs the max-heap, then removes the largest element to extract the next largest value.

Write a program to help Aarav determine and print the second largest element, or display a message if it does not exist.

Input Format

The first line contains an integer n, representing the number of elements in the heap.

The second line contains n space-separated integers representing the elements of the heap.

Output Format

The output prints the second-largest element in the heap.

If there is no second-largest element, print the message:

"There is no second largest element in the heap."

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 4
100 70 10 30

Output: 70

Answer

```
// You are using GCC
#include <stdio.h>
```

```
void swap(int *a, int *b) {
    int temp = *a;
    *a = *b;
    *b = temp;
}
```

```
void maxHeapify(int heap[], int n, int i) {
    int largest = i;
    int left = 2 * i + 1;
    int right = 2 * i + 2;

    if (left < n && heap[left] > heap[largest])
        largest = left;

    if (right < n && heap[right] > heap[largest])
        largest = right;

    if (largest != i) {
```

```
        swap(&heap[i], &heap[largest]);
        maxHeapify(heap, n, largest);
    }
}
```

```
void buildMaxHeap(int heap[], int n) {
    for (int i = n / 2 - 1; i >= 0; i--) {
        maxHeapify(heap, n, i);
    }
}
```

```
int extractMax(int heap[], int *n) {
    if (*n <= 0) return -1;

    int max = heap[0];
    heap[0] = heap[*n - 1];
    (*n)--;
    maxHeapify(heap, *n, 0);

    return max;
}
```

```
int main() {
    int n;
    scanf("%d", &n);
```

```
    int heap[20];
    for (int i = 0; i < n; i++) {
        scanf("%d", &heap[i]);
    }
```

```
    // Build max heap
    buildMaxHeap(heap, n);
```

```
    // Extract the largest element
    int largest = extractMax(heap, &n);
```

```
    // If there are more elements, extract the second largest
    if (n > 0) {
        int secondLargest = extractMax(heap, &n);
        printf("%d\n", secondLargest);
    } else {
```

```
        printf("There is no second largest element in the heap.\n");
    }

    return 0;
}
```

Status : Correct

Marks : 10/10

2. Problem Statement

Elijah wants to verify if an array represents a valid min-heap. Implement a program to check whether a given array satisfies the min-heap property.

In a min-heap, for every node i , the value at that node must be less than or equal to the values of its child nodes (if they exist). Write a program to read an array and determine whether it maintains the min-heap structure.

Input Format

The first line contains an integer n , representing the number of elements.

The next line contains n space-separated integers representing the elements to be inserted into the min heap.

Output Format

The first line prints "The array represents a valid min-heap" if the array satisfies the min-heap property. Otherwise, print "The array does not represent a valid min-heap".

The second line prints the space-separated elements of the array as given in the input.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 3
1 2 3

Output: The array represents a valid min-heap
1 2 3

Answer

```
// You are using GCC
#include <stdio.h>
#include <stdbool.h>

// Function to check if array maintains min-heap property
bool isValidMinHeap(int arr[], int n) {
    for (int i = 0; i < n; i++) {
        int left = 2 * i + 1;
        int right = 2 * i + 2;

        // Check left child
        if (left < n && arr[i] > arr[left]) {
            return false;
        }

        // Check right child
        if (right < n && arr[i] > arr[right]) {
            return false;
        }
    }
    return true;
}

int main() {
    int n;
    scanf("%d", &n);

    int arr[10];
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    // Check if it's a valid min-heap
    if (isValidMinHeap(arr, n)) {
        printf("The array represents a valid min-heap\n");
    } else {
        printf("The array does not represent a valid min-heap\n");
    }
}
```

```
// Print the array
for (int i = 0; i < n; i++) {
    printf("%d ", arr[i]);
}
printf("\n");

return 0;
}
```

Status : Correct

Marks : 10/10

3. Problem Statement

Given a list of daily transaction amounts over N days. Each transaction can be positive (deposit) or negative (withdrawal).

You must efficiently:

Find the sum of transactions between two given days. Update the transaction amount for any specific day.

To perform these operations efficiently, implement a Segment Tree.

Input Format

The first line contains two integers, N and Q, representing the number of days and number of operations.

The second line contains N space-separated integers representing the transaction amount for each day.

The next Q lines contain operations in one of the following formats:

- S L R represents Query sum from index L to R
- U i val Updates index i with new value val

All indices are 0-based.

Output Format

For each query S L R, print the sum of transactions in that range on a new line.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 6 5
10 -5 20 15 -10 5
S 0 2
U 1 7
S 0 2
S 3 5
S 0 5

Output: 25
37
10
47

Answer

```
// You are using GCC
#include <stdio.h>
```

```
int segmentTree[100]; // Segment tree array (size 4*n is sufficient for n<=20)
```

```
// Function to build the segment tree for sum queries
```

```
void build(int arr[], int node, int start, int end) {
    if (start == end) {
        segmentTree[node] = arr[start];
        return;
    }
```

```
    int mid = (start + end) / 2;
    build(arr, 2 * node + 1, start, mid);
    build(arr, 2 * node + 2, mid + 1, end);
```

```
// For sum query
```

```
segmentTree[node] = segmentTree[2 * node + 1] + segmentTree[2 * node + 2];
}
```

```
// Function to update a value at index idx
```

```
void update(int node, int start, int end, int idx, int val) {
    if (start == end) {
```

```

        segmentTree[node] = val;
        return;
    }

    int mid = (start + end) / 2;
    if (idx <= mid) {
        update(2 * node + 1, start, mid, idx, val);
    } else {
        update(2 * node + 2, mid + 1, end, idx, val);
    }

    segmentTree[node] = segmentTree[2 * node + 1] + segmentTree[2 * node + 2];
}

```

```

// Function to query sum in range [l, r]
int query(int node, int start, int end, int l, int r) {
    if (r < start || l > end) {
        return 0; // Return 0 for out of range
    }

```

```

    if (l <= start && end <= r) {
        return segmentTree[node];
    }

```

```

    int mid = (start + end) / 2;
    int leftSum = query(2 * node + 1, start, mid, l, r);
    int rightSum = query(2 * node + 2, mid + 1, end, l, r);

    return leftSum + rightSum;
}

```

```

int main() {
    int N, Q;
    scanf("%d %d", &N, &Q);

```

```

    int arr[20];
    for (int i = 0; i < N; i++) {
        scanf("%d", &arr[i]);
    }

```

```

    // Build segment tree
    build(arr, 0, 0, N - 1);

```



```

// Process operations
for (int i = 0; i < Q; i++) {
    char op;
    scanf(" %c", &op);

    if (op == 'S') {
        int L, R;
        scanf("%d %d", &L, &R);
        int sum = query(0, 0, N - 1, L, R);
        printf("%d\n", sum);
    }
    else if (op == 'U') {
        int idx, val;
        scanf("%d %d", &idx, &val);
        update(0, 0, N - 1, idx, val);
    }
}

return 0;
}

```

Status : Correct

Marks : 10/10

4. Problem Statement

Given a string *s* and a string array dictionary, return the longest string in the dictionary that can be formed by deleting some of the given string characters.

If there is more than one possible result, return the longest word with the smallest lexicographical order.

Example

Input:

s = "abpcplea"

dictionary = ["ale", "apple", "monkey", "plea"]

Output:

"apple"

Input Format

The first line of input consists of the string.

The second line of input consists of the array size.

The following lines of input consist of the array elements, each in a separate line.

Output Format

The output prints the longest string in the dictionary that can be formed by deleting some of the given string characters.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: abpcplea

4

ale

apple

monkey

plea

Output: apple

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
#include <string.h>
```

```
#include <stdbool.h>
```

```
// Function to check if a word can be formed from s by deleting characters
```

```
bool isSubsequence(const char *s, const char *word) {
```

```
    int i = 0, j = 0;
```

```
    int s_len = strlen(s);
```

```
    int word_len = strlen(word);
```

```

    while (i < s_len && j < word_len) {
        if (s[i] == word[j]) {
            j++;
        }
        i++;
    }

    return j == word_len;
}

```

```

int main() {
    char s[1001];
    scanf("%s", s);

    int n;
    scanf("%d", &n);

    char dictionary[1000][1001];
    for (int i = 0; i < n; i++) {
        scanf("%s", dictionary[i]);
    }

    char result[1001] = "";
    int maxLen = 0;

    for (int i = 0; i < n; i++) {
        if (isSubsequence(s, dictionary[i])) {
            int currentLen = strlen(dictionary[i]);

            if (currentLen > maxLen) {
                maxLen = currentLen;
                strcpy(result, dictionary[i]);
            }
            else if (currentLen == maxLen) {
                // If same length, choose lexicographically smaller
                if (strcmp(dictionary[i], result) < 0) {
                    strcpy(result, dictionary[i]);
                }
            }
        }
    }
}

```

```
    printf("%s\n", result);  
    return 0;  
}
```

Status : Correct

Marks : 10/10

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2028_NeoColab_Competitive Coding II_L9

VTU_Week 11_MCQ

Attempt : 1

Total Mark : 15

Marks Obtained : 15

Section 1 : MCQ

1. Consider the following two sequences:

$X = \langle B, C, D, C, A, B, C \rangle$ $Y = \langle C, A, D, B, C, B \rangle$

The length of longest common subsequence of X and Y is:

Answer

4

Status : Correct

Marks : 1/1

2. Which of the following problems should be solved using dynamic programming?

Answer

Longest common subsequence

Status : Correct

Marks : 1/1

3. Which of the following problems is closely related to the Subsets Sum problem and can also be solved using dynamic programming?

Answer

Coin Change Problem

Status : Correct

Marks : 1/1

4. Consider two strings $A = \text{"qpqr"}$ and $B = \text{"pqprqp"}$. Let x be the length of the longest common subsequence (not necessarily contiguous) between A and B and let y be the number of such longest common subsequences between A and B .

Then $x + 10y = \underline{\hspace{2cm}}$.

Answer

34

Status : Correct

Marks : 1/1

5. Consider the strings "PQRSTPQRS" and "PRATPBRQRPS" . What is the length of the longest common subsequence?

Answer

7

Status : Correct

Marks : 1/1

6. Which of the following is/are property/properties of a dynamic programming problem?

Answer

Both optimal substructure and overlapping subproblems

Status : Correct

Marks : 1/1

7. If an optimal solution can be created for a problem by constructing optimal solutions for its subproblems, the problem possesses _____ property.

Answer

Optimal substructure

Status : Correct

Marks : 1/1

8. In dynamic programming, the technique of storing the previously calculated values is called _____.

Answer

Memoization

Status : Correct

Marks : 1/1

9. Which of the following is an example of a problem that can be solved using memoization in dynamic programming?

Answer

Computing the edit distance between two strings

Status : Correct

Marks : 1/1

10. When a top-down approach to dynamic programming is applied to a problem, it usually _____.

Answer

decreases the time complexity and increases the space complexity

Status : Correct

Marks : 1/1

11. Which of the following problems is NOT solved using dynamic programming?

Answer

Fractional knapsack problem

Status : Correct

Marks : 1/1

12. Which of the following is NOT a characteristic of dynamic programming?

Answer

Solving problems in a sequential manner.

Status : Correct

Marks : 1/1

13. What happens when a top-down approach of dynamic programming is applied to any problem?

Answer

It increases the space complexity and decreases the time complexity.

Status : Correct

Marks : 1/1

14. The Fibonacci sequence is often used to illustrate dynamic programming concepts. What is the time complexity of a naive recursive implementation of Fibonacci numbers?

Answer

$O(2^n)$

Status : Correct

Marks : 1/1

15. What are the different techniques to solve dynamic programming problems:

Answer

Memoization and Bottom-Up

Status : Correct

Marks : 1/1

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2028_NeoColab_Competitive Coding II_L9

VTU_Week 11_COD

Attempt : 1

Total Mark : 50

Marks Obtained : 50

Section 1 : Coding

1. Problem Statement

Given an integer n , representing the total number of stairs, the task is to determine the number of distinct ways to reach the top when at each move you can climb 1, 2, or 3 steps.

Write a program using the Dynamic Programming (Bottom-Up Approach) algorithm to ensure that the total number of possible ways to climb the stairs is correctly calculated and displayed.

Examples:

Input: 4

Output: 7

Explanation: There are seven ways: {1, 1, 1, 1}, {1, 2, 1}, {2, 1, 1}, {1, 1, 2}, {2,

2}, {3, 1}, {1, 3}.

Input: 3

Output: 4

Explanation: There are four ways: {1, 1, 1}, {1, 2}, {2, 1}, {3}.

Input Format

The input contains a single integer n, representing the total number of stairs.

Output Format

The output prints a single integer representing the total number of distinct ways to reach the top.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 4

Output: 7

Answer

```
// You are using GCC
#include <stdio.h>
```

```
int main() {
    int n;
    scanf("%d", &n);

    if (n == 0) {
        printf("1\n");
        return 0;
    }
    if (n == 1) {
        printf("1\n");
        return 0;
    }
    if (n == 2) {
        printf("2\n");
```

```

    return 0;
}

int a = 1; // ways[0]
int b = 1; // ways[1]
int c = 2; // ways[2]
int d;

for (int i = 3; i <= n; i++) {
    d = a + b + c;
    a = b;
    b = c;
    c = d;
}

printf("%d\n", c);

return 0;
}

```

Status : Correct

Marks : 10/10

2. Problem Statement

Given a non-negative integer n , the task is to compute the n -th Lucas number using the Dynamic Programming (Bottom-Up Approach) algorithm. The Lucas sequence follows the recurrence relation:

$L(0) = 2$, $L(1) = 1$, and $L(n) = L(n-1) + L(n-2)$ for $n \geq 2$.

Write a program to ensure that the n -th Lucas number is correctly calculated and displayed.

Input Format

The input contains a single integer n , representing the position in the Lucas sequence.

Output Format

The output prints a single integer representing the n -th Lucas number.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 3

Output: 4

Answer

```
// You are using GCC
#include <stdio.h>
```

```
int main() {
    int n;
    scanf("%d", &n);

    // Handle base cases
    if (n == 0) {
        printf("2\n");
        return 0;
    }
    if (n == 1) {
        printf("1\n");
        return 0;
    }

    int dp[n + 1];
    dp[0] = 2;
    dp[1] = 1;

    for (int i = 2; i <= n; i++) {
        dp[i] = dp[i - 1] + dp[i - 2];
    }

    printf("%d\n", dp[n]);

    return 0;
}
```

Status : Correct

Marks : 10/10

3. Problem Statement:

Dr. Ananya is a computational biologist working on analyzing DNA sequences. She needs to identify the longest palindromic subsequence within a DNA string, which may represent certain symmetric patterns important for genetic research.

Given a string S representing a DNA sequence (composed of letters A, C, G, T), find the length of the longest subsequence of S that reads the same forwards and backwards.

Note:

A subsequence is formed by deleting zero or more characters without changing the order of the remaining characters. The DNA sequences can be long, and an efficient solution using dynamic programming is required to handle them within computational limits.

Input Format

The input displays a single line containing the DNA string S.

Output Format

The output displays a single integer representing the length of the longest palindromic subsequence in S.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: madam

Output: 5

Answer

```
// You are using GCC
#include <stdio.h>
#include <string.h>

#define MAX 101
```

```
int max(int a, int b) {  
    return (a > b) ? a : b;  
}
```

```
int main() {  
    char s[MAX];  
    scanf("%s", s);
```

```
    int n = strlen(s);  
    int dp[MAX][MAX];
```

```
    // Initialize dp array with 0
```

```
    for (int i = 0; i < n; i++) {  
        for (int j = 0; j < n; j++) {  
            dp[i][j] = 0;
```

```
        }  
    }
```

```
    // Base case: single characters are palindromes of length 1
```

```
    for (int i = 0; i < n; i++) {  
        dp[i][i] = 1;  
    }
```

```
    // Build the dp table for substrings of increasing length
```

```
    for (int len = 2; len <= n; len++) {  
        for (int i = 0; i <= n - len; i++) {  
            int j = i + len - 1; // Ending index
```

```
            if (s[i] == s[j]) {  
                if (len == 2) {  
                    dp[i][j] = 2;  
                } else {  
                    dp[i][j] = dp[i + 1][j - 1] + 2;
```

```
                } else {  
                    dp[i][j] = max(dp[i + 1][j], dp[i][j - 1]);
```

```
            }  
        }  
    }
```

```
    // Result is the LPS length for the whole string  
    printf("%d\n", dp[0][n - 1]);
```

```
return 0;
```

Status : Correct

Marks : 10/10

4. Problem Statement

Arjun is tracking the stock prices for the last N days. He wants to invest in a way that he buys and sells stocks alternately to maximize his profit. He can either buy or skip on a day, but after buying he must sell before buying again. Find the maximum profit he can achieve from the stock prices. Help him to implement the task.

Formula

Let $dp[i][0]$ = maximum profit on day i if not holding a stock

Let $dp[i][1]$ = maximum profit on day i if holding a stock

Transition:

$dp[i][0] = \max(dp[i-1][0], dp[i-1][1] + \text{prices}[i])$ either skip or sell today

$dp[i][1] = \max(dp[i-1][1], dp[i-1][0] - \text{prices}[i])$ either keep holding or buy today

Base case:

$dp[0][0] = 0$

$dp[0][1] = -\text{prices}[0]$

Result:

$\text{max_profit} = dp[N-1][0]$

Input Format

The first line of input consists of Integer N representing the number of days.

The second line of input consists of N Integers $\text{prices}[i]$ representing the stock price on day i.

Output Format

The output prints: "Maximum profit:" max_profit

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 3

1 2 3

Output: Maximum profit: 2

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
int max(int a, int b) {  
    return (a > b) ? a : b;  
}
```

```
int main() {  
    int n;  
    scanf("%d", &n);
```

```
    int prices[n];  
    for (int i = 0; i < n; i++) {  
        scanf("%d", &prices[i]);  
    }
```

```
    // dp[i][0] - not holding stock  
    // dp[i][1] - holding stock  
    int dp[n][2];
```

```
    // Base case for day 0  
    dp[0][0] = 0;           // Not holding on day 0  
    dp[0][1] = -prices[0];  // Holding on day 0 (bought on day 0)
```

```
    // Fill the dp table  
    for (int i = 1; i < n; i++) {  
        dp[i][0] = max(dp[i-1][0], dp[i-1][1] + prices[i]); // Skip or sell  
        dp[i][1] = max(dp[i-1][1], dp[i-1][0] - prices[i]); // Skip or buy
```



```
}  
// Maximum profit is when not holding stock on last day  
int maxProfit = dp[n-1][0];  
  
printf("Maximum profit: %d\n", maxProfit);  
  
return 0;  
}
```

Status : Correct

Marks : 10/10

5. Problem Statment

To celebrate Diwali, Dhoni wants to buy a variety of crackers. He noticed four different types of crackers on the menu list, each with a price. He's now trying to figure out how many different ways he can get the crackers for Rs.N.

Rockets - 1 RsSparklers - 2 RsChakkars - 3 RsFlower pots - 5 Rs

Example

Input

4

Output

The total number of ways to get crackers is 4

Explanation

1st way -> 4 rockets

2nd way -> 2 Sparklers

3rd way -> 1 Sparkler, 2 rockets

4th way -> 1 Chakkar, 1 rocket

Input Format

The input consists of a single integer N, representing the total amount (in rupees) Dhoni wants to spend on crackers.

Output Format

The output prints "The total number of ways to get crackers is X" Where X representing the total number of different ways Dhoni can buy crackers such that the total cost equals N.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 5

Output: The total number of ways to get crackers is 6

Answer

```
// You are using GCC
#include <stdio.h>
```

```
int main() {
    int n;
    scanf("%d", &n);
```

```
    // Denominations: 1, 2, 3, 5
    int coins[] = {1, 2, 3, 5};
    int numCoins = 4;
```

```
    // dp array to store number of ways for each amount
    long long dp[n + 1];
```

```
    // Initialize dp array
    for (int i = 0; i <= n; i++) {
        dp[i] = 0;
    }
```

```
    dp[0] = 1; // One way to make amount 0
```

```
    // Fill dp array
    for (int i = 0; i < numCoins; i++) {
        for (int j = coins[i]; j <= n; j++) {
            dp[j] += dp[j - coins[i]];
        }
    }
```

```
printf("The total number of ways to get crackers is %lld\n", dp[n]);  
    return 0;  
}
```

Status : Correct

Marks : 10/10

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2028_NeoColab_Competitive Coding II_L9

VTU_Week 11_CY

Attempt : 1

Total Mark : 40

Marks Obtained : 21.5

Section 1 : Coding

1. Problem Statement

Given a set of coin denominations and a target value V , determine the minimum number of coins required to make the value V .

If it's not possible to make the value with the given coins, return -1.

Input Format

The first line contains an integer n , representing the number of coin denominations.

The second line contains n space-separated integers, representing the coin denominations.

The third line contains an integer amount, representing the target value V .

Output Format

The output prints the minimum number of coins required to make the target value.

Print -1 if the value cannot be formed using the given denominations.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 3

1 2 5

11

Output: 3

Answer

// You are using GCC

#include <stdio.h>

#include <limits.h>

int main() {

int n;

scanf("%d", &n);

int coins[n];

for (int i = 0; i < n; i++) {

scanf("%d", &coins[i]);

}

int amount;

scanf("%d", &amount);

// Initialize dp array with a large value (infinity)

int dp[amount + 1];

for (int i = 0; i <= amount; i++) {

dp[i] = INT_MAX;

}

dp[0] = 0; // 0 coins needed to make amount 0

// Fill dp array

for (int i = 1; i <= amount; i++) {

```

for (int j = 0; j < n; j++) {
    if (coins[j] <= i && dp[i - coins[j]] != INT_MAX) {
        if (dp[i - coins[j]] + 1 < dp[i]) {
            dp[i] = dp[i - coins[j]] + 1;
        }
    }
}

// Check if amount can be formed
if (dp[amount] == INT_MAX) {
    printf("-1\n");
} else {
    printf("%d\n", dp[amount]);
}

return 0;
}

```

Status : Correct

Marks : 10/10

2. Problem Statement:

Rajesh runs a delivery company and needs to deliver n packages to different cities. Each package i has:

A weight $wt[i]$ A profit $val[i]$ if delivered on time A delivery deadline $deadline[i]$ (in hours from now) Rajesh has a truck with maximum weight capacity W and can only deliver one package at a time. He can choose the order of deliveries, but if a package is delivered after its deadline, he earns zero profit for it.

Task: Determine the maximum total profit Rajesh can earn by selecting which packages to deliver and in what order, without exceeding the truck's capacity at any point.

Input Format

First line: integer n — number of packages

Second line: n integers — values of each package (profit)

Third line: n integers — weights of each package

Fourth line: n integers — deadlines of each package

Fifth line: integer W — truck capacity

Output Format

The output prints the single integer maximum total profit.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 4
100 200 150 120
10 20 15 15
5 3 4 2
30

Output: 470

Answer

```
// You are using GCC
#include <stdio.h>
#include <stdlib.h>

#define MAXN 20

int val[MAXN], wt[MAXN], deadline[MAXN];
int n, W;

typedef struct {
    int val, wt, deadline;
} Item;

// Sort by profit descending
int cmp(const void *a, const void *b) {
    Item *x = (Item *)a;
    Item *y = (Item *)b;
    return y->val - x->val;
```

```
}
```

```
int main() {  
    scanf("%d", &n);
```

```
    for(int i = 0; i < n; i++) scanf("%d", &val[i]);  
    for(int i = 0; i < n; i++) scanf("%d", &wt[i]);  
    for(int i = 0; i < n; i++) scanf("%d", &deadline[i]);
```

```
    scanf("%d", &W);
```

```
    int maxProfit = 0;  
    int totalMask = 1 << n;
```

```
    for(int mask = 0; mask < totalMask; mask++) {
```

```
        int totalWeight = 0;  
        Item items[MAXN];  
        int k = 0;
```

```
        // build subset  
        for(int i = 0; i < n; i++) {  
            if(mask & (1<<i)) {  
                totalWeight += wt[i];
```

```
                items[k].val = val[i];  
                items[k].wt = wt[i];  
                items[k].deadline = deadline[i];  
                k++;
```

```
            }  
        }
```

```
        if(totalWeight > W) continue;
```

```
        // sort by profit (greedy scheduling)  
        qsort(items, k, sizeof(Item), cmp);
```

```
        int slots[51] = {0}; // max deadline = 50  
        int profit = 0;
```

```
        for(int i = 0; i < k; i++) {  
            for(int t = items[i].deadline; t >= 1; t--) {
```



```

        if(slots[t] == 0) {
            slots[t] = 1;
            profit += items[i].val;
            break;
        }
    }
}

if(profit > maxProfit)
    maxProfit = profit;
}

printf("%d\n", maxProfit);

return 0;
}

```

Status : Wrong

Marks : 0/10

3. Problem Statement

You are given n items, each with a certain weight and value. Your task is to select items to put in a knapsack of capacity W so that the total value is maximized. You cannot break an item; you must either take the whole item or leave it.

Formally, given two integer arrays, $val[0..n-1]$ and $wt[0..n-1]$, representing the values and weights of n items respectively, and an integer W representing the knapsack capacity, determine the maximum value subset of items such that the total weight does not exceed W .

Solve this problem using dynamic programming.

Input Format

The first line contains an integer n , the number of items.

The second line contains n integers, the values of each item separated by spaces.

The third line contains n integers, the weights of each item separated by spaces.

The fourth line contains an integer W, the maximum capacity of the knapsack.

Output Format

Print a single integer representing the maximum value that can be carried in the knapsack.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 2

50 60

5 10

10

Output: 60

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
int max(int a, int b) {  
    return (a > b) ? a : b;  
}
```

```
int main() {  
    int n;  
    scanf("%d", &n);
```

```
    int val[n], wt[n];
```

```
    for (int i = 0; i < n; i++) {  
        scanf("%d", &val[i]);  
    }
```

```
    for (int i = 0; i < n; i++) {  
        scanf("%d", &wt[i]);  
    }
```

```

int W;
scanf("%d", &W);

// Create DP table
int dp[n + 1][W + 1];

// Initialize DP table
for (int i = 0; i <= n; i++) {
    for (int w = 0; w <= W; w++) {
        if (i == 0 || w == 0) {
            dp[i][w] = 0;
        } else if (wt[i - 1] <= w) {
            dp[i][w] = max(dp[i - 1][w], dp[i - 1][w - wt[i - 1]] + val[i - 1]);
        } else {
            dp[i][w] = dp[i - 1][w];
        }
    }
}

printf("%d\n", dp[n][W]);

return 0;
}

```

Status : Correct

Marks : 10/10

4. Problem Statement

Imagine you are a tourist in Moscow and have booked a bus tour to see some amazing attractions just outside of town. The bus first drives around town for a while (a long while, since Moscow is a big city) picking up people at their respective hotels. It then proceeds to the amazing attraction, and after a few hours go back into the city, again driving to each hotel, this time to drop people off.

For some reason, whenever you do this, your hotel is always the first to be visited for pickup and the last to be visited for drop-off, meaning that you have to suffer through two not-so-amazing sightseeing tours of all the local hotels.

This is clearly not what you want to do (unless for some reason you are really into hotels), so let's fix it. We will develop some software to enable the sightseeing company to route its bus tours more fairly—though it may sometimes mean a longer total distance for everyone, fair is fair, right?

For this problem, there is a starting location (the sightseeing company headquarters), h hotels that need to be visited for pickups and dropoffs, and a destination location (the amazing attraction). We need to find a route that goes from the headquarters, through all the hotels, to the attraction, then back through all the hotels again (possibly in a different order), and finally back to the headquarters.

To guarantee that none of the tourists (and, in particular, you) is forced to suffer through two full tours of the hotels, we require that every hotel that is visited among the first $h/2$ hotels on the way to the attraction is also visited among the first $h/2$ hotels on the way back.

Subject to these restrictions, we would like to make the complete bus tour as short as possible. Note that these restrictions may force the bus to drive past a hotel without stopping there (this is not considered visiting) and then visit it later, as illustrated in the sample input.

Input Format

The first line of each test case consists of two integers n and m , where n is the number of locations (hotels, headquarters, attraction) and m is the number of pairs of locations between which the bus can travel.

The n different locations are numbered from 0 to $n-1$, where 0 is the headquarters, 1 through $n-2$ are the hotels, and $n-1$ is the attraction. Assume that there is at most one direct connection between any pair of locations and it is possible to travel from any location to any other location (but not necessarily directly).

Following the first line are m lines, each containing three integers u , v , and t , indicating that the bus can go directly between locations u and v in t seconds (in either direction).

Output Format

For each test case, print the case number and the minimum total travel time of

the complete tour.

The output should follow this format: "Case X: Y"

where:

X is the test case number starting from 1.

Y is the shortest possible total time (in seconds) for the bus tour.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 5 4

0 1 10

1 2 20

2 3 30

3 4 40

4 6

0 1 1

0 2 1

0 3 1

1 2 1

1 3 1

2 3 1

Output: Case 1: 300

Case 2: 6

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
#include <string.h>
```

```
#include <limits.h>
```

```
#define MAXN 20
```

```
#define INF 1000000000
```

```
int dist[MAXN][MAXN];
```

```
int dp[1<<18][18];
```

```

// TSP: start visit nodes[] end
int tsp(int start, int end, int nodes[], int k) {
    if(k == 0) return dist[start][end];

    int maxMask = 1 << k;

    for(int i = 0; i < maxMask; i++)
        for(int j = 0; j < k; j++)
            dp[i][j] = INF;

    for(int i = 0; i < k; i++)
        dp[1<<i][i] = dist[start][nodes[i]];

    for(int mask = 1; mask < maxMask; mask++) {
        for(int u = 0; u < k; u++) {
            if(!(mask & (1<<u))) continue;

            for(int v = 0; v < k; v++) {
                if(mask & (1<<v)) continue;

                int newMask = mask | (1<<v);
                int val = dp[mask][u] + dist[nodes[u]][nodes[v]];
                if(val < dp[newMask][v])
                    dp[newMask][v] = val;
            }
        }
    }

    int ans = INF;
    for(int i = 0; i < k; i++) {
        int val = dp[maxMask-1][i] + dist[nodes[i]][end];
        if(val < ans) ans = val;
    }

    return ans;
}

```

```

int main() {
    int n, m, caseNo = 1;

    while(scanf("%d %d", &n, &m) == 2) {

```

```

for(int i = 0; i < n; i++)
    for(int j = 0; j < n; j++)
        dist[i][j] = (i == j) ? 0 : INF;

```

```

for(int i = 0; i < m; i++) {
    int u, v, t;
    scanf("%d %d %d", &u, &v, &t);
    if(t < dist[u][v]) {
        dist[u][v] = dist[v][u] = t;
    }
}

```

```

// Floyd-Warshall
for(int k = 0; k < n; k++)
    for(int i = 0; i < n; i++)
        for(int j = 0; j < n; j++)
            if(dist[i][j] > dist[i][k] + dist[k][j])
                dist[i][j] = dist[i][k] + dist[k][j];

```

```

int h = n - 2;
int hotels[18];
for(int i = 0; i < h; i++) hotels[i] = i + 1;

```

```

int half = h / 2;
int totalMask = 1 << h;
int answer = INF;

```

```

for(int mask = 0; mask < totalMask; mask++) {
    if(__builtin_popcount(mask) != half) continue;

```

```

    int S[18], T[18];
    int s = 0, t = 0;

```

```

    for(int i = 0; i < h; i++) {
        if(mask & (1<<i)) S[s++] = hotels[i];
        else T[t++] = hotels[i];
    }

```

```

    int cost =
        tsp(0, -1, S, s) + // 0 S
        tsp(S[s-1], -1, T, t) + // S T

```

```

        dist[T[t-1]][n-1] +      // T attraction
        tsp(n-1, -1, S, s) +      // return: attraction S
        tsp(S[s-1], -1, T, t) +    // S T
        dist[T[t-1]][0];          // T 0

    if(cost < answer) answer = cost;
}

printf("Case %d: %d\n", caseNo++, answer);
}

return 0;
}

```

Status : Partially correct

Marks : 1.5/10

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2028_NeoColab_Competitive Coding II_L9

VTU_Week 11_PAH

Attempt : 1

Total Mark : 40

Marks Obtained : 40

Section 1 : Coding

1. Problem Statement

Write a program to implement an optimal binary search tree using dynamic programming.

Given a sorted array key $[0 \dots n-1]$ and an array freq $[0 \dots n-1]$ of frequency counts, where freq $[i]$ is the number of searches for key $[i]$, Construct a binary search tree of all keys such that the total cost of all the searches is as small as possible.

Note: Cost of a BST node: level of that node * frequency. The level of the root is 1.

Input Format

The first line of the input is the value of n.

The second line of input is the keys separated by a single space.

The last line of input is the frequency separated by a single space.

Output Format

The output prints the cost of the optimal binary search tree.

Sample Test Case

Input: 3

10 12 20

34 8 50

Output: Cost of Optimal BST is 142

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
#include <limits.h>
```

```
// Function to find the sum of frequencies from i to j
```

```
int sum(int freq[], int i, int j) {
```

```
    int s = 0;
```

```
    for (int k = i; k <= j; k++)
```

```
        s += freq[k];
```

```
    return s;
```

```
}
```

```
// Function to find optimal BST cost using dynamic programming
```

```
int optimalBST(int keys[], int freq[], int n) {
```

```
    // Create a 2D table to store costs of optimal BST
```

```
    int dp[n][n];
```

```
    // For single keys, cost is frequency
```

```
    for (int i = 0; i < n; i++)
```

```
        dp[i][i] = freq[i];
```

```
    // Consider chains of increasing length
```

```
    for (int length = 2; length <= n; length++) {
```

```
        for (int i = 0; i <= n - length; i++) {
```

```
            int j = i + length - 1;
```

```
            dp[i][j] = INT_MAX;
```

```

        // Try making each key in the range as root
        for (int r = i; r <= j; r++) {
            // Calculate cost when keys[r] becomes root
            int cost = (r > i ? dp[i][r - 1] : 0) +
                (r < j ? dp[r + 1][j] : 0) +
                sum(freq, i, j);

            if (cost < dp[i][j])
                dp[i][j] = cost;
        }
    }
}

return dp[0][n - 1];
}

int main() {
    int n;
    scanf("%d", &n);

    int keys[n];
    int freq[n];

    // Read keys
    for (int i = 0; i < n; i++) {
        scanf("%d", &keys[i]);
    }

    // Read frequencies
    for (int i = 0; i < n; i++) {
        scanf("%d", &freq[i]);
    }

    int result = optimalBST(keys, freq, n);
    printf("Cost of Optimal BST is %d\n", result);

    return 0;
}

```

Status : Correct

Marks : 10/10

2. Problem Statement

Given the weights and values of n items, put these items in a knapsack of capacity W to get the maximum total value in the knapsack. In other words, given two integer arrays, $val[0..n-1]$ and $wt[0..n-1]$ which represent values and weights associated with n items, respectively. Also given an integer W which represents knapsack capacity, find out the maximum value subset of $val[]$ such that the sum of the weights of this subset is smaller than or equal to W . You cannot break an item, either pick the complete item or don't pick it.

Solve this problem using dynamic programming.

Input Format

The first line of input represents the number of items.

The second line of input consists of the values of each item separated by spaces.

The third line of input consists of the weights of each item separated by spaces.

The fourth line of input consists of an integer, which represents the maximum capacity of the knapsack.

Output Format

The output prints the maximum value of items that can be held by the knapsack.

Sample Test Case

Input: 3
60 100 120
10 20 30
50

Output: 220

Answer

```
// You are using GCC
#include <stdio.h>
```

```
// Function to find maximum of two integers
int max(int a, int b) {
```

```

    return (a > b) ? a : b;
}

// Function to solve 0/1 knapsack problem using dynamic programming
int knapsack(int W, int wt[], int val[], int n) {
    int dp[n + 1][W + 1];

    // Build table dp[][] in bottom-up manner
    for (int i = 0; i <= n; i++) {
        for (int w = 0; w <= W; w++) {
            if (i == 0 || w == 0) {
                dp[i][w] = 0;
            }
            else if (wt[i - 1] <= w) {
                dp[i][w] = max(val[i - 1] + dp[i - 1][w - wt[i - 1]], dp[i - 1][w]);
            }
            else {
                dp[i][w] = dp[i - 1][w];
            }
        }
    }

    return dp[n][W];
}

int main() {
    int n;
    scanf("%d", &n);

    int val[n];
    int wt[n];

    // Read values
    for (int i = 0; i < n; i++) {
        scanf("%d", &val[i]);
    }

    // Read weights
    for (int i = 0; i < n; i++) {
        scanf("%d", &wt[i]);
    }
}

```

```
// Read knapsack capacity
int W;
scanf("%d", &W);

int result = knapsack(W, wt, val, n);
printf("%d\n", result);

return 0;
}
```

Status : Correct

Marks : 10/10

3. Problem Statement

Given an array, the task is to find the sum of the maximum sum by alternating subsequences starting with the first element. Here alternating sequence means first decreasing, then increasing, then decreasing,... For example, 10, 5, 14, and 3 are alternating sequences.

Note: The reverse sequence type (increasing—decreasing—increasing—...) is not considered alternating here.

Example

Input:

6

4 3 8 5 3 8

Output:

28

Explanation:

The alternating subsequence (starting with the first element) that has the maximum sum is {4, 3, 8, 5, 8}

Input Format

The first line contains an integer n, representing the number of elements in the array.

The second line contains n space-separated integers, representing the elements of the array.

Output Format

The output displays a single integer representing the maximum sum of an alternating subsequence from the given array.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 6

4 3 8 5 3 8

Output: 28

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
#include <limits.h>
```

```
// Function to find maximum of two integers
```

```
int max(int a, int b) {  
    return (a > b) ? a : b;  
}
```

```
// Recursive function to find maximum sum alternating subsequence
```

```
int solve(int arr[], int n, int index, int prev, int isDecreasing) {  
    if (index == n) {  
        return 0;  
    }  
}
```

```
// Option 1: Skip current element
```

```
int skip = solve(arr, n, index + 1, prev, isDecreasing);
```

```
// Option 2: Take current element if it satisfies alternating condition
```

```
int take = 0;
```

```
if (prev == -1) {
```

```
    // First element - must take it
```

```

        take = arr[index] + solve(arr, n, index + 1, index, 1);
    }
    else {
        if (isDecreasing) {
            // Looking for a decreasing element
            if (arr[index] < arr[prev]) {
                take = arr[index] + solve(arr, n, index + 1, index, 0);
            }
        }
        else {
            // Looking for an increasing element
            if (arr[index] > arr[prev]) {
                take = arr[index] + solve(arr, n, index + 1, index, 1);
            }
        }
    }

    return max(skip, take);
}

int main() {
    int n;
    scanf("%d", &n);

    int arr[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    // Start with first element and in decreasing phase
    int result = arr[0] + solve(arr, n, 1, 0, 1);

    printf("%d\n", result);

    return 0;
}

```

Status : Correct

Marks : 10/10

4. Problem Statement

A taxi driver wants to maximize the cab fare by choosing the longest possible travel path through a city represented as a square matrix.

Each cell in the matrix represents the kilometers travelled at that location.

The taxi can move only in two directions:

Right $(i, j+1)$ Down $(i+1, j)$

The driver will move to the next cell only if the difference between the current value and the next value is exactly +1 or -1.

Rules:

Movement is allowed only right or down. The absolute difference between the current cell value and the next cell value must be 1. If both right and down moves satisfy the condition, the driver always prefers the right move. The trip continues until no valid move is possible.

Your task is to determine and print the longest trip path starting from the first cell (0,0).

Input Format

The first line contains an integer N , representing the size of the square matrix.

The next N lines contain N space-separated integers, representing the matrix values.

Output Format

The output prints the longest trip path starting from the first element of the matrix in the format "Longest trip path $a \rightarrow b \rightarrow c \rightarrow d$ " where a, b, c, d are the values visited in the path.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 3

```
2 3 1
0 4 5
8 7 6
```

Output: Longest trip path 2->3->4->5->6

Answer

// You are using GCC

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#define MAX 50
```

```
int N;
```

```
int matrix[MAX][MAX];
```

```
int path[MAX * 2]; // Store the path values
```

```
int maxPath[MAX * 2]; // Store the longest path found
```

```
int maxLength = 0;
```

// Function to check if a move is valid

```
int isValidMove(int currentRow, int currentCol, int nextRow, int nextCol) {
```

```
    if (nextRow >= N || nextCol >= N) {
```

```
        return 0;
```

```
    }
```

```
    int diff = matrix[currentRow][currentCol] - matrix[nextRow][nextCol];
```

```
    return (diff == 1 || diff == -1);
```

```
}
```

// DFS function to find the longest path

```
void dfs(int row, int col, int pathIndex) {
```

```
    path[pathIndex] = matrix[row][col];
```

```
    pathIndex++;
```

// Check if current path is longer than previously found

```
if (pathIndex > maxLength) {
```

```
    maxLength = pathIndex;
```

```
    for (int i = 0; i < pathIndex; i++) {
```

```
        maxPath[i] = path[i];
```

```
    }
```

```
}
```

// Check right move

```
int rightValid = isValidMove(row, col, row, col + 1);
```

```

// Check down move
int downValid = isValidMove(row, col, row + 1, col);

if (rightValid) {
    // Prefer right move
    dfs(row, col + 1, pathIndex);
}
else if (downValid) {
    // Only go down if right is not valid
    dfs(row + 1, col, pathIndex);
}
// If no valid moves, path ends here
}

int main() {
    scanf("%d", &N);

    for (int i = 0; i < N; i++) {
        for (int j = 0; j < N; j++) {
            scanf("%d", &matrix[i][j]);
        }
    }

    // Start from (0,0)
    dfs(0, 0, 0);

    // Print the longest path
    printf("Longest trip path ");
    for (int i = 0; i < maxLength; i++) {
        if (i > 0) {
            printf("->");
        }
        printf("%d", maxPath[i]);
    }
    printf("\n");

    return 0;
}

```

Status : Correct

Marks : 10/10

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2028_NeoColab_Competitive Coding II_L9

VTU_Week 12_MCQ

Attempt : 1

Total Mark : 15

Marks Obtained : 13

Section 1 : MCQ

1. The first step in the naive greedy algorithm is _____.

Answer

analysing the zero flow

Status : Correct

Marks : 1/1

2. Suppose you have coins of denominations 1,3 and 4. You use a greedy algorithm, in which you choose the largest denomination coin which is not greater than the remaining sum. For which of the following sums, will the algorithm doesn't produce an optimal answer?

Answer

10

Status : Correct

Marks : 1/1

3. Which data structure is most commonly used to efficiently implement a greedy algorithm that repeatedly selects the minimum (or maximum) element at each step?

Answer

Heap

Status : Correct

Marks : 1/1

4. What is a greedy algorithm?

Answer

An algorithm that makes the best choice at each step, without considering the overall impact

Status : Correct

Marks : 1/1

5. Which of the following problems is typically solved using a greedy algorithm?

Answer

Coin change problem.

Status : Wrong

Marks : 0/1

6. What is the primary advantage of greedy algorithms over other approaches for certain problems?

Answer

Greedy algorithms are easy to implement

Status : Correct

Marks : 1/1

7. Which of the following is a disadvantage of greedy algorithm?

Answer

It may not always find the globally optimal solution

Status : Correct

Marks : 1/1

8. Which of the following approaches should be used to find the solution to the activity selection problem?

Answer

Greedy approach

Status : Correct

Marks : 1/1

9. Suppose two activities A and B, have start and finish times as S_A , F_A and S_B , F_B respectively. Both activities are said to be compatible, under which of the following conditions?

Answer

$S_A \leq F_B$ or $S_B \leq F_A$

Status : Correct

Marks : 1/1

10. In the activity selection problem, if two activities have the same finish time, which one is selected?

Answer

The one with the later start time

Status : Wrong

Marks : 0/1

11. Consider the following number of activities with their start and finish time given below. In which sequence will the activity be selected in order to maximize the number of activities, without any conflicts?

Answer

A1, A2, A5

Status : Correct

Marks : 1/1

12. Which of the following is a characteristic of the greedy algorithm?

Answer

It makes the best choice at each step by considering only the local problem state.

Status : Correct

Marks : 1/1

13. What is the primary reason why the greedy algorithm does not work for some optimization problems?

Answer

It does not reconsider previous decisions and makes choices based only on the current state.

Status : Correct

Marks : 1/1

14. In the greedy approach using a stack, what condition do we check before popping from the stack

Answer

If the top is greater than current digit and $k > 0$

Status : Correct

Marks : 1/1

15. Why is sorting not used in this greedy solution?

Answer

All of the mentioned options

Status : Correct

Marks : 1/1

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2028_NeoColab_Competitive Coding II_L9

VTU_Week 12_COD

Attempt : 1

Total Mark : 50

Marks Obtained : 50

Section 1 : Coding

1. Problem Statement

Given an array of non-negative integers, the task is to find the minimum number of elements such that their sum should be greater than the sum of the rest of the elements of the array.

Example:

Input: arr[] = [3 , 1 , 7, 1]

Output: 1

Explanation: Smallest subset is {7}. Sum of this subset is greater than the sum of all other elements left after removing subset {7} from the array

Input: arr[] = [2 , 1 , 2]

Output: 2

Explanation: Smallest subset is {2 , 1}. Sum of this subset is greater than the sum of all other elements left after removing subset {2 , 1} from the array

Input Format

The first line contains an Integer n, representing the size of the array.

The second line contains n space-separated integers, representing the elements of the array.

Output Format

The output prints a single integer representing the minimum number of elements whose sum is greater than the sum of the remaining elements.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 4

3 1 7 1

Output: 1

Answer

```
// You are using GCC
#include <iostream>
#include <vector>
#include <algorithm>
#include <numeric>
```

```
using namespace std;
```

```
int main() {
    int n;
    cin >> n;

    vector<int> arr(n);
    for(int i = 0; i < n; i++) {
        cin >> arr[i];
    }
}
```

```

// Sort in descending order
sort(arr.begin(), arr.end(), greater<int>());

// Calculate total sum
int totalSum = accumulate(arr.begin(), arr.end(), 0);

int subsetSum = 0;
int count = 0;

// Add largest elements until subset sum > remaining sum
for(int i = 0; i < n; i++) {
    subsetSum += arr[i];
    count++;

    int remainingSum = totalSum - subsetSum;

    if(subsetSum > remainingSum) {
        break;
    }
}

cout << count << endl;

return 0;
}

```

Status : Correct

Marks : 10/10

2. Problem Statement

Given an array `prices[]` of `n` different candies, where `prices[i]` represents the price of the `i`th candy and all prices are distinct. The store has an offer where for every candy you buy, you can get up to `k` other different candies for free. Find the minimum and maximum amount of money needed to buy all the candies.

Note: In both cases, you must take the maximum number of free candies possible during each purchase.

Examples:

Input: prices[] = [3, 2, 1, 4], k = 2

Output: [3, 7]

Explanation: For minimum cost, buy the candies costing 1 and 2. for maximum cost, buy the candies costing 4 and 3.

Input: prices[] = [3, 2, 1, 4, 5], k = 4

Output: [1, 5]

Explanation: For minimum cost, buy the candy costing 1. for maximum cost, buy the candy costing 5.

Input Format

The first line contains an Integer n, representing the number of candies.

The second line contains n space-separated integers, representing the prices of the candies.

The third line contains an integer k, representing the number of free candies per purchase.

Output Format

The output prints two integers separated by a space:

minCost maxCost

representing the minimum and maximum money needed.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 4

3 2 1 4

2

Output: 3 7

Answer

```
// You are using GCC
#include <stdio.h>
#include <stdlib.h>

// Comparison function for sorting in ascending order
int compare(const void *a, const void *b) {
    return *(int*)a - *(int*)b;
}

int main() {
    int n;
    scanf("%d", &n);

    int prices[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &prices[i]);
    }

    int k;
    scanf("%d", &k);

    // Sort the prices in ascending order
    qsort(prices, n, sizeof(int), compare);

    // Number of candies we need to purchase
    int purchases = (n + k) / (k + 1); // Ceiling division

    // Minimum cost: buy the cheapest 'purchases' candies
    int minCost = 0;
    for (int i = 0; i < purchases; i++) {
        minCost += prices[i];
    }

    // Maximum cost: buy the most expensive 'purchases' candies
    int maxCost = 0;
    for (int i = n - 1; i >= n - purchases; i--) {
        maxCost += prices[i];
    }

    printf("%d %d\n", minCost, maxCost);
}
```

```
    return 0;  
}
```

Status : Correct

Marks : 10/10

3. Problem Statement

Given a fraction represented by two integers — a numerator n and a denominator d — express the fraction as a sum of unique unit fractions (fractions with numerator 1). This representation is known as the Egyptian Fraction.

Write a program to ensure that the given fraction is converted into its Egyptian Fraction form.

Input Format

The input contains two space-separated integers n and d , representing the numerator and denominator of the fraction.

Output Format

The output prints the Egyptian Fraction representation of the given fraction in the form of unit fractions separated by $+$.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 6 14

Output: $1/3 + 1/11 + 1/231$

Answer

```
// You are using GCC  
#include <stdio.h>
```

```
int main() {  
    int n, d;  
    scanf("%d %d", &n, &d);
```

```

int first = 1; // To handle + sign properly

while (n > 0) {
    // Find the smallest denominator for unit fraction
    int x = (d + n - 1) / n; // Ceil(d/n)

    // Print the unit fraction
    if (first) {
        printf("1/%d", x);
        first = 0;
    } else {
        printf(" + 1/%d", x);
    }

    // Update the fraction: n/d - 1/x = (n*x - d) / (d*x)
    n = n * x - d;
    d = d * x;

    // Simplify the fraction by dividing numerator and denominator by GCD
    int gcd = 1;
    for (int i = 1; i <= n && i <= d; i++) {
        if (n % i == 0 && d % i == 0) {
            gcd = i;
        }
    }
    n /= gcd;
    d /= gcd;
}

printf("\n");
return 0;
}

```

Status : Correct

Marks : 10/10

4. Problem Statement

Emily is building a data compression feature in her file organizing system. She wants filenames to have no repeated characters and appear in the smallest possible order so users can locate files easily. For example, if a

corrupted filename is "cbacdcabc", she wants it cleaned up as "acdb" — which preserves one instance of each letter, keeps relative order, and is the smallest alphabetically.

This function helps ensure filenames are clean, minimal, and alphabetically organized!

Example 1:

Input: s = "bcabc"

Output: "ABC"

Example 2:

Input: s = "cbacdcabc"

Output: "acdb"

Input Format

The first line of input contains a string num representing a non-negative integer.

The second line contains an integer k representing how many digits to remove.

Output Format

The output prints the string representing the smallest possible number after removing exactly k digits.

Leading zeros should be removed, except if the number is zero itself.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: bcabc

Output: abc

Answer

```
// You are using GCC
#include <stdio.h>
```

```
#include <string.h>
#include <stdbool.h>
```

```
#define MAX 26
```

```
int main() {
    char s[10001];
    scanf("%s", s);
```

```
    int len = strlen(s);
    int count[MAX] = {0};
    bool inStack[MAX] = {false};
    char stack[MAX + 1];
    int top = -1;
```

```
    // Count frequency of each character
    for (int i = 0; i < len; i++) {
        count[s[i] - 'a']++;
    }
```

```
    // Process each character
    for (int i = 0; i < len; i++) {
        char current = s[i];
        int idx = current - 'a';
```

```
        // Decrease count of current character
        count[idx]--;
```

```
        // If already in stack, skip
        if (inStack[idx]) {
            continue;
        }
```

```
        // While stack not empty and current char is smaller than top
        // and top character will appear again later
        while (top >= 0 && current < stack[top] && count[stack[top] - 'a'] > 0) {
            inStack[stack[top] - 'a'] = false;
            top--;
        }
```

```
        // Push current character
        top++;
```



```
    stack[top] = current;
    inStack[idx] = true;
}

// Add null terminator
stack[top + 1] = '\0';

printf("%s\n", stack);

return 0;
}
```

Status : Correct

Marks : 10/10

5. Problem Statement

Given the arrival and departure times of all trains that reach a railway station, find the minimum number of platforms required for the station so that no train is kept waiting. Consider that all the trains arrive on the same day and leave on the same day.

Arrival and departure times can never be the same for a train, but we can have the arrival time of one train equal to the departure time of the other. At any given instance, the same platform cannot be used for both the departure of a train and the arrival of another train. In such cases, we need different platforms.

Example 1

Input:

N=6

arr[] = {0900, 0940, 0950, 1100, 1500, 1800}, dep[] = {0910, 1200, 1120, 1130, 1900, 2000}

Output: 3

Explanation: A minimum of three platforms are required to safely arrive and depart all trains.

Input Format

The first line consists of an integer N, which represents the total number of trains.

The second line consists of N space-separated integers representing the arrival time.

The third line consists of N space-separated integers representing the departure time.

Output Format

The output displays an integer which is the minimum number of platforms required to safely arrive and depart the train.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 6

0900 0940 0950 1100 1500 1800

0910 1200 1120 1130 1900 2000

Output: 3

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
// Comparison function for sorting
```

```
int compare(const void *a, const void *b) {
```

```
    return *(int*)a - *(int*)b;
```

```
}
```

```
int main() {
```

```
    int n;
```

```
    scanf("%d", &n);
```

```
    int arr[n], dep[n];
```

```

for (int i = 0; i < n; i++) {
    scanf("%d", &arr[i]);
}

for (int i = 0; i < n; i++) {
    scanf("%d", &dep[i]);
}

// Sort arrival and departure times
qsort(arr, n, sizeof(int), compare);
qsort(dep, n, sizeof(int), compare);

int platforms = 1; // Minimum platforms needed
int result = 1;    // Result to return
int i = 1;         // Pointer for arrival
int j = 0;         // Pointer for departure

// Process all events in chronological order
while (i < n && j < n) {
    // If next train arrives before current train departs
    if (arr[i] <= dep[j]) {
        platforms++;
        i++;

        // Update result if needed
        if (platforms > result) {
            result = platforms;
        }
    } else {
        // A train departs, free a platform
        platforms--;
        j++;
    }
}

printf("%d\n", result);

return 0;
}

```

Status : Correct

Marks : 10/10

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2028_NeoColab_Competitive Coding II_L9

VTU_Week 12_CY

Attempt : 1

Total Mark : 40

Marks Obtained : 40

Section 1 : Coding

1. Problem Statement:

Raju has a hot dog of a certain length that he wishes to divide strategically to maximize the number of friends he can serve. The effectiveness of a given partition is determined by the product of its segment lengths. Raju's goal is to determine the optimal way to cut the hot dog such that this product is maximized, thereby maximizing the number of friends he can serve.

Example

Input:

4

Output:

Maximum friends served 4

Explanation:

For a hot dog of length 4, possible partitions and their corresponding products are:

$$1, 3 \quad (1 \times 3) = 3$$

$$2, 2 \quad (2 \times 2) = 4 \text{ (Maximum)}$$

$$3, 1 \quad (3 \times 1) = 3$$

$$1, 1, 2 \quad (1 \times 1 \times 2) = 2$$

$$1, 2, 1 \quad (1 \times 2 \times 1) = 2$$

$$2, 1, 1 \quad (2 \times 1 \times 1) = 2$$

$$1, 1, 1, 1 \quad (1 \times 1 \times 1 \times 1) = 1$$

Thus, the optimal partition results in a maximum product of 4, serving the highest number of friends.

Input Format

The input consists of a single integer n , representing the length of the hot dog.

Output Format

The output prints the maximum number of friends served.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 4

Output: Maximum friends served 4

Answer

```
// You are using GCC
#include <stdio.h>
```

```
int main() {
    int n;
    scanf("%d", &n);
```

```

int result;

if (n == 1) {
    result = 0; // Can't break into positive integers
} else if (n == 2) {
    result = 1; // 1×1 = 1
} else if (n == 3) {
    result = 2; // 2×1 = 2
} else {
    int quotient = n / 3;
    int remainder = n % 3;

    if (remainder == 0) {
        result = 1;
        for (int i = 0; i < quotient; i++) {
            result *= 3;
        }
    } else if (remainder == 1) {
        result = 1;
        for (int i = 0; i < quotient - 1; i++) {
            result *= 3;
        }
        result *= 4; // Replace one 3+1 with 2×2 = 4
    } else { // remainder == 2
        result = 1;
        for (int i = 0; i < quotient; i++) {
            result *= 3;
        }
        result *= 2;
    }
}

printf("Maximum friends served %d\n", result);

return 0;
}

```

Status : Correct

Marks : 10/10

2. Problem Statement

Imagine you are given a set of items, each with a weight, a value, and a penalty value. You need to solve the Fractional Knapsack problem with the objective of maximizing the total value while minimizing penalties. Implement a greedy algorithm to choose items that best meet this objective.

Example:

Input:

2

5 20 2

10 30 5

15

Output:

Maximum total value while minimizing penalties: 14.00

Explanation:

Sort the items based on the modified value-to-weight ratios (value - penalty) / weight:

Item 1: $(7 - 1) / 2 = 3$

Item 2: $(12 - 4) / 3 = 2$

Start filling the knapsack:

Add all of Item 1 (Weight: 2, Value: $7 - 1 = 6$)

Add 1 unit of Item 2 (Weight: 3, Value: $12 - 4 = 8$)

The total value is $6 + 8 = 14$.

So, the correct output for the given input is 14.

Example:

Input:

2

5 20 2

10 30 5

15

Output:

Maximum total value while minimizing penalties: 43.00

Explanation:

Sort the items based on the modified value-to-weight ratios (value - penalty) / weight:

Item 2: $(30 - 5) / 10 = 2.5$

Item 1: $(20 - 2) / 5 = 3.6$

Start filling the knapsack:

Add all of Item 1 (Weight: 5, Value: $20 - 2 = 18$)

Add 1 unit of Item 2 (Weight: 10, Value: $30 - 5 = 25$)

Now, let's calculate the total value of the knapsack:

Total Weight = $5 + 10 = 15$ (fits within the knapsack's capacity)

Total Value = $18 + 25 = 43$

Input Format

The first line contains an integer 'N' representing the number of items.

The next 'N' lines contain information for each item:

Three integers 'weight', 'value', and 'penalty', are separated by spaces, where 'weight' is the weight of the item, 'value' is the value of the item, and 'penalty' is the penalty for taking the item.

The last line contains an integer 'knapsackWeight' representing the weight limit

of the knapsack.

Output Format

The output prints "Maximum total value while minimizing penalties: " followed by the maximum total value achievable while minimizing penalties, rounded to two decimal places.

Refer to the sample outputs for the formatting specifications.

Sample Test Case

Input: 2

27 1

3 12 4

5

Output: Maximum total value while minimizing penalties: 14.00

Answer

// You are using GCC

#include <stdio.h>

#include <stdlib.h>

typedef struct {

int weight;

int value;

int penalty;

double modifiedValue;

double ratio;

} Item;

int compare(const void *a, const void *b) {

Item *itemA = (Item *)a;

Item *itemB = (Item *)b;

if (itemB->ratio > itemA->ratio) return 1;

if (itemB->ratio < itemA->ratio) return -1;

return 0;

}

int main() {

int n;

```

scanf("%d", &n);

Item items[n];

for (int i = 0; i < n; i++) {
    scanf("%d %d %d", &items[i].weight, &items[i].value, &items[i].penalty);
    items[i].modifiedValue = items[i].value - items[i].penalty;
    items[i].ratio = items[i].modifiedValue / items[i].weight;
}

int knapsackWeight;
scanf("%d", &knapsackWeight);

// Sort items by ratio in descending order
qsort(items, n, sizeof(Item), compare);

double totalValue = 0.0;
int currentWeight = 0;

// Fill knapsack greedily
for (int i = 0; i < n; i++) {
    if (currentWeight + items[i].weight <= knapsackWeight) {
        // Take full item
        totalValue += items[i].modifiedValue;
        currentWeight += items[i].weight;
    } else {
        // Take fractional part
        int remaining = knapsackWeight - currentWeight;
        totalValue += items[i].ratio * remaining;
        break; // Knapsack is full
    }
}

printf("Maximum total value while minimizing penalties: %.2f\n", totalValue);

return 0;
}

```

Status : Correct

Marks : 10/10

3. Problem Statement

There are n participants in a race. Each participant is assigned a performance score given in the integer array `performance`.

You are responsible for awarding medals to these participants, subject to the following conditions:

Each participant must receive at least one medal.

Participants with a higher performance score than their neighbors must receive more medals than their neighbors.

Return the minimum number of medals you need to distribute to the participants.

Example 1:

Input: `ratings = [1,0,2]`

Output: 5

Explanation:

Participant 1 gets 1 medal.

Participant 2 gets 1 medal (since their performance is the lowest).

Participant 3 gets 2 medals (since their performance is higher than Participant 2).

The total number of medals required is 5.

Example 2:

Input: `ratings = [1,2,2]`

Output: 4

Explanation:

Participant 1 gets 1 medal.

Participant 2 gets 2 medals (since their performance is better than Participant 1).

Participant 3 gets 1 medal (because their performance is equal to

Participant 2, and they don't need more medals).

The total number of medals required is 4.

Input Format

The first line consists of an integer n representing the number of participants.

The second line consists of n space-separated integers representing the performance scores of the participants.

Output Format

An integer representing the minimum number of medals required to satisfy the conditions.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 3

1 0 2

Output: 5

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
int main() {
```

```
    int n;
```

```
    scanf("%d", &n);
```

```
    int ratings[n];
```

```
    for (int i = 0; i < n; i++) {
```

```
        scanf("%d", &ratings[i]);
```

```
    }
```

```
    // Initialize medals array with 1 medal for each participant
```

```
    int medals[n];
```

```
    for (int i = 0; i < n; i++) {
```

```

    medals[i] = 1;
}

// Left to right pass
for (int i = 1; i < n; i++) {
    if (ratings[i] > ratings[i - 1]) {
        medals[i] = medals[i - 1] + 1;
    }
}

// Right to left pass
for (int i = n - 2; i >= 0; i--) {
    if (ratings[i] > ratings[i + 1]) {
        if (medals[i] <= medals[i + 1]) {
            medals[i] = medals[i + 1] + 1;
        }
    }
}

// Calculate total medals
int total = 0;
for (int i = 0; i < n; i++) {
    total += medals[i];
}

printf("%d\n", total);

return 0;
}

```

Status : Correct

Marks : 10/10

4. Problem Statement

You are planning a circular road trip that passes through several gas stations. Each station provides a certain amount of fuel, and traveling from one station to the next consumes a specific amount of fuel.

Your goal is to determine a starting gas station index from which you can begin your journey such that you complete the entire circuit in a clockwise

direction without running out of fuel at any point.

Given two integer arrays:

$gas[i]$ represents the fuel available at the i -th station,

$cost[i]$ represents the fuel required to travel from station i to station $i+1$ (circularly),

Return the starting station index that allows completing the full trip.

If no such station exists, return -1.

Example 1

Input:

5

1 2 3 4 5

3 4 5 1 2

Output:

3

Explanation:

Start at station 3 (index 3) and fill up with 4 units of gas. Your tank = $0 + 4 = 4$

Travel to station 4. Your tank = $4 - 1 + 5 = 8$

Travel to station 0. Your tank = $8 - 2 + 1 = 7$

Travel to station 1. Your tank = $7 - 3 + 2 = 6$

Travel to station 2. Your tank = $6 - 4 + 3 = 5$

Travel to station 3. The cost is 5. Your gas is just enough to travel back to station 3.

Therefore, return 3 as the starting index.

Example 2

Input

3

2 3 4

3 4 3

Output:

-1

Explanation:

You can't start at station 0 or 1, as there is not enough gas to travel to the next station.

Let's start at station 2 and fill up with 4 units of gas. Your tank = $0 + 4 = 4$

Travel to station 0. Your tank = $4 - 3 + 2 = 3$

Travel to station 1. Your tank = $3 - 3 + 3 = 3$

You cannot travel back to station 2, as it requires 4 units of gas but you only have 3.

Therefore, you can't travel around the circuit once no matter where you start.

Input Format

The first line of input consists of an integer N , representing the number of gas stations.

The second line of input consists of N space-separated integers representing the amount of gas available at each station.

The third line of input consists of N space-separated integers representing the cost of traveling from each station to the next.

Output Format

The output prints an integer representing the index of the starting gas station that allows you to complete the circuit, or -1 if no such station exists.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 5

1 2 3 4 5

3 4 5 1 2

Output: 3

Answer

// You are using GCC

#include <stdio.h>

```
int main() {
    int n;
    scanf("%d", &n);

    int gas[n], cost[n];

    for (int i = 0; i < n; i++) {
        scanf("%d", &gas[i]);
    }

    for (int i = 0; i < n; i++) {
        scanf("%d", &cost[i]);
    }

    int totalGas = 0, totalCost = 0;
    for (int i = 0; i < n; i++) {
        totalGas += gas[i];
        totalCost += cost[i];
    }

    // If total gas is less than total cost, impossible
    if (totalGas < totalCost) {
        printf("-1\n");
        return 0;
    }

    int start = 0;
    int tank = 0;
```



```
for (int i = 0; i < n; i++) {  
    tank += gas[i] - cost[i];  
  
    // If tank becomes negative, we can't start from current start  
    if (tank < 0) {  
        start = i + 1;  
        tank = 0;  
    }  
}  
  
printf("%d\n", start);  
  
return 0;  
}
```

Status : Correct

Marks : 10/10

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2028_NeoColab_Competitive Coding II_L9

VTU_Week 12_PAH

Attempt : 1

Total Mark : 40

Marks Obtained : 40

Section 1 : Coding

1. Problem Statement

Given an amount, find the minimum number of notes of different denominations that sum up to the given amount. Starting from the highest denomination note, try to accommodate as many notes as possible for a given amount.

We may assume that we have infinite supply of notes of values {2000, 500, 200, 100, 50, 20, 10, 5, 1}

Examples:

Input: 800

Output: Currency Count

500 : 1

200 : 1

100 : 1

Input: 2456

Output: Currency Count

2000 : 1

200 : 2

50 : 1

5 : 1

1 : 1

Input Format

The input contains an integer representing the amount to be broken down.

Output Format

The output prints "Currency Count" followed by each denomination used in the format:

"note_value : count"

(Only print denominations that are used.)

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 800

Output: Currency Count

500 : 1

200 : 1

100 : 1

Answer

```
// You are using GCC
#include <stdio.h>
```

```

int main() {
    int amount;
    scanf("%d", &amount);

    // Denominations in descending order
    int denominations[] = {2000, 500, 200, 100, 50, 20, 10, 5, 1};
    int numDenominations = sizeof(denominations) / sizeof(denominations[0]);

    printf("Currency Count\n");

    for (int i = 0; i < numDenominations; i++) {
        if (amount >= denominations[i]) {
            int count = amount / denominations[i];
            printf("%d : %d\n", denominations[i], count);
            amount = amount % denominations[i];
        }
    }

    return 0;
}

```

Status : Correct

Marks : 10/10

2. Problem Statement

Given a lock which is made up of n-different circular rings and each ring has 0-9 digit printed serially on it. Initially all n-rings together show a n-digit integer but there is particular code only which can open the lock.

You can rotate each ring any number of time in either direction. You have to find the minimum number of rotation done on rings of lock to open the lock.

Examples:

Input: Input = 2345, Unlock code = 5432

Output: Rotations required = 8

Explanation: 1st ring is rotated thrice as 2->3->4->5

2nd ring is rotated once as 3->4

3rd ring is rotated once as 4->3

4th ring is rotated thrice as 5->4->3->2

Input: Input = 1919, Unlock code = 0000

Output: Rotations required = 4

Explanation: 1st ring is rotated once as 1->0

2nd ring is rotated once as 9->0

3rd ring is rotated once as 1->0

4th ring is rotated once as 9->0

Input Format

The first line contains an integer representing the current input on the lock.

The second line contains an integer representing the unlock code.

Output Format

The output prints "Minimum Rotation = X" where X is the minimum total number of rotations needed.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 28756

98234

Output: Minimum Rotation = 12

Answer

```
// You are using GCC
```

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <string.h>
```

```
int main() {
```

```

char current[100], unlock[100];
scanf("%s", current);
scanf("%s", unlock);

int len = strlen(current);
int totalRotations = 0;

for (int i = 0; i < len; i++) {
    int currDigit = current[i] - '0';
    int targetDigit = unlock[i] - '0';

    int forward = (targetDigit - currDigit + 10) % 10;
    int backward = (currDigit - targetDigit + 10) % 10;
    int minRotation = (forward < backward) ? forward : backward;
    totalRotations += minRotation;
}

printf("Minimum Rotation = %d\n", totalRotations);

return 0;
}

```

Status : Correct

Marks : 10/10

3. Problem Statement:

Sarah is organizing a race in her local community park, where participants must jump from one marker to the next. Each participant starts at the first marker and can make jumps to other markers based on the distance provided for each marker. Sarah wants to calculate the minimum number of jumps it would take for a participant to reach the final marker.

You are tasked with writing a program that helps Sarah determine the minimum number of jumps required for a participant to reach the last marker.

Each element in the array represents the maximum number of steps a participant can jump forward from that marker. For example, if you're at marker i , you can jump to any marker within the range $[i+1, i + \text{nums}[i]]$.

Write a program to calculate the minimum number of jumps to reach the last marker.

Example 1:

Input: nums = 2 3 1 1 4

Output: 2

Explanation: The minimum number of jumps to reach the last index is 2.
Jump 1 step from index 0 to 1, then 3 steps to the last index.

Example 2:

Input: nums = 2 3 0 1 4

Output: 2

Input Format

The input consists of space-separated integers representing the jump lengths from each index.

Output Format

The output prints an integer, representing the minimum number of jumps required to reach the last index.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 2 3 1 1 4

Output: 2

Answer

```
// You are using GCC
#include <stdio.h>
```

```
int main() {
    int nums[10000];
    int n = 0;
```

```

// Read space-separated integers until newline
while (scanf("%d", &nums[n]) == 1) {
    n++;
    if (getchar() == '\n') break;
}

if (n <= 1) {
    printf("0\n");
    return 0;
}

int jumps = 0;
int currentEnd = 0;
int farthest = 0;

for (int i = 0; i < n - 1; i++) {
    // Update farthest we can reach from current position
    if (i + nums[i] > farthest) {
        farthest = i + nums[i];
    }

    // If we've reached the end of current jump range
    if (i == currentEnd) {
        jumps++;
        currentEnd = farthest;

        // If we can already reach the last index, break
        if (currentEnd >= n - 1) {
            break;
        }
    }
}

printf("%d\n", jumps);

return 0;
}

```

Status : Correct

Marks : 10/10

4. Problem Statement

Shoot the Kuruvi (Balloons)

There are some spherical balloons taped onto a flat wall that represent the XY-plane. The balloons are represented as a 2D integer array points, where $\text{points}[i] = [\text{xstart}, \text{xend}]$ denotes a balloon whose horizontal diameter stretches between xstart and xend . You do not know the exact y-coordinates of the balloons.

Arrows can be shot up directly vertically (in the positive y-direction) from different points along the x-axis. A balloon with xstart and xend is burst by an arrow shot at x if $\text{xstart} \leq x \leq \text{xend}$. There is no limit to the number of arrows that can be shot. A shot arrow keeps traveling up infinitely, bursting any balloons in its path.

Given the number of array points, followed by the array points. Return the minimum number of arrows that must be shot to burst all balloons.

Example

Input

4

10 16

2 8

1 6

7 12

Output

2

Explanation

The balloons can be burst by 2 arrows:

Shoot an arrow at $x = 6$, bursting the balloons $[2,8]$ and $[1,6]$.

Shoot an arrow at $x = 11$, bursting the balloons $[10,16]$ and $[7,12]$.

Input

4

1 2

3 4

5 6

7 8

Output

4

Explanation

One arrow needs to be shot for each balloon for a total of 4 arrows.

Input

4

1 2

2 3

3 4

4 5

Output

2

Explanation

The balloons can be burst by 2 arrows:

Shoot an arrow at $x = 2$, bursting the balloons $[1,2]$ and $[2,3]$.

Shoot an arrow at $x = 4$, bursting the balloons $[3,4]$ and $[4,5]$.

Input Format

The first line of input consists of a single integer n , representing the Number of balloons

The next n lines contain two integers x_{start} and x_{end} , representing the start and

end positions of each balloon

Output Format

The output prints a single integer, which represents the minimum number of arrows needed to burst all the balloons.

Refer to the output for the formatting specifications.

Sample Test Case

Input: 4

1 2

2 3

3 4

4 5

Output: 2

Answer

// You are using GCC

#include <stdio.h>

#include <stdlib.h>

typedef struct {

int start;

int end;

} Balloon;

int compare(const void *a, const void *b) {

Balloon *balloonA = (Balloon *)a;

Balloon *balloonB = (Balloon *)b;

return balloonA->end - balloonB->end;

}

int main() {

int n;

scanf("%d", &n);

Balloon balloons[n];

for (int i = 0; i < n; i++) {

```
        scanf("%d %d", &balloons[i].start, &balloons[i].end);
    }

    // Sort balloons by their end coordinate
    qsort(balloons, n, sizeof(Balloon), compare);

    int arrows = 1;
    int arrowPos = balloons[0].end;

    for (int i = 1; i < n; i++) {
        // If current balloon starts after the arrow position,
        // we need a new arrow
        if (balloons[i].start > arrowPos) {
            arrows++;
            arrowPos = balloons[i].end;
        }
    }

    printf("%d\n", arrows);

    return 0;
}
```

Status : Correct

Marks : 10/10